

Who Gets to Live in Productive Cities? Family Transfers, Housing Costs, and Spatial Sorting

Weikang Hu

The Chinese University of Hong Kong, Shenzhen

2001 Longxiang Road, Longgang District, Shenzhen, China

Corresponding author: weikanghu@link.cuhk.edu.cn

September 17, 2026

Abstract

High housing costs may limit young workers' access to productive cities, especially by suppressing homeownership. Family transfers may thus play a crucial role in shaping where young households live and whether they rent or purchase housing. I develop a quantitative life-cycle spatial equilibrium model in which parental transfers affect those decisions as well as mortgage and saving choices. Calibrated to urban China, the model implies that transfers increase the share of young households living in the high-productivity region by 3.3 percent and the share owning there by 39.3 percent, relative to an equilibrium without inter vivos transfers. The smaller residence response reflects the availability of rental housing, which allows young workers to access productive labor markets without financing a down payment. Transfers therefore have modest aggregate output effects but large distributional consequences. They increase young households' mean home equity by RMB 75,900 and raise equilibrium house prices. Welfare gains are concentrated among young and lower-talent households receiving parental transfers, while higher purchase prices can reduce welfare for buyers with little family support. Transfers broaden ownership in productive cities among lower-talent children, yet widen ownership and wealth gaps by parental wealth. Policy experiments show that family transfers partly substitute for credit expansion and rental assistance at entry margins, while remaining important for housing-wealth accumulation.

Keywords: spatial sorting; housing tenure; intergenerational transfers; family finance; housing affordability; China.

JEL Classification: D64; E21; J61; R21; R23.

1 Introduction

High-productivity cities offer higher wages, stronger career prospects, and access to denser labor markets, while their high housing costs can restrict workers' access to these opportunities and generate aggregate productivity losses (Hsieh and Moretti, 2019). Young workers are particularly constrained because they typically have limited liquid wealth to meet large down-payment requirements. When their labor income and accumulated savings are insufficient to cover these upfront costs, parental transfers can bridge the financing gap. Such support may shape where young adults live and work, whether they become homeowners, and how much wealth they accumulate over the life cycle.

Family support for housing is widespread across countries. In the United States, parental financial assistance helps young adults meet down-payment requirements and enter homeownership (Lee et al., 2020). Intergenerational support is also an important source of housing finance in China, where housing prices in major cities are high relative to household earnings (Chen and Wen, 2017, Fang et al., 2016, Meng et al., 2023b). Across these settings, parental resources supplement young households' own income and savings at a stage when housing decisions require substantial financial commitments.

In this paper, I examine how family wealth shapes young households' access to productive labor markets, homeownership within those markets, and subsequent wealth accumulation. I trace the macroeconomic implications for equilibrium housing prices and rents, aggregate productivity, and the distribution of wealth and welfare. The aggregate consequences of parental support are ambiguous. Transfers can enable financially constrained households to enter productive locations or purchase housing after entering. Transfer-financed demand can also raise purchase prices and disadvantage prospective buyers who receive little or no family support. Moreover, when rental housing is available, an inability to buy does not necessarily exclude a household from the productive city: a young adult who cannot finance ownership may continue to live and work there as a renter. The effects of family finance therefore depend on households' joint location and tenure choices and the resulting adjustment of housing markets.

To study these mechanisms, I develop a six-stage overlapping-generations model with two regions that differ in productivity and housing costs. Region A represents large, high-productivity cities, while Region B offers lower earnings and more affordable housing. Households differ in talent, geographic origins, family resources, and their financial and housing positions. They choose their region of residence, housing tenure and size, liquid saving, and mortgage borrowing. Altruistic parents aged 50–59 choose a one-time transfer to their young-adult children aged 20–29 in response to the children's earning capacity

and housing opportunities, subject to their own resource constraints. The model additionally incorporates mortgage amortization, home-equity accumulation, age-dependent migration costs, terminal estates, and stochastic inheritance. Purchase prices and rents are determined in equilibrium, with separate owner and rental markets whose relative supplies respond to market conditions.

The quantitative analysis uses urban Chinese data from the China Household Finance Survey and related housing-market sources to discipline earnings heterogeneity, regional housing costs, and age-specific location–tenure profiles. I quantify the role of parental support by comparing the benchmark with a counterfactual stationary equilibrium without inter vivos transfers. I evaluate four policies with and without family transfers to assess their economic effects and whether family finance complements or substitutes for public support: lower down-payment requirements, rental assistance, a talent subsidy, and a home-purchase voucher.

The main finding is that family transfers expand young households’ ownership in the productive region substantially more than their residence there. Among households aged 20–29, transfers increase the share living in Region A by 3.3 percent, or 1.13 percentage points, and the share owning a home there by 39.3 percent, or 4.04 percentage points, relative to the no-transfer equilibrium. An aggregate decomposition attributes approximately 90 percent of the ownership response to the change in ownership conditional on Region A residence. Renting provides access to the region without requiring a down payment, allowing young households to continue working there when purchasing is beyond their financial means.

The spatial response is concentrated among financially constrained households near the lower end of the talent distribution. In the benchmark economy, parental support enables more of these households to locate in Region A, increasing its working-age population. The expansion is accompanied by a decline in average effective productivity among Region A workers, since the additional residents have lower effective productivity than those who can finance access without parental assistance. This composition change partly offsets the effective-labor gain from the larger working population. Together with the availability of rental housing, it limits the aggregate-output increase to 0.22 percent.

Earlier ownership strengthens household balance sheets and improves welfare. Transfers raise mean home equity among households aged 20–29 by RMB 75,900 and reduce their share with nonpositive net worth by 50.3 percent, or 25.15 percentage points, relative to the no-transfer equilibrium. Young households also hold more liquid assets and less mortgage debt. These initial differences carry into subsequent stages through saving and mortgage repayment. The no-transfer economy requires a uniform 10.78-percent increase

in consumption and housing services to attain benchmark welfare. The largest welfare gains from family transfers accrue to young and lower-talent households whose housing choices are most sensitive to liquid resources.

The distributional implications differ across children's own earning capacity and parental wealth. Transfers relax financing constraints for low- and middle-talent young adults, narrowing their ownership gap with high-talent households and reducing inequality in young households' home equity. Access to this assistance, however, is uneven across families. Wealthier parents are more likely to give and provide larger amounts, enabling their children to finance purchases and accumulate housing equity that would be difficult to obtain from their own earnings alone. This creates a dynastic advantage in housing outcomes and a stronger initial position for subsequent wealth accumulation. The Region A ownership gap between children in the highest and lowest parental-wealth quartiles rises from 0.63 percentage points without transfers to 11.16 points when transfers are available.

Nonrecipients are also affected by family support through equilibrium purchase prices and rents. Transfer-supported households shift toward ownership, putting upward pressure on purchase prices and reducing pressure on the rental market. In Region A, the purchase price is 4.12 percent higher in the benchmark than in the no-transfer equilibrium, while rent is slightly lower. Prospective buyers without parental assistance consequently face higher purchase costs and larger required down payments. Nonrecipients who rent in Region A benefit from lower rental expenses, while existing owners hold more valuable housing assets.

The policy experiments show that public housing and place-based policies interact with family finance differently across spatial access, purchase entry, and housing investment. Policies that relax upfront financing constraints or reduce the cost of accessing productive cities can partly substitute for parental resources in location and purchase decisions. Yet family transfers remain important for households' leverage, housing investment, and equity accumulation after entry. By contrast, policies targeted toward higher-earning households generate location and ownership responses that depend little on family support, because intended beneficiaries face weaker financing constraints. Across policies, equilibrium increases in housing demand and fiscal costs can shift part of the burden to households that receive no direct support.

The paper contributes to three strands of research. The first examines housing costs, worker mobility, and the spatial allocation of labor (Behrens et al., 2014, Hsieh and Moretti, 2019, Bryan and Morten, 2019). Closely related quantitative studies evaluate housing-affordability policies and the regional effects of mortgage-credit conditions (Favilukis

et al., 2023, Mabilie, 2023). Research on China also links housing costs and collateral constraints to household migration (Garriga et al., 2023, Meng et al., 2023a, He et al., 2026). I introduce endogenous parental transfers as an additional source of finance for young households' migration and housing purchases. Parents choose support in response to their adult children's earning capacity and location-tenure opportunities, connecting family resources to the allocation of workers across regions and to the equilibrium incidence of housing and place-based policies.

Second, the paper extends research on parental assistance and homeownership among young households. Empirical studies examine how transfers affect purchase timing, down payments, and the size and financing of the dwelling acquired (Engelhardt and Mayer, 1998, Guiso and Jappelli, 2002, Lee et al., 2020, Meng et al., 2023b). Brandsaas (2025) studies endogenous parental transfers and housing-policy effects in a life-cycle framework. I introduce adult location choice across regions with different productivity and housing costs, making the spatial sorting of labor an outcome of intergenerational finance. The spatial setting allows me to assess how family finance affects aggregate productivity and the distribution of housing wealth and welfare across regions and households.

Third, the paper relates to the broader literature on intergenerational resources, economic opportunity, and wealth inequality. This literature examines how family background shapes education, occupational choice, and adult economic outcomes, and how transfers and bequests transmit wealth across generations (De Nardi, 2004, Nishiyama, 2002, Benhabib et al., 2011, Adermon et al., 2018, Abbott et al., 2019, Boar and Lashkari, 2021, Biasi, 2023). Related housing research studies the transmission of parental housing wealth to children's homeownership and wealth, as well as the effects of local housing markets on children's education and future income (Daysal et al., 2023, Gilraine et al., 2023). Chyn and Daruich (2025) evaluate housing vouchers and place-based subsidies in a general-equilibrium model with neighborhood effects, endogenous location choice, and child development. I examine direct monetary support when adult children enter labor and housing markets and trace its consequences for their subsequent balance sheets. The framework provides a housing-finance channel through which parental resources shape adult economic status and permits a joint assessment of disparities associated with children's earning capacity and those associated with family background.

The remainder of the paper is organized as follows. Section 2 presents the model, and Section 3 describes the calibration. Section 4 characterizes the benchmark equilibrium. Section 5 quantifies the general-equilibrium effects of inter vivos family transfers. Section 6 evaluates the four policy experiments and their interaction with private family finance.

Section 7 concludes.

2 An Intergenerational Spatial Model of Housing and Talent Allocation

This section develops a six-stage overlapping-generations model with two regions that incorporates household location and tenure choice, mortgage borrowing, inter vivos family transfers, and stochastic inheritance, with house prices and rents determined in equilibrium. Region A represents large cities with higher total factor productivity (TFP) but higher housing costs. Region B represents smaller, less productive cities where lower housing costs make homeownership more affordable. This two-region structure abstracts from the full geography of many cities, but it captures the central trade-off between productive labor-market opportunities and housing affordability.

The economy is populated by overlapping generations of heterogeneous households that differ in talent and family resources. A model period spans ten years, and each household lives for six periods, corresponding to ages 20–29, 30–39, 40–49, 50–59, 60–69, and 70–79. Households supply labor inelastically from ages 20–69 and retire at ages 70–79. The family link becomes active when a household enters the 50–59 age group and is paired with a child aged 20–29. After receiving its inheritance and observing the child’s talent, the altruistic parent chooses a monetary transfer. The child inherits its origin region from the parent’s beginning-of-period location, begins adult life without housing or mortgage debt, and makes an independent location and tenure choice. Households exit at the end of the 70–79 stage and leave terminal net estates. These estates determine the common inheritance distribution described below. There is no mortality before the terminal age. Population is stationary, with equal mass in each age group and total population normalized to one.

At each age, a household selects a location–tenure alternative: renting or owning in Region A or Region B. An incumbent homeowner may instead keep the existing house. Conditional on this discrete choice, the household chooses consumption, housing size, next-period liquid assets, and, when purchasing a house, mortgage borrowing. These decisions depend on talent, the household’s current balance sheet, housing prices and rents, and any family transfer or inheritance received. Home purchases must satisfy the household budget and loan-to-value and payment-to-income limits. Family resources can therefore affect both tenure and location by helping finance a down payment or entry into the high-cost Region A.

2.1 Preferences

Let $g \in \mathcal{G} \equiv \{1, \dots, 6\}$ index ten-year age groups, ordered from ages 20–29 ($g = 1$) to ages 70–79 ($g = 6$). Let $j \in \mathcal{J} \equiv \{rB, oB, rA, oA\}$ index location–tenure choices, where r and o denote renting and owning and A and B denote the two regions. Incumbent homeowners may also keep their current house.

Conditional on consumption c , housing services h , and alternative j , the deterministic flow utility of an age-group- g household is

$$u^g(c, h, j) = \frac{\mathcal{C}(c, h)^{1-\gamma}}{1-\gamma} + \xi_j^g, \quad (1)$$

where the consumption–housing composite is

$$\mathcal{C}(c, h) = ((1 - \alpha_h)c^\epsilon + \alpha_h h^\epsilon)^{\frac{1}{\epsilon}}. \quad (2)$$

The parameter γ controls curvature, α_h is the weight on housing, and the elasticity of substitution between consumption and housing is $1/(1 - \epsilon)$. For each age group g , the alternative-specific constants are collected in

$$\boldsymbol{\xi}^g \equiv (\xi_{rB}^g, \xi_{oB}^g, \xi_{rA}^g, \xi_{oA}^g) = (0, \xi_{oB}^g, \xi_{rA}^g, \xi_{oA}^g). \quad (3)$$

Only relative utility levels are identified, so renting in Region B is normalized to zero. The remaining terms are deterministic, age-specific residual utility shifters. After accounting for consumption and housing services, prices, household states, Region A entry costs, recurring rental frictions, and continuation values, they absorb the average nonpecuniary attractiveness of each location–tenure combination, including unmodeled amenities, tenure preferences, and location-specific ownership considerations. They are common within each age group, vary freely across age groups, and are disciplined by the corresponding age-specific location–tenure shares. In particular, ξ_{oA}^g directly captures the residual attractiveness of owning in Region A compared with renting in Region B. A homeowner who keeps the current house receives ξ_{oB}^g in Region B or ξ_{oA}^g in Region A.

In each period, a household draws an idiosyncratic taste shock ν_j for each feasible location–tenure alternative. These shocks are independently drawn from a Type-I extreme-value distribution with scale σ_ν .

Households in two age groups have additional intergenerational utility terms. In age group $g = 4$ (50–59), the parent places weight α_p on the value of its linked child, evaluated at the family transfer chosen by the parent; hence, α_p governs the strength of the altruistic

transfer motive. In age group $g = 6$ (70–79), the household has a warm-glow bequest motive over its terminal net estate e :

$$\omega_b \frac{(e + \underline{e})^{1-\gamma}}{1 - \gamma}, \quad (4)$$

where ω_b governs the strength of the warm-glow motive and $\underline{e} > 0$ is a small shift. Terminal estates also determine the stochastic inheritance distribution described below.

2.2 Endowments, Income and Intergenerational Links

A household enters the labor market with idiosyncratic productivity z , which remains fixed over its life cycle. As a benchmark, child productivity is drawn independently of parental productivity (Galor and Tsiddon, 1997, Benhabib et al., 2011, Colas et al., 2021). Appendix E.3 relaxes this assumption by allowing positive intergenerational persistence in talent.

During working ages $g = 1, \dots, 5$, gross labor income in region ℓ is $y_\ell^g(z) = w \text{TFP}_\ell G_g z$, where w is the monetary wage per productivity-adjusted efficiency unit over a ten-year model period. The term G_g captures the life-cycle productivity profile, and TFP_ℓ captures regional productivity; I normalize $\text{TFP}_B = 1$. Retirement income is $y^{\text{pen}}(z) = \rho_{\text{pen}} w G_4 z$. The pension base $w G_4 z$ excludes regional productivity differences, so payments do not depend on the household's past or current location. Pension payments enter government expenditure and are financed from labor-tax revenue.

Within each linked parent–child pair, the altruistic parent chooses a one-time monetary transfer $f \geq 0$ in response to the child's earning capacity and housing opportunities, subject to its own resource constraints. The transfer increases the entrant's resources one-for-one and reduces the parent's resources by the same amount.

Inheritance is drawn once from the common stationary distribution of terminal estates, independently of the recipient's productivity and earlier transfer receipt. Appendix A.3 describes its construction. Figure 1 summarizes the life-cycle timing.

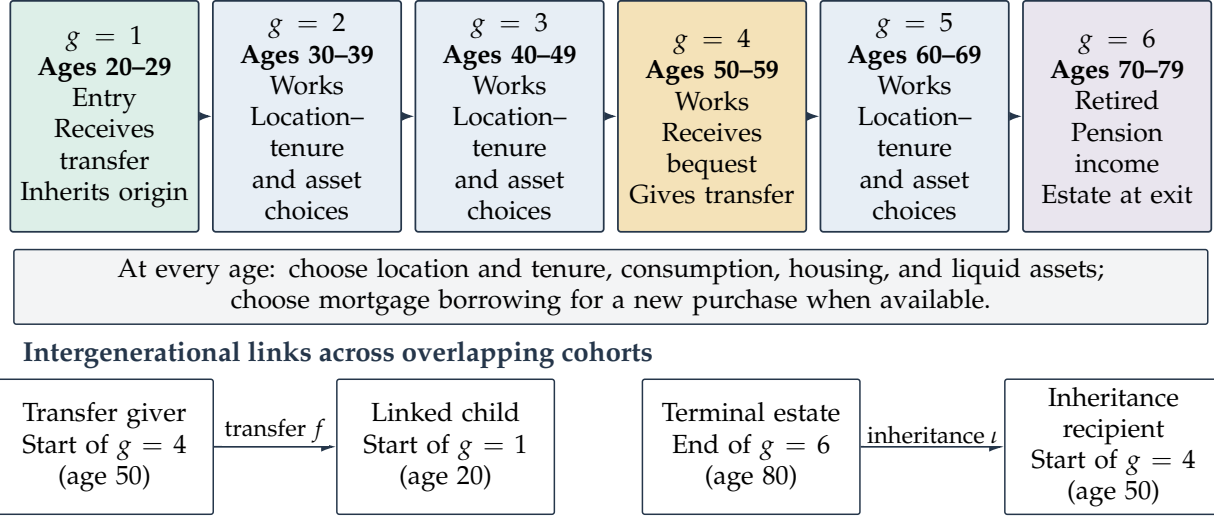


Figure 1: Six-period life cycle and intergenerational links

Notes: Each model period is ten years. A household at $g = 4$ gives transfer f to its linked $g = 1$ child. Inheritance ι is drawn from the stationary distribution of terminal estates. Estates are left at the end of $g = 6$, at the age-80 boundary, while inheritance is received at the beginning of $g = 4$, at age 50. This timing preserves the same 30-year generational gap as the inter vivos transfer link.

2.3 Housing, Mobility, and Mortgage Contracts

In each region $\ell \in \{A, B\}$, households take the purchase price $P_{\ell,t}$ and rent $R_{\ell,t}$ as given. A renter pays $R_{\ell,t}h$ for current housing services. A buyer pays $P_{\ell,t}h$ and carries the house and any mortgage into the next age. An incumbent owner may instead keep the current house, leaving its size fixed and ruling out a new mortgage.

Let ℓ_- , h_- , and m denote the location, size, and outstanding mortgage balance of a house owned at the beginning of the period. Housing depreciates at the model-period rate δ_h . If the household sells the house, the resources released by the sale are its home equity,

$$\mathcal{E}(\ell_-, h_-, m) = (1 - \delta_h)P_{\ell_-,t}h_- - m. \quad (5)$$

Sale equity can finance renting or another purchase. A keeper does not realize this equity and instead makes the scheduled mortgage payment. The feasible state space excludes negative home equity.

To capture institutional barriers such as China's *Hukou* system, a household moving from Region B to either rA or oA pays the age-specific entry cost $\kappa_{A,g}^{\text{entry}}$; an age-1 entrant's previous location is its parental origin. Choosing rA also incurs the recurring friction $\kappa_{rA,g}^{\text{rent}}$. This wedge captures costs associated with renters' weaker access to housing-linked

schools, health care, and other local services, which often depends on residence permits and registered leases (Zhang and Chen, 2018, General Office of the State Council of China, 2016, General Office of the Guangzhou Municipal Government, 2017). The combined cost is

$$\kappa^g(j; \ell_-) = \kappa_{A,g}^{\text{entry}} \mathbb{1}\{\ell_- = B, j \in \{rA, oA\}\} + \kappa_{rA,g}^{\text{rent}} \mathbb{1}\{j = rA\}. \quad (6)$$

The entry cost is paid whenever a household moves from B to A, including re-entry after a previous departure; the rental friction is paid in every period spent renting in A. Both frictions enter the budget as resource costs. A continuing homeowner in Region A pays neither.

Households can save in a liquid asset x and earn the gross return R_s ; $x' \geq 0$ rules out unsecured borrowing. A new purchase can be financed with mortgage M , subject to LTV and PTI limits:

$$0 \leq M \leq \theta_{\text{LTV}} P_{\ell,t} h, \quad (7)$$

$$\Pi_{K_g}(M) \leq \theta_{\text{PTI}} y_{\ell}^g(z), \quad (8)$$

where K_g is the available mortgage term and $\Pi_{K_g}(M)$ is the payment over the current model period. The LTV limit determines the minimum equity contribution, while the PTI limit ties debt service to gross labor income. Within these limits, the household budget determines whether liquid wealth, disposable income, sale equity, and net family transfers can cover the down payment, entry cost, and debt service while maintaining positive consumption and nonnegative saving.

Mortgages are fully amortizing. Let q denote the number of ten-year payment blocks remaining at the beginning of the period. A mortgage balance m requires the payment $\Pi_q(m) = \pi_q m$ and evolves according to

$$m' = \vartheta_q m, \quad q' = \max\{q - 1, 0\}, \quad (9)$$

where π_q and ϑ_q are determined by the mortgage rate and remaining maturity. A new mortgage begins with $q = K_g$ and balance M . When $q = 1$, it is repaid within the period. Selling repays the outstanding balance, whereas keeping preserves the scheduled amortization and rules out refinancing. New contracts are available only with a new purchase. Appendix A.2 derives the payment and remaining-balance coefficients.

2.4 Household Decisions over the Life Cycle

The household problem has a common location–tenure decision at every age and age-specific departures generated by transfers, inheritance, retirement, and bequests. Conditional on a discrete alternative, households optimize consumption, housing, liquid saving, and, for a new purchase, mortgage debt. In a stationary equilibrium, prices and distributions are time invariant; calendar-time subscripts are suppressed in the household problem below.

At ages 20–29, the household state is

$$s^1 = (z, f, \ell^{\text{org}}), \quad (10)$$

where z is lifetime productivity, f is the parental transfer received upon entry, and ℓ^{org} is the parent’s beginning-of-period region. The entrant begins without liquid assets, housing, or mortgage debt. For ages 30–79, the state is

$$s^g = (z, x, \ell_-, \pi_-, h_-, m, q; z_c \mathbb{1}\{g = 4\}), \quad (11)$$

where x denotes liquid assets; (ℓ_-, π_-) previous location and tenure; and (h_-, m, q) the housing and mortgage position of an incumbent owner. At ages 50–59, the household also observes the productivity z_c of its linked adult child entering ages 20–29.

The feasible location–tenure alternatives are

$$\mathcal{J}_g(s^g) = \begin{cases} \{rB, oB, rA, oA\}, & g = 1 \text{ or } \pi_- = r, \\ \{rB, oB, rA, oA, k\ell_-\}, & g \geq 2 \text{ and } \pi_- = o, \end{cases} \quad (12)$$

where $k\ell_-$ denotes keeping the current house. Entrants and renters choose among the four location–tenure alternatives; incumbent owners can also keep.

Let $v_j^g(s^g)$ denote the value conditional on alternative j after optimizing its controls. After observing the location–tenure taste shocks ν , the household chooses its preferred alternative and has value

$$V^g(s^g, \nu) = \max_{j \in \mathcal{J}_g(s^g)} \{v_j^g(s^g) + \nu_j\}. \quad (13)$$

Before the shocks are realized, its expected value is, up to an additive constant,

$$\bar{V}^g(s^g) = \sigma_\nu \log \left[\sum_{j \in \mathcal{J}_g(s^g)} \exp \left(\frac{v_j^g(s^g)}{\sigma_\nu} \right) \right], \quad (14)$$

and the probability of choosing alternative j is

$$\Pr(j \mid s^g) = \frac{\exp(v_j^g(s^g)/\sigma_\nu)}{\sum_{k \in \mathcal{J}_g(s^g)} \exp(v_k^g(s^g)/\sigma_\nu)}. \quad (15)$$

Disposable household income in region ℓ is

$$Y_\ell^{g,\text{disp}}(z) = \begin{cases} (1 - \tau_l)y_\ell^g(z) + T, & g \leq 5, \\ y^{\text{pen}}(z) + T, & g = 6. \end{cases} \quad (16)$$

Here $y_\ell^g(z) = w \text{TFP}_\ell G_g z$ is gross labor income, $y^{\text{pen}}(z) = \rho_{\text{pen}} w G_4 z$ is pension income, and T is the universal government transfer. Let $f_g^{\text{in}} = f$ at ages 20–29 and zero otherwise, and let $f_g^{\text{out}} = f$ at ages 50–59 and zero otherwise. If an incumbent owner sells, $\mathcal{E}(s^g)$ denotes the home equity in (5); it is zero for entrants and incumbent renters.

For a renter choosing region $\ell \in \{A, B\}$, the budget constraint is

$$\begin{aligned} c + x' + f_g^{\text{out}} + R_\ell h + \kappa^g(r\ell; \ell_-) &= Y_\ell^{g,\text{disp}}(z) + R_s x + f_g^{\text{in}} + \mathcal{E}(s^g), \\ x' &\geq 0, \quad h \geq \underline{h}. \end{aligned} \quad (17)$$

Here $\underline{h}$ denotes the minimum dwelling size. Because renters cannot originate a mortgage and must choose nonnegative liquid assets, they cannot borrow.

For a household purchasing a new house in region ℓ , the budget is

$$\begin{aligned} c + x' + f_g^{\text{out}} + P_\ell h - M + \Pi_{K_g}(M) + \kappa^g(o\ell; \ell_-) &= Y_\ell^{g,\text{disp}}(z) + R_s x + f_g^{\text{in}} + \mathcal{E}(s^g), \\ x' &\geq 0, \quad h \geq \underline{h}. \end{aligned} \quad (18)$$

The new mortgage is subject to

$$\begin{aligned} 0 &\leq M \leq \theta_{\text{LTV}} P_\ell h, \\ \Pi_{K_g}(M) &\leq \theta_{\text{PTI}} y_\ell^g(z). \end{aligned} \quad (19)$$

The first restriction imposes the LTV ceiling; the second limits debt service relative to gross labor income. Family transfers expand the resources available in the purchase budget but are excluded from the gross-income base used to calculate the PTI borrowing limit.

An incumbent owner that keeps its current house faces

$$c + x' + f_g^{\text{out}} + \Pi_q(m) = Y_{\ell_-}^{g,\text{disp}}(z) + R_s x + f_g^{\text{in}}, \quad x' \geq 0. \quad (20)$$

The keeper neither realizes sale equity nor pays rent, a purchase price, or a mobility cost; it retains its housing size, and its mortgage follows the scheduled transition in Appendix A.2.

For ages 20–49 and 60–69, conditional values have the common form

$$v_j^g(s^g) = \max_{\mathbf{a}_j \in \Gamma_j^g(s^g)} \left\{ u^g(c, h, j) + \beta \mathbb{E} \left[\bar{V}^{g+1}(s^{g+1}) \right] \right\}. \quad (21)$$

For renters, new buyers, and keepers, respectively, $\mathbf{a}_j$ is (c, h, x') , (c, h, x', M) , and (c, x') ; Γ_j^g imposes the corresponding budget, financing constraints, and state transition. Although equation (21) has the same recursive form at these ages, their feasible sets differ. At ages 20–29, f is a predetermined transfer received from the parent, $x = \mathcal{E} = 0$, and ℓ^{org} determines whether the Region A entry cost applies. No family transfer enters the budget at ages 30–49 or 60–69. At ages 40–49, the expectation also integrates over inheritance received upon entry into ages 50–59.

Upon entering ages 50–59, the household receives an inheritance ι , which augments its liquid-asset state:

$$x_4 = x'_3 + \frac{\iota}{R_s}. \quad (22)$$

Thus, inheritance raises current resources one-for-one.

The parent observes the productivity z_c of its linked child entering ages 20–29. The child's origin is the parent's beginning-of-period region ℓ_- and is unaffected by the parent's current location choice. The altruistic parent chooses a one-time monetary transfer $f \geq 0$ jointly with its own location, housing, and saving decisions. It anticipates the child's optimal response given z_c , f , and ℓ_- , averaging over the child's idiosyncratic location–tenure taste shocks. Conditional on its own alternative j , the parent chooses the action-specific controls $\mathbf{a}_j$ and transfer f :

$$v_j^4(s^4) = \max_{(\mathbf{a}_j, f) \in \Gamma_j^4(s^4)} \left\{ u^4(c, h, j) + \beta \mathbb{E} \left[\bar{V}^5(s^5) \right] + \alpha_p \bar{V}^1(z_c, f, \ell_-) \right\}, \quad f \geq 0. \quad (23)$$

The altruistic term $\alpha_p \bar{V}^1(z_c, f, \ell_-)$ places weight α_p on the child's optimized expected lifetime utility at the chosen transfer. Parental giving expands the child's budget while reducing the resources available for the parent's own consumption, housing, and saving.

At ages 70–79, pension income replaces labor income and is independent of retirement

location. New mortgage borrowing is unavailable, so a purchase must be financed from liquid resources and realized home equity. The continuation value is replaced by the warm-glow utility in (4), evaluated at the terminal estate

$$e = x' + \mathbb{1}\{\pi' = o\} [(1 - \delta_h)P_{\ell'}h' - m']. \quad (24)$$

Terminal estates generate the inheritance distribution received at ages 50–59. Appendix A.1 gives detailed Bellman problems and budgets for entrants, moving and keeping owners, transfer-giving parents, and retirees.

2.5 Location and Tenure Trade-Offs

Location and tenure are jointly determined by the return to talent and household financing capacity. Region A's productivity premium raises the payoff to locating there, especially for high-talent workers, while its higher housing and entry costs make that choice depend on liquid assets, home equity, inheritance, and family transfers. Renting provides access to Region A's labor market without the down-payment and mortgage requirements, while Region A renters bear a recurring friction and do not accumulate housing equity. Ownership requires more upfront financing but builds equity. Family transfers can therefore change both spatial sorting and tenure by relaxing an entrant's financing constraint, while reducing the donor's resources for consumption, housing, and saving.

2.6 Firms

2.6.1 Consumption-Good Producer

The homogeneous consumption good is freely traded across regions and is the numeraire. In region $\ell \in \{A, B\}$, a representative competitive firm uses a linear technology. Let $\mu_{\ell,t}^g$ denote the measure of age- g workers residing in region ℓ . These measures are expressed relative to the total population: each age group has mass $1/6$ across the two regions, so age weights are already incorporated. Regional effective labor and output are

$$L_{\ell,t} = \sum_{g=1}^5 \int G_g z d\mu_{\ell,t}^g(z), \quad Y_{\ell,t} = \text{TFP}_{\ell} L_{\ell,t}. \quad (25)$$

Only ages $g = 1, \dots, 5$ supply labor. Linearity, competition, and positive production imply a normalized wage of one per productivity-adjusted efficiency unit. This normalization is distinct from the monetary wage scale w in household budgets, which is fixed in the calibration.

2.6.2 Housing Supply

Residential investment in region $\ell \in \{A, B\}$ follows the reduced-form supply schedule

$$I_{\ell,t} = \bar{I}_\ell P_{\ell,t}^{\rho_\ell}, \quad (26)$$

where $I_{\ell,t}$ is newly constructed floor space, $\bar{I}_\ell$ is a region-specific supply shifter, and ρ_ℓ is the price elasticity of housing supply. Allowing both parameters to vary by region captures differences in land availability and construction constraints. The housing stock evolves as

$$H_{\ell,t} = (1 - \delta_h)H_{\ell,t-1} + I_{\ell,t}. \quad (27)$$

Owner-occupied and rental housing constitute separate market segments, but the allocation of the regional stock across them can respond to relative returns (Mabille, 2023, Greenwald and Guren, 2025). Let $s_{\ell,t}^r$ denote the rental share of the housing stock and let $(P_{\ell,0}, R_{\ell,0}, s_{\ell,0}^r)$ denote its reference-equilibrium values. The rental share satisfies

$$\log\left(\frac{s_{\ell,t}^r}{1 - s_{\ell,t}^r}\right) = \log\left(\frac{s_{\ell,0}^r}{1 - s_{\ell,0}^r}\right) + \chi_\ell \left[\log\left(\frac{R_{\ell,t}}{R_{\ell,0}}\right) - \log\left(\frac{P_{\ell,t}}{P_{\ell,0}}\right) \right], \quad (28)$$

and segment supplies are

$$H_{\ell,t}^{\text{rent}} = s_{\ell,t}^r H_{\ell,t}, \quad H_{\ell,t}^{\text{own}} = (1 - s_{\ell,t}^r) H_{\ell,t}. \quad (29)$$

An increase in rent relative to the purchase price shifts housing toward the rental segment. The parameter χ_ℓ governs this reallocation: when it is zero, the owner–rental split is fixed; positive values permit partial conversion while preserving segmentation between the two markets. Appendix A.4 gives the closed-form rental-share rule, the reference-equilibrium supply backout, and the segment-market residuals used in the equilibrium computation.

2.7 Government

The government levies a proportional tax τ_l on the gross labor income of working-age households. Retirees do not pay labor-income tax and receive the pension $y^{\text{pen}}(z)$ defined above. Every household also receives the uniform lump-sum transfer T_t . Family transfers and inheritances are private and are neither taxed nor counted as government expenditure in the baseline.

Let $\mu_{\ell,t}^g$ denote the measure of age- g households residing in region ℓ . The government

does not issue debt, and its period budget constraint is

$$\tau_l \sum_{g=1}^5 \sum_{\ell \in \{A,B\}} \int y_\ell^g(z) d\mu_{\ell,t}^g(z) = \sum_{\ell \in \{A,B\}} \int y^{\text{pen}}(z) d\mu_{\ell,t}^6(z) + T_t \sum_{g=1}^6 \sum_{\ell \in \{A,B\}} \int d\mu_{\ell,t}^g(z). \quad (30)$$

Thus, T_t adjusts so that labor-tax revenue equals spending on pensions and universal transfers.

2.8 Equilibrium

The model is solved as a stationary recursive equilibrium, so prices, rents, policies, and distributions are time invariant. Because production is linear and labor supply is inelastic, regional effective labor is determined by households' location choices; there is no separate labor-demand margin.

In the housing market, stationarity requires residential investment to replace depreciated housing, which implies

$$H_\ell = \frac{\bar{I}_\ell P_\ell^{\rho_\ell}}{\delta_h}. \quad (31)$$

Let $\mu_{o\ell}^g$ and $\mu_{r\ell}^g$ denote the stationary measures of owners and renters in region ℓ . The owner-occupied and rental segments clear separately:

$$\sum_{g=1}^6 \int h d\mu_{o\ell}^g = (1 - s_\ell^r) H_\ell, \quad (32)$$

$$\sum_{g=1}^6 \int h d\mu_{r\ell}^g = s_\ell^r H_\ell, \quad \ell \in \{A, B\}, \quad (33)$$

where s_ℓ^r follows (28). The four prices (P_A, P_B, R_A, R_B) clear the owner and rental segments in both regions.

The gross liquid-asset return and mortgage-contract terms are exogenous. During the sample period, banks priced household deposits and mortgages within national interest-rate benchmarks and prudential rules, rather than through local housing-market clearing (People's Bank of China, 2024). The financial sector elastically absorbs household saving and supplies mortgage credit at these rates; consequently, aggregate liquid assets need not equal aggregate mortgage debt. Their difference is reported as net financial assets, but it is not an equilibrium condition. The uniform transfer T adjusts to satisfy the government budget constraint (30).

Let $\mathcal{P}$ collect the household policy functions and location–tenure choice probabilities, let μ denote the distribution over age and household states, and let $\mathcal{B}$ denote the inheri-

tance distribution. The policies and the demographic, productivity, and intergenerational transitions described above induce a law of motion $\mathcal{T}$ for the cross-sectional distribution. Stationarity requires

$$\mu = \mathcal{T}(\mu; \mathcal{P}, \mathcal{B}). \quad (34)$$

The inheritance distribution must also be internally consistent: the discrete distribution from which households entering age group $g = 4$ draw must coincide with the corresponding discretization of terminal net estates generated by age-6 households under μ and $\mathcal{P}$.

A stationary recursive equilibrium consists of household value and policy functions $\mathcal{P}$, a stationary distribution μ , an inheritance distribution $\mathcal{B}$, regional housing stocks and rental shares, and (P_A, P_B, R_A, R_B, T) such that households optimize, competitive production satisfies (25), all four housing-market segments clear, the government budget balances, and the cross-sectional and inheritance distributions are stationary and mutually consistent.

3 Calibration

I calibrate the stationary equilibrium using the 2017 China Household Finance Survey (CHFS), a large-scale household survey with information on demographics, labor income, location, housing tenure, dwelling characteristics, home values, and rents. I restrict the sample to urban households because the model describes housing accessed through market prices and mortgage contracts and an earnings process that maps naturally to urban labor markets; rural housing is often obtained through village-allocated homestead land. Following the conventional city-tier classification used in Chinese policy and housing-market discussions, Region A comprises Tier 1 and Tier 2 cities, while Region B contains the remaining urban cities. Appendix B documents the sample, complete city classification, and construction of each target.

A model period is ten years, with six age groups spanning ages 20–29 through 70–79. I match the corresponding cross-sectional moments in the stationary equilibrium; the terminal survey group pools respondents aged 70 and older.

Internally calibrated parameters are chosen using a simulated method of moments, externally assigned parameters come from the literature or institutional sources, and parameters with direct empirical counterparts are estimated outside the model. Monetary quantities are expressed in units of 10,000 RMB, with housing prices measured per square meter. Table 1 reports each parameter’s value, classification, and empirical discipline.

Table 1: Calibration: Main parameters

Parameter	Explanation	Value	Source/target
Panel A. Preferences			
Internally calibrated			
$\{\xi_{oB}^g, \xi_{rA}^g, \xi_{oA}^g\}_{g=1}^6$	Age-specific choice constants	Appendix Table B.5	Age-choice shares
σ_ν	Taste-shock scale	0.0146	Working-age A/B income ratio: 2.0
Externally assigned			
β	Ten-year discount factor	0.5987	Annual factor: 0.95
γ	Risk aversion	2.000	Mabille (2023)
ϵ	CES curvature	-2.000	Substitution elasticity: 1/3
α_h	Housing weight	0.500	Equal CES weights
α_p	Parental altruism	0.400	Boar and Lashkari (2021), etc.
ω_b	Warm-glow weight	0.100	De Nardi (2004)
Panel B. Labor income			
Estimated from data and normalizations			
$\{G_g\}_{g=1}^6$	Age-efficiency profile	(0.844, 1.000, 0.968, 0.911, 0.670, 0)	CHFS regression
TFP_A, TFP_B	Region productivity	1.37, 1	CHFS regression
w^{ann}	Annual earnings scale	5.000	Mean $z = 1$; RMB 50,000
Panel C. Mobility costs			
Externally assigned			
$\{\kappa_{A,g}^{\text{entry}}\}_{g=1}^6$	Region A entry cost	(4, 5, 6, 7, 8, 9)	Morten and Oliveira (2016)
$\{\kappa_{rA,g}^{\text{rent}}\}_{g=1}^6$	Region A rental friction	(2, 2, 2.5, 3, 3, 3)	Talent sorting discipline
Panel D. Housing market			
Internally calibrated			
$\Omega_{A,0}, \Omega_{B,0}$	Owner-space shares	0.562, 0.793	CHFS prices and rents
$\bar{I}_A, \bar{I}_B$	Construction shifters	1.386, 9.497	CHFS prices and rents
Externally assigned			
ρ_A, ρ_B	Supply elasticities	0.65, 0.95	Huang (2022)
χ_A, χ_B	Owner-rental elasticities	1.00, 1.00	Unit-elastic benchmark
δ_h	Ten-year depreciation	0.140	Kaplan et al. (2020)
Panel E. Financial market			
Externally assigned			
r_s^{ann}	Liquid-asset return	0.0175	One-year RMB deposit rate
r_m^{ann}	Mortgage rate	0.0501	PBC, 2017:Q3
θ_{LTV}	LTV limit	0.70	30% down payment
θ_{PTI}	PTI limit	0.50	Mortgage regulation

Continued on next page

Table 1: Calibration: Main parameters (continued)

Parameter	Explanation	Value	Source/target
$\{K_g\}_{g=1}^6$	Mortgage term (ten-year blocks)	(3, 3, 3, 2, 1, 0)	30-year maximum
Panel F. Fiscal system			
Externally assigned			
ρ_{pen}	Pension replacement rate	0.50	Fang and Feng (2020); IMF (2022)
τ_l	Labor-income tax rate	0.10	Urban average tax rate
Internally calibrated			
T	Universal transfer	0.4339	Balanced government budget

Notes: Parameters are internally calibrated, estimated from data and normalizations, or externally assigned. Monetary values are in units of 10,000 RMB. Appendix Table B.4 reports inheritance discretization and secondary institutional parameters.

3.1 Preferences

The consumption–housing composite assigns equal CES weights to consumption and housing, $\alpha_h = 0.5$. Because the aggregator is not Cobb–Douglas, this parameter is a utility weight rather than a fixed housing expenditure share. I set $\epsilon = -2$, implying an intratemporal elasticity of substitution $1/(1 - \epsilon) = 1/3$. I set relative risk aversion to $\gamma = 2$, as in Mabilie (2023). I set the annual discount factor to 0.95, within the range used in quantitative life-cycle housing models (Campbell and Cocco, 2003, Favilukis et al., 2017), yielding the ten-year model-period value $\beta = 0.95^{10} = 0.5987$. I assign the parental altruism weight $\alpha_p = 0.4$, guided by the range used in quantitative models of intergenerational transfers. Boar and Lashkari (2021), for example, adopt a weight of 0.5. I set the warm-glow weight on terminal estates to $\omega_b = 0.10$, following the standard bequest-motive specification in De Nardi (2004). Appendix E.2 examines alternative consumption–housing curvature and parental altruism.

The deterministic location–tenure constants are calibrated separately for each age group. I normalize the constant for renting in Region B to zero and choose ξ_{oB}^g , ξ_{rA}^g , and ξ_{oA}^g to reproduce the joint age-specific shares of renting and owning in the two regions. Because the four shares sum to one, they provide three independent moments for each age group. The calibrated constants capture residual differences in the nonpecuniary attractiveness of the three alternatives relative to renting in Region B after accounting for the explicit income, housing-cost, mobility-cost, and financing channels in the model. Appendix Table B.5 reports their calibrated values and the corresponding target shares.

3.2 Labor Income

Using winsorized annual labor income for working-age CHFS respondents, I estimate the age-efficiency profile and the distribution of permanent productivity. I normalize efficiency at ages 30–39 to one; Table 1 reports the age profile. I set $G_6 = 0$ because the final age group is retired.

Weighted maximum likelihood gives the Fréchet shape $\theta_z = 1.9266$ and scale $s_z = 0.5430$ for normalized residual productivity. A separate weighted log-earnings regression with age controls gives $TFP_A = 1.37$ and $TFP_B = 1$. Conditional on this regional productivity gap, I set $\sigma_\nu = 0.0146$ to match a working-age average gross-income ratio of 2.0 between Regions A and B, the model’s sorting target.

I set the pension replacement rate to $\rho_{\text{pen}} = 0.50$, close to China’s 46 percent average replacement ratio in 2017 and within the 40–60 percent adequacy range (Fang and Feng, 2020, IMF, 2022). I set a uniform proportional labor-income tax rate of $\tau_l = 0.10$, representing a common tax burden on gross labor earnings. The rate remains fixed across counterfactuals.

Appendix B.3 documents the earnings normalization, talent-grid construction and fit, and supporting pension and labor-tax details. Appendix E.3 relaxes the benchmark assumption of independent talent draws across generations.

3.3 Mobility Costs

The CHFS does not separately identify the resource cost of entering Region A from unobserved location preferences. I therefore assign the entry costs externally and use the age-specific location–tenure constants to absorb the remaining differences in nonpecuniary attractiveness. In a structural model of internal migration, Morten and Oliveira (2016) estimate that the mean observed cost of moving between municipalities is 0.8–1.2 times the mean annual wage, with most of the cost taking the form of a fixed component. Guided by this evidence, I set $(\kappa_{A,1}^{\text{entry}}, \dots, \kappa_{A,6}^{\text{entry}}) = (4, 5, 6, 7, 8, 9)$. Because the baseline annual wage is $w^{\text{ann}} = 5$, these costs equal 0.8–1.8 times baseline annual earnings. The values for the first three age groups span the range estimated by Morten and Oliveira (2016); the increasing profile at later ages captures the decline in mobility over the life cycle, consistent with dynamic migration evidence that moving costs rise with age (Kennan and Walker, 2011). As specified in the household problem, the cost is paid only when a household whose previous region—or inherited origin at entry—is Region B chooses either rA or oA .

I set the recurring Region A rental friction to $(\kappa_{rA,1}^{\text{rent}}, \dots, \kappa_{rA,6}^{\text{rent}}) = (2, 2, 2.5, 3, 3, 3)$, or

0.4–0.6 times baseline annual earnings. The friction is lower than the one-time entry cost at every age and is paid only when a household chooses rA . It summarizes recurring resource costs associated with renting in Region A, motivated by unequal access to housing-linked local public services (Zhang and Chen, 2018, General Office of the State Council of China, 2016, General Office of the Guangzhou Municipal Government, 2017). The recurring friction is needed for the benchmark to reproduce the observed strength of positive selection into Region A renting relative to Region B ownership. I use the robust ordering $E[z \mid rA] > E[z \mid oB]$ as a sorting discipline, without targeting the exact gap because conditional means vary with the survey unit of observation. Among young household heads, mean productivity is 1.490 for Region A renters and 1.367 for Region B owners, a 9.0 percent difference; the corresponding model means are 0.999 and 0.907, a 10.1 percent difference. Appendix B.4 reports the construction and an alternative all-worker sample. Appendix E.5 shows that removing the friction compresses the model gap to 2.3 percent.

3.4 Housing Market

I jointly calibrate the reference owner-space shares $(\Omega_{A,0}, \Omega_{B,0})$ and construction shifters $(\bar{I}_A, \bar{I}_B)$ to four household-weighted CHFS purchase-price and rent targets: $P_{A,0} = 1.837$, $P_{B,0} = 0.475$, $R_{A,0} = 0.311$, and $R_{B,0} = 0.158$. Purchase prices are in units of 10,000 RMB per square meter and rents are in the same units over ten years; thus, $P_{A,0} = 1.837$ is RMB 18,370 per square meter. The calibration gives $\Omega_{A,0} = 0.562$, $\Omega_{B,0} = 0.793$, $\bar{I}_A = 1.386$, and $\bar{I}_B = 9.497$. Appendix B.5 documents the moment construction and reference-equilibrium backout.

Because comparable region-specific estimates of owner–rental conversion are unavailable for China, I set $\chi_A = \chi_B = 1$. This unit-elastic benchmark allows partial reallocation between owner and rental use, placing it between the fixed-share case in Mabilie (2023) and a frictionless rental market; the role of market segmentation is emphasized by Greenwald and Guren (2025).

I assign housing-supply elasticities externally and set $\rho_A = 0.65$ and $\rho_B = 0.95$, guided by Huang (2022), who reports average elasticities of 0.676 in more regulated Chinese cities and 0.999 in less regulated cities. The lower value for Region A captures its tighter land and regulatory constraints. Appendix E.4 examines alternative housing-supply and owner–rental conversion elasticities.

I set annual housing depreciation to 1.5%, consistent with the benchmark used by Kaplan et al. (2020), implying a ten-year depreciation rate of $\delta_h = 1 - (1 - 0.015)^{10} = 0.140$. I set the minimum dwelling size to $\underline{h} = 10 \text{ m}^2$, about one quarter of average urban per-capita housing space in the CHFS.

3.5 Financial Market

Households save through a nonnegative liquid asset and cannot issue unsecured debt. I set its annual net return to $r_s^{\text{ann}} = 1.75\%$, corresponding to the one-year RMB deposit rate posted by the Bank of China during the sample period (Bank of China, 2015). The gross return over one ten-year model period is therefore $R_s = (1.0175)^{10} = 1.1894$.

Home purchases can be financed with fully amortizing mortgages. I set the annual mortgage rate to $r_m^{\text{ann}} = 5.01\%$, the weighted-average rate on personal housing loans reported by the PBC for 2017:Q3 (People’s Bank of China, 2017). The loan-to-value limit is $\theta_{\text{LTV}} = 0.70$, requiring a down payment of at least 30% of the purchase price, and the mortgage payment cannot exceed $\theta_{\text{PTI}} = 0.50$ of gross labor income. The latter follows the Chinese regulatory ceiling documented by Fang et al. (2016).

Households in the first three age groups may obtain a mortgage with a maximum term of 30 years. The maximum term falls to 20 years at ages 50–59 and 10 years at ages 60–69, while retirees in the final age group cannot originate a new mortgage. The 30-year maximum is consistent with standard residential mortgage contracts in China (Bank of China, 2008). The payment and remaining-balance coefficients π_q and ϑ_q are implied mechanically by the mortgage rate and remaining maturity and are not independently calibrated. Appendix A.2 derives these coefficients and reports their numerical values.

Table 2 compares the targeted age-specific location–tenure shares, housing prices and rents, and working-age regional income ratio with their model counterparts.

Table 2: Targeted Moments and Model Fit

Age group	rB		oB		rA		oA	
	Target	Model	Target	Model	Target	Model	Target	Model
<i>Panel A. Location–tenure shares (percent)</i>								
20–29	26.00	24.31	39.00	40.35	22.05	21.03	12.95	14.31
30–39	9.75	9.00	55.25	55.52	14.00	13.19	21.00	22.29
40–49	10.50	9.54	59.50	59.61	9.00	8.53	21.00	22.31
50–59	10.50	10.28	59.50	59.39	9.00	9.02	21.00	21.32
60–69	9.75	9.72	55.25	55.15	10.50	10.56	24.50	24.57
70–79	9.75	9.82	55.25	55.31	10.50	10.55	24.50	24.31
<i>Panel B. Prices and spatial sorting</i>								
Moment	Target					Model		
Region A purchase price	1.837					1.837		
Region B purchase price	0.475					0.475		
Region A ten-year rent	0.311					0.311		
Region B ten-year rent	0.158					0.158		
Working-age average gross-income ratio, A/B	2.000					1.989		

Notes: The four alternatives are renting in Region B (rB), owning in Region B (oB), renting in Region A (rA), and owning in Region A (oA). Purchase prices and ten-year rents are in units of 10,000 RMB per square meter. The income ratio is computed among working-age households. All moments reported in the table enter the calibration.

4 Steady-State Results

This section describes the calibrated steady state. I first summarize life-cycle location and tenure patterns, then document spatial sorting and the allocation of intergenerational transfers, and finally examine housing affordability and household balance-sheet dynamics.

4.1 Life-Cycle Location and Tenure Patterns

The calibrated age profiles display a comparatively stable regional allocation alongside a pronounced rise in homeownership early in adult life. Figure 2 shows that homeownership in either region rises from 54.7% at ages 20–29 to 77.8% at ages 30–39, while Region A residence remains around 35% over these two stages. Homeownership remains high at later ages, with comparatively modest changes in regional residence. The age-specific location–tenure constants are calibrated to these shares, so the figure summarizes the benchmark allocation.

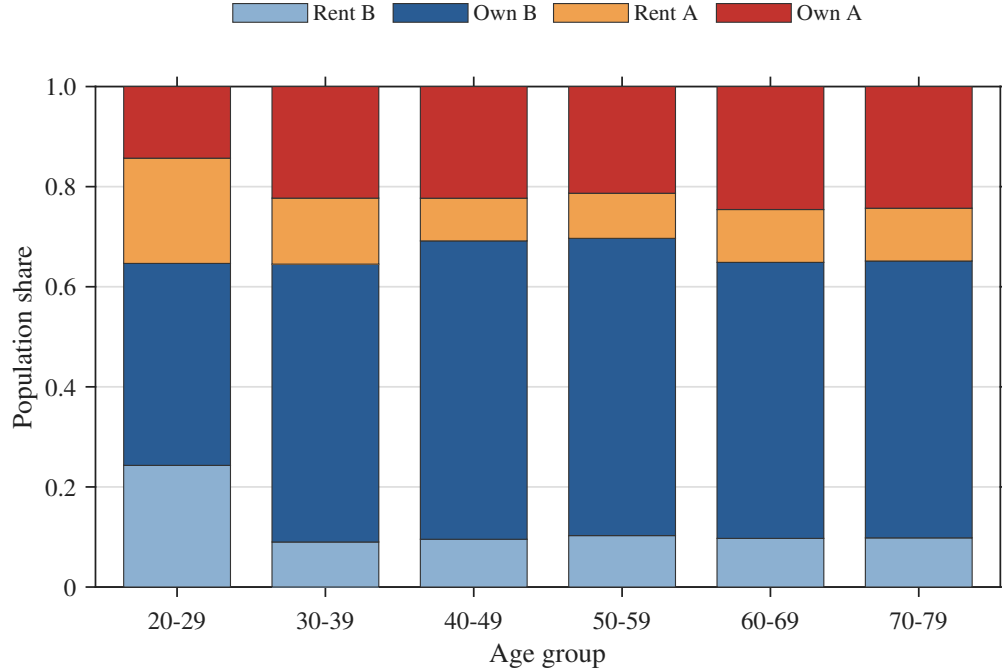


Figure 2: Population Distribution by Age and Location-Tenure

Notes: Each bar reports the model-implied distribution of an age group across renting in Region B (rB), owning in Region B (oB), renting in Region A (rA), and owning in Region A (oA). The four components sum to one within each age group.

4.2 Talent and Location-Tenure Sorting

This subsection shows how access to the high-productivity region and the choice between renting and owning there vary with household talent. The calibrated economy exhibits positive sorting into Region A at every age, with renting serving as an intermediate margin for young households.

Figure 3 shows positive spatial sorting among young households. The probability of living in Region A rises from 20.7% in the lowest talent bin to 51.1% in the highest. Renting in Region A rises from 11.9% to a peak of 24.9% around the middle of the talent distribution and then declines to 16.1%, while ownership in Region A increases steadily from 8.9% to 35.0%. Region A renting is most prevalent among intermediate-talent young households, whereas ownership becomes increasingly prevalent at higher talent levels. Appendix Figure C.1 reports all four location-tenure shares across talent for each age group, while Appendix Figure C.2 isolates Region A residence and its rental and ownership components.

Appendix Table C.1 shows that mean talent among young households is 0.999 for Region A renters, compared with 0.907 for Region B owners and 1.419 for Region A owners. Renting in A provides access to higher-paying employment without the upfront cost of

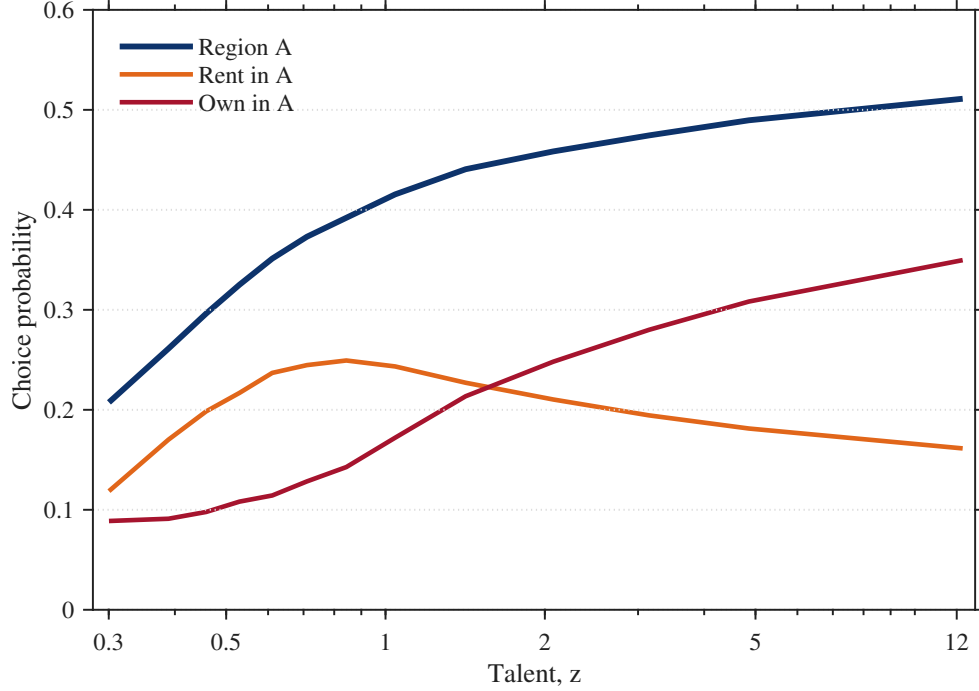


Figure 3: Region A Location Choices by Talent, Ages 20–29

Notes: Each line reports the model-implied choice probability conditional on talent among households aged 20–29. Region A includes both renters and owners in Region A. The talent axis is shown on a logarithmic scale.

buying there, while cheaper housing in B makes ownership affordable at lower incomes. Ownership alone therefore does not indicate greater earning capacity across regions: location and tenure must be considered together.

4.3 Intergenerational Transfers and Location–Tenure Sorting

Parental transfers connect young households' housing opportunities to both their own earning capacity and their family's resources. Parents aged 50–59 choose a one-time transfer to their children aged 20–29, balancing the child's financing needs against their own consumption, housing, and saving. In the baseline, 52.9 percent of young households receive a positive transfer, and the average transfer across all young households is approximately RMB 188,000. Support is directed disproportionately toward children with lower talent, while higher-talent parents provide larger amounts. Figure 4 reports these recipient and giver profiles. Lower child earnings increase the need for help with housing finance, while higher parental earnings expand the resources available to provide it. The highest-talent 1 percent of parents supply 15.8 percent of total transfers.

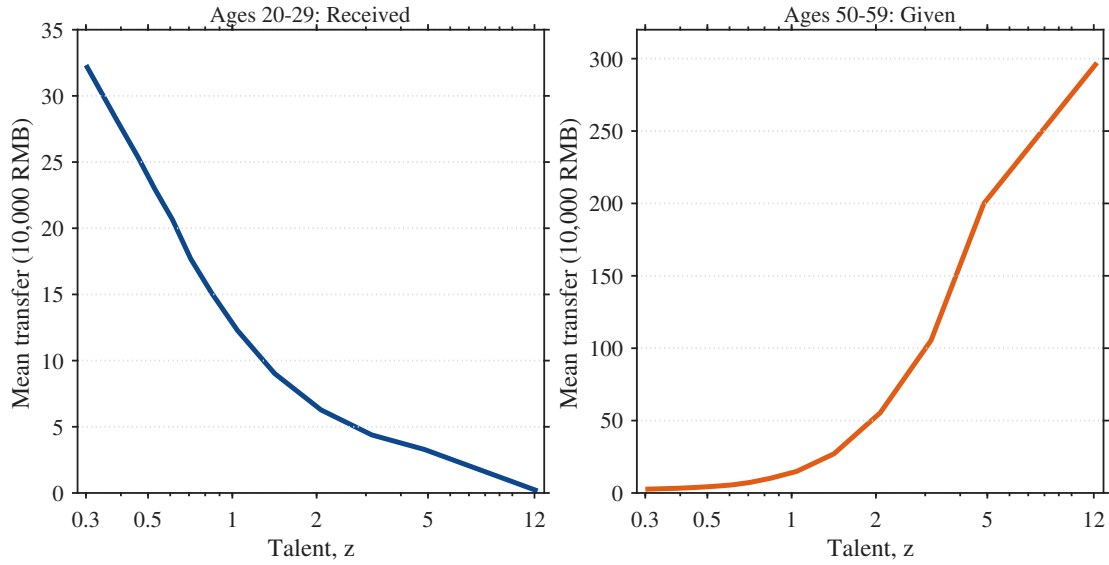


Figure 4: Intergenerational Transfers by Recipient and Giver Talent

Notes: The left panel reports mean transfers received by households aged 20–29 conditional on the recipient’s talent. The right panel reports mean transfers given by households aged 50–59 conditional on the giver’s talent. Means include zero transfers. Monetary values are expressed in units of 10,000 RMB, and the talent axes are shown on logarithmic scales.

Figure 5 compares the Region A residence and ownership probabilities of young households whose parents belong to the lowest and highest wealth quartiles, conditional on the child’s talent. Parental wealth is beginning-of-stage liquid assets plus housing resale value net of depreciation and outstanding mortgage debt, measured before giving. Parental wealth is associated with substantial differences in both outcomes at the lower end of the talent distribution. At the lowest talent node, approximately 33 percent of children in the highest parental-wealth quartile reside in Region A, compared with 15 percent in the lowest quartile. Their probabilities of owning there are approximately 20 and 2 percent, respectively. Young households with similar potential earnings therefore enter adulthood with markedly different location and housing opportunities.

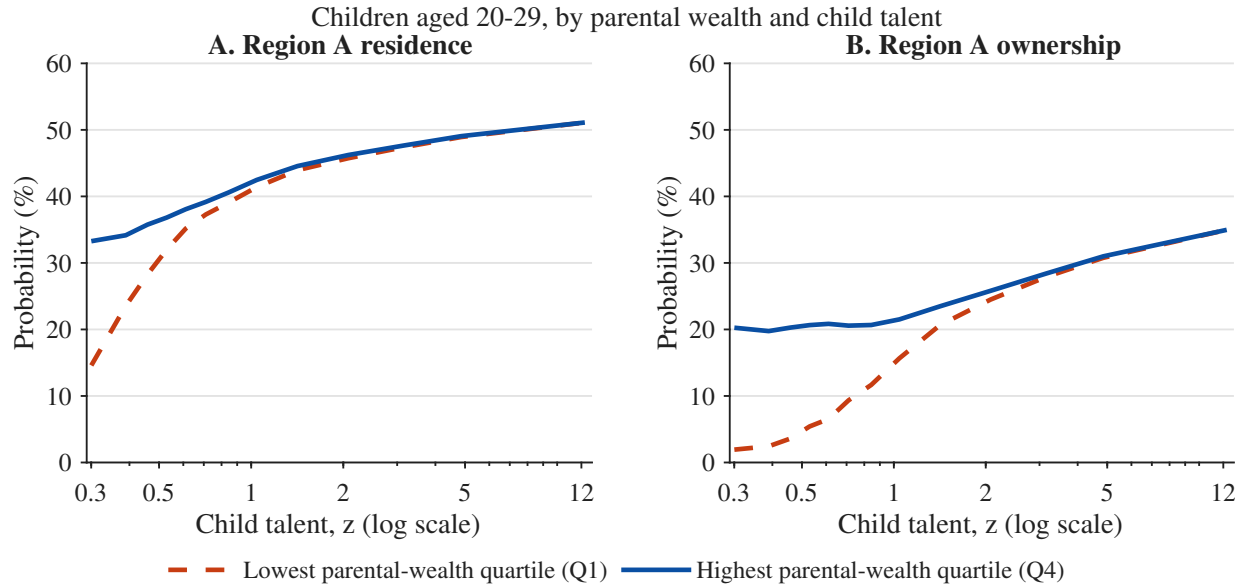


Figure 5: Parental Wealth and Young Households' Region A Residence and Ownership

Notes: The figure reports baseline choice probabilities for households aged 20–29, conditional on child talent and parental-wealth quartile. Solid blue lines correspond to the highest parental-wealth quartile, Q4, and dashed lines to the lowest quartile, Q1. Panel A reports the probability of residing in Region A, including both renters and owners. Panel B reports the probability of owning in Region A among all children in the indicated talent–parental-wealth group, not conditional on Region A residence. The talent axis is logarithmic, and probabilities are expressed in percent.

The parental-wealth gap narrows as the child's talent increases, but it closes more quickly for residence than for ownership. At intermediate talent levels, children from the two wealth groups have similar probabilities of living in Region A, while those from wealthier families remain more likely to own there. Renting allows young households to access the productive region without financing a purchase, whereas parental resources can support ownership within that location. At high talent levels, both residence and ownership probabilities are nearly identical across the two parental-wealth groups. The baseline thus displays a dynastic advantage concentrated among children whose own earning capacity provides less financial flexibility.

To examine the monetary channel underlying these patterns, Figure 6 varies the transfer received by children at the 10th, 50th, and 90th percentiles of the talent distribution. Prices, rents, the universal government transfer, continuation values, and parental-origin weights remain fixed at their baseline values. The figure traces how financial assistance changes the location–tenure choices available to children of a given talent. Appendix Figure C.3 reports the corresponding policy functions over the full talent–transfer grid.

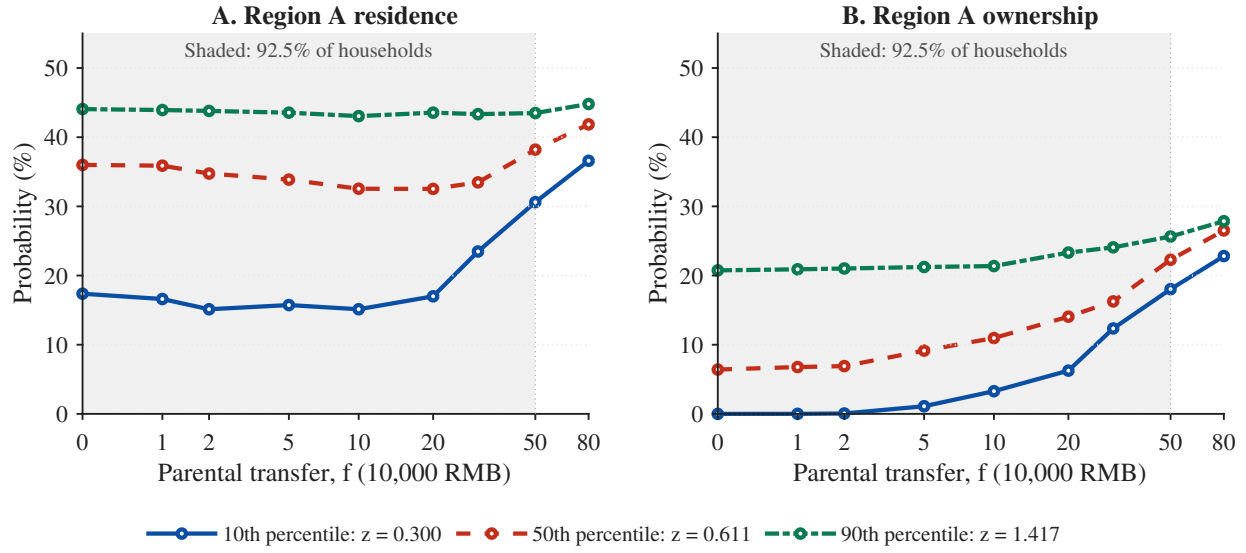


Figure 6: Parental Transfers and Region A Choices, Ages 20–29

Notes: Panel A shows the probability of living in Region A and Panel B the probability of owning there, at each talent level. Blue, red, and green curves denote the 10th, 50th, and 90th talent percentiles, respectively. The horizontal axis reports parental transfers in 10,000 RMB on a $\log(1 + f)$ scale; the vertical axis reports probabilities in percent.

The residence response is nonmonotone because financial support changes the attractiveness of ownership in both regions. For the low-talent household, a transfer of RMB 20,000 raises the probability of ownership in Region B from 30.9 to 41.9 percent, while Region A residence falls from 17.4 to 15.1 percent. The four-choice decomposition in Appendix Figure C.4 shows that this increase in Region B ownership is accompanied by declines in renting in both regions. At this level of support, the principal adjustment is toward relatively affordable ownership in B. Larger transfers also support purchases in Region A. At a transfer of RMB 500,000, the low-talent household's probability of residing in A reaches 30.6 percent, and its probability of owning there rises to 18.0 percent from approximately zero without a transfer.

The distinction between residence and ownership is also visible at higher talent levels. For the middle-talent household, ownership in A increases as support rises, while residence initially declines before increasing at larger transfers. For the high-talent household, residence changes little over the displayed transfer range, although additional support still increases ownership. Family finance can therefore change where a young household works, the tenure through which it accesses that labor market, and the location in which it acquires housing assets. The next subsection examines how these financing differences are reflected in young buyers' down payments, mortgage positions, and home equity.

4.4 Housing Affordability and Household Balance Sheets

Housing in Region A imposes a substantially larger financing burden on young households. Table 3 reports housing and financing outcomes for ages 20–29 across the four location–tenure alternatives. Because one model stage spans ten years, the headline affordability ratios compare purchase costs, rent, and mortgage payments with gross income earned over the same ten-year period. Appendix Table C.2 places these housing and financing outcomes in life-cycle context.

Table 3: Age-1 Housing Affordability and Transfer Financing

Moment	rB	oB	rA	oA
Panel A. Housing and financing				
Mean housing size (m ²)	57.28	47.58	50.57	37.67
House value	—	22.59	—	69.22
House value / 10-year income	—	0.590	—	0.844
Down payment / 10-year income	—	0.388	—	0.463
Rent / 10-year income	0.235	—	0.272	—
Mortgage payment / 10-year income	—	0.130	—	0.246
Origination LTV	—	0.364	—	0.469
Mean transfer	14.55	19.75	14.01	30.60
Transfer / down payment	—	1.330	—	0.806
Mean home equity	0	16.29	0	43.79
Panel B. Constraint incidence				
Pr(LTV binds)	—	0.032	—	0.290
Pr(PTI binds)	—	0	—	0

Notes: The table reports stationary age-1 means, weighted by state mass and location–tenure choice probabilities. Housing size is in square meters; house value, transfers, and home equity are in units of 10,000 RMB. Affordability ratios use weighted group totals and ten-year gross income. Origination debt is recovered from the post-amortization balance, and home equity is measured at the end of age 1. Mean transfers include zero receipts. Transfers are fungible, so transfer-to-down-payment ratios can exceed one. Panel B reports exact or near-exact LTV and PTI binding incidence at the selected discrete controls.

Region A ownership is considerably more expensive than Region B ownership. The mean age-1 house value is 69.22 in Region A, compared with 22.59 in Region B. The house-value-to-income ratio is correspondingly higher (0.844 versus 0.590), as are the down-payment burden (0.463 versus 0.388) and mortgage-payment burden (0.246 versus 0.130). The LTV ceiling binds for 29.0% of oA purchases but only 3.2% of oB purchases, whereas the PTI ceiling does not bind at the optimized grid points.

Family support is substantial relative to young buyers’ upfront financing needs. Mean transfers among Region A owners are 30.60, compared with 19.75 among Region B owners. For Region A owners, the ratio of mean transfers to mean down payments is 80.6%. These resources can finance the purchase alongside consumption and saving.

Appendix Figure C.5 reports the baseline life-cycle balance sheets. Between ages 20–29 and 50–59, mean mortgage debt declines from 6.18 to 0.54 as mortgages amortize,

while home equity rises from 12.84 to 31.13. End-of-stage debt is zero at ages 60–69 and 70–79, reflecting the model’s mortgage maturities and restrictions on late-life borrowing. Section 5 compares these profiles with the no-transfer equilibrium.

5 The Role of Family Transfers

To quantify the equilibrium role of inter vivos family transfers, I compare the baseline economy B with a counterfactual stationary equilibrium B^0 in which the age-4-to-age-1 monetary transfer is constrained to zero in every state. Parents remain altruistic, but they can no longer implement this concern through an inter vivos payment. Terminal estates and the inheritance mechanism remain active, and their stationary distribution is recomputed. All structural parameters are held fixed, and the stationary equilibrium is re-solved, incorporating both life-cycle distributional changes and general-equilibrium housing-market feedback. Unless otherwise stated, effects are reported as the difference between the transfer economy and the no-transfer economy, $B - B^0$. Percentage changes use the no-transfer equilibrium as the reference.

5.1 Spatial Access versus Ownership Access

Family transfers expand young households’ ownership in the productive region substantially more than their residence there. Among households aged 20–29, transfers raise the share living in Region A by 1.13 percentage points, or 3.3 percent relative to the no-transfer equilibrium. The share owning a home in Region A increases by 4.04 percentage points, or 39.3 percent. Ownership also expands in Region B, bringing the increase in homeownership among young households to 8.45 percentage points.

Table 4 separates the increase in Region A ownership into a spatial component, associated with the change in Region A residence, and a tenure component, associated with the change in ownership conditional on residence. For young households, the tenure component accounts for 3.64 percentage points, or 90.2 percent of the ownership response, while the spatial component accounts for 0.40 points. Appendix D.1 presents the decomposition. The larger tenure adjustment is also reflected in the share of young households renting in Region A, which is 2.91 percentage points lower in the transfer economy. Renting provides access to the productive region without requiring a down payment, allowing ownership to respond much more strongly than regional residence.

Figure 7 shows that the spatial response is concentrated among young households with low talent. At the lowest talent node, transfers raise Region A residence by 5.10 percentage points and Region A ownership by 8.88 points. The residence effect becomes

Table 4: Residence–Ownership Decomposition of the Family-Transfer Effect

Group	$\Delta \Pr(A)$	$\Delta \Pr(\text{own} \mid A)$	$\Delta \Pr(oA)$	Spatial component	Tenure component	Tenure share
Age 20–29	1.13	10.46	4.04	0.40	3.64	90.2%
Working ages	1.41	4.39	2.29	0.85	1.44	62.7%
All households	1.24	3.13	1.81	0.77	1.03	57.3%

Notes: Changes are $B - B^0$. All changes and components are in percentage points. Appendix D.1 defines the components and tenure share. Working ages comprise ages 20–69. All statistics use stationary population weights.

small as talent increases, while the ownership response remains appreciable over a broader range of lower- and middle-talent households. At high talent levels, both responses are small. The curves compare conditional choice probabilities across the two stationary equilibria, averaging over the remaining household states within each economy.

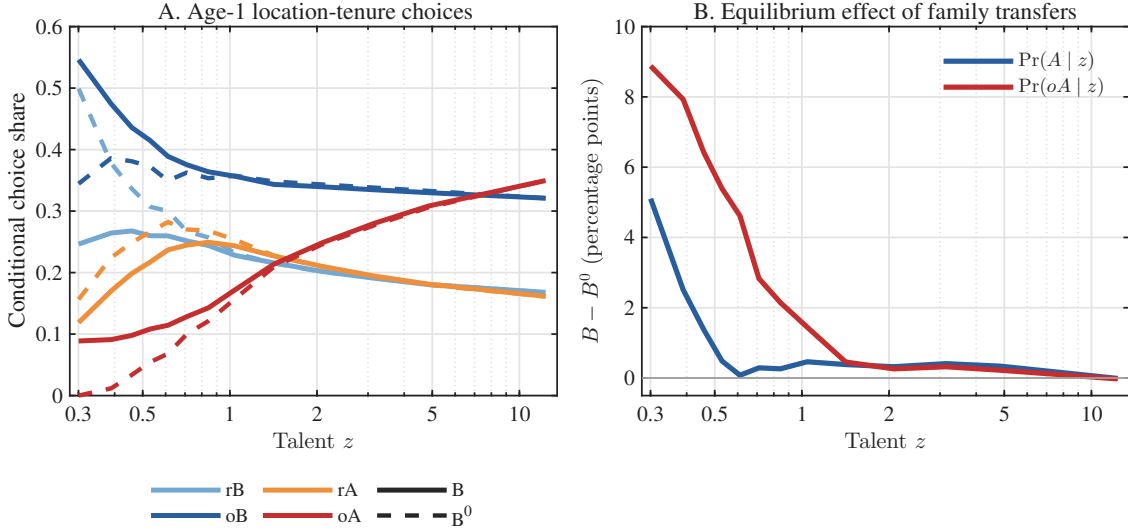


Figure 7: Family Transfers and Age-1 Location–Tenure Choices

Notes: Panel A reports age-1 choice probabilities conditional on talent. Colors identify renting and owning in Regions B and A; solid lines denote the baseline equilibrium B , and dashed lines denote the no-transfer equilibrium B^0 . Panel B reports $B - B^0$ in percentage points, where Region A residence includes both rA and oA . Each curve averages over the remaining age-1 states using the stationary distribution of the corresponding equilibrium. The talent axis is logarithmic.

The responses at later ages are more heterogeneous. Appendix Figure D.1 shows that higher Region A ownership among lower-talent households persists at ages 30–39 and 40–49. From ages 50–59 onward, however, the direction of the ownership response differs more sharply across talent. Lower-talent households continue to have higher Region A ownership and lower Region B ownership in the transfer economy. Among middle- and higher-talent households, Region A ownership is instead lower, accompanied by a partial offsetting increase in ownership in the less expensive Region B. Homeownership in either

region also falls for these households, particularly at ages 60–69 and 70–79.

These patterns are consistent with households' changing position in the intergenerational transfer system. Lower-talent households receive more assistance when young and give relatively little later in life. Middle- and higher-talent households receive less support at entry but devote more resources to their own children at ages 50–59. The financing advantage associated with receiving transfers therefore coexists with a resource cost when households become donors.

Despite the reversals among some older households, transfers increase both Region A residence and ownership in the aggregate. Across working ages 20–69, the Region A residence share rises by 1.41 percentage points and the ownership share by 2.29 points. Across the full population, the corresponding increases are 1.24 percentage points, or 3.82 percent, and 1.81 percentage points, or 9.16 percent. Table 4 shows that ownership conditional on residence accounts for 62.7 percent of the working-age ownership response and 57.3 percent of the all-household response. The spatial component is therefore relatively more important for the broader population than for young entrants, although tenure remains the larger component.

5.2 Ownership Selection and the Spatial Allocation of Labor

Family transfers weaken the dependence of Region A ownership on young households' own earning capacity. Panel A of Table 5 shows that, among households aged 20–29 in the bottom 80 percent of the talent distribution, the probability of owning in Region A rises from 6.84 percent without transfers to 11.79 percent with transfers. The corresponding probability for the top 20 percent changes only slightly, from 24.02 to 24.37 percent. Weighting these responses by the groups' population shares, households in the bottom 80 percent account for 98.3 percent of the net increase in young Region A ownership. The expansion is therefore concentrated among households whose own earnings provide less capacity to finance a purchase.

This expansion changes the composition of young Region A homeowners. The bottom 80 percent of the talent distribution account for 65.94 percent of these owners in B , compared with 53.24 percent in B^0 . Mean annual gross labor income among young A owners is consequently 19.9 percent lower in the transfer economy, at approximately RMB 82,000 rather than RMB 102,400. Parental resources enable households with lower own earnings to purchase in Region A even after housing prices adjust.

Table 5: Ownership Selection, Effective Labor, and Aggregate Output

Outcome	B : transfers	B^0 : no transfers	Transfer effect ($B - B^0$)
Panel A. Young-household ownership and homeowner composition			
Region A ownership among bottom-80 households	11.79%	6.84%	+4.96 pp
Region A ownership among top-20 households	24.37%	24.02%	+0.35 pp
Bottom-80 share among Region A owners	65.94%	53.24%	+12.70 pp
Mean annual labor income among Region A owners (RMB)	82,035	102,443	−19.9%
Panel B. Working-age effective labor and output			
Working-age mass in A	0.2786	0.2668	+0.0118
Mean $G_g z$ in A	1.1081	1.1380	−0.0299
Effective labor in A	0.3087	0.3036	+0.0051
Residence-mass component	—	—	+0.0132
Composition component	—	—	−0.0081
Effective labor in B	0.4234	0.4285	−0.0051
Output in A	0.4229	0.4159	+0.0069
Output in B	0.4234	0.4285	−0.0051
Aggregate output	0.8463	0.8444	+0.0019 (+0.22%)

Notes: Panel A concerns ages 20–29; Panel B concerns working ages 20–69. Effects are $B - B^0$, with percentage changes relative to B^0 ; differences are calculated before rounding. In Panel A, ownership rates use the respective talent group’s population as the denominator, whereas the bottom-80 share uses all young Region A owners. Income is annual gross labor income, excluding family and government transfers. Population-weighted bottom-80 ownership changes account for 98.3 percent of the net increase in young Region A ownership. In Panel B, working-age mass is relative to the full stationary population. Since working ages comprise 5/6 of that population, the 1.18-percentage-point mass increase corresponds to the 1.41-point residence-share increase among working-age households in Table 4. Effective labor and its components are in efficiency units; regional output equals regional TFP times effective labor.

The reduction in mean talent among Region A owners persists at later ages but becomes smaller, as reported in Appendix Table D.1. Changes in homeowner composition, however, need not translate into comparable changes in the regional workforce. A household that rents rather than owns in Region A supplies the same effective labor at a given age and talent. The implications for production therefore depend on the accompanying changes in regional residence and the age–talent composition of workers, including both renters and owners.

To quantify these effects, I aggregate labor supplied by households aged 20–69. Regional effective labor can be written as $L_A = M_A \overline{Gz}_A$, where M_A is the working-age population mass residing in Region A and $\overline{Gz}_A$ is its mean age-adjusted productivity. I decompose the change in effective labor into a residence-mass component and a composition

component. The first evaluates the change in worker mass at mean efficiency averaged across the two equilibria; the second evaluates the change in mean efficiency at the average worker mass. This symmetric decomposition adds exactly to the change in Region A effective labor. Appendix D.1 provides the formal expressions.

Panel B of Table 5 shows that transfers increase Region A's working-age population mass from 0.2668 to 0.2786, measured relative to the full stationary population. Mean age-adjusted productivity among these workers falls from 1.1380 to 1.1081. The larger working population contributes 0.0132 efficiency units to Region A labor supply, while the composition change contributes -0.0081 , offsetting 61.6 percent of the residence-mass contribution. Effective labor in Region A consequently increases by 0.0051 efficiency units. The population expansion thus raises the amount of labor employed in the productive region, but its contribution is moderated by the lower average efficiency of the resulting workforce.

Total effective labor is unchanged because the population age profile, talent distribution, and individual labor supply are fixed. The increase in Region A effective labor is therefore matched by an equal decline in Region B. With regional productivity equal to 1.37 in A and 1 in B, this reallocation raises output in A by 0.0069 and lowers output in B by 0.0051, yielding a net aggregate-output gain of approximately 0.0019, or 0.22 percent. Weaker earnings selection into Region A ownership therefore coexists with a positive output effect: production benefits from the net movement of effective labor toward the more productive region, while the larger tenure adjustment changes who owns its housing assets.

5.3 Housing Markets and Life-Cycle Balance Sheets

Family transfers shift young households toward ownership and increase the resources available to finance purchases. Table 6 shows that purchase prices are 4.12 percent higher in Region A and 7.84 percent higher in Region B in the transfer economy. Through the regional supply schedules, these price differences support greater residential investment and larger housing stocks. Rents adjust much less: Region A rent is 0.42 percent lower with transfers, while Region B rent is 0.48 percent higher.

The allocation of housing between owner and rental use also adjusts. Because purchase prices increase more than rents, the rent-to-price ratio is lower in both regions when transfers are available. The rental share of the housing stock is consequently 43.76 percent in Region A in B , compared with 44.86 percent in B^0 ; the corresponding shares in Region B are 20.70 and 21.89 percent. Family support thus changes both the size of the regional housing stock and its allocation across tenure segments. These market adjust-

ments extend the effects beyond transfer recipients. Higher purchase prices increase the resources required by prospective buyers, including those without parental assistance, while the small decline in Region A rent lowers rental expenditure for a given dwelling. Nevertheless, homeownership among young households remains higher in B : the lower purchase prices in the no-transfer equilibrium do not offset the loss of family finance.

Table 6: Housing-Market and Age-1 Balance-Sheet Effects of Family Transfers

Outcome	B : transfers	B^0 : no transfers	Transfer effect ($B - B^0$)
Panel A. Housing markets and fiscal balance			
P_A	1.837	1.765	+4.12%
P_B	0.475	0.440	+7.84%
R_A	0.311	0.312	-0.42%
R_B	0.158	0.157	+0.48%
Rental share of housing stock in A	43.76%	44.86%	-1.10 pp
Rental share of housing stock in B	20.70%	21.89%	-1.18 pp
Universal transfer T	0.434	0.428	+1.32%
Panel B. End-of-age-1 household balance sheets			
Liquid assets	5.32	0.09	+5.24
Housing wealth	19.02	12.12	+6.89
Mortgage debt	6.18	6.87	-0.69
Home equity	12.84	5.25	+7.59
Net financial assets	-0.86	-6.79	+5.93

Notes: Effects are $B - B^0$; percentage changes use B^0 as the reference, and differences are calculated before rounding. Purchase prices are in RMB 10,000 per square meter; rents are the corresponding ten-year payments per square meter. Rental-stock shares measure floor space, not household population shares; their effects are percentage points. T is in RMB 10,000 per household per stage. Panel B reports within-age, end-of-stage means in RMB 10,000 per young household, including renters, whose housing wealth and home equity are zero. Net financial assets equal household liquid assets minus mortgage debt. Young-household choice-share levels appear in Panel A of Appendix Table D.2.

The expansion of ownership is accompanied by stronger young-household balance sheets. By the end of ages 20–29, households in B hold, on average, RMB 52,400 more liquid assets and RMB 75,900 more home equity than households in B^0 , while carrying RMB 6,900 less mortgage debt. Transfers therefore support greater ownership without a corresponding increase in average borrowing. Young households can contribute more equity to their purchases while retaining more liquidity. Housing wealth and home equity are valued at each economy’s equilibrium prices, so their differences reflect both housing holdings and property valuations.

The financing comparison among young Region A buyers reinforces this result. Mean

origination LTV is 46.9 percent with transfers, compared with 68.8 percent without them, as reported in Panel B of Appendix Table D.2. Together with the ownership-selection results in Section 5.2, this shows that the transfer economy accommodates lower-earning young homeowners who nevertheless use less leverage. Parental resources expand purchasing capacity beyond what the child's own earnings would support and allow a larger share of the purchase to be financed with equity. The conditional buyer statistics reflect both financing decisions and the composition of households selecting into Region A ownership.

Figure 8 shows that the early financing differences persist into subsequent stages. At ages 30–39 and 40–49, when households receive no new inter vivos transfer, mean home equity remains approximately RMB 90,000 higher in B , while mortgage debt is approximately RMB 30,000 lower. Earlier assistance therefore has consequences beyond the initial purchase stage through the housing and financial positions carried forward, subsequent saving, and mortgage repayment.

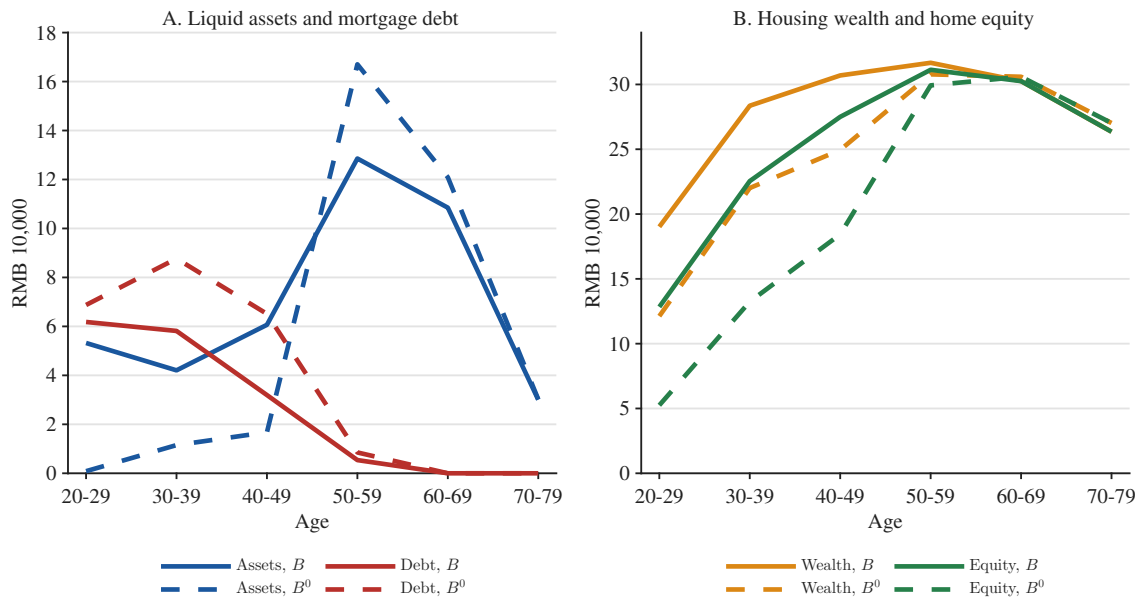


Figure 8: Life-Cycle Balance Sheets With and Without Family Transfers

Notes: Panel A reports liquid assets and mortgage debt; Panel B reports housing wealth and home equity. Solid lines denote B and dashed lines B^0 . Series are within-age, end-of-stage population means in two stationary equilibria, measured in RMB 10,000. Zero mortgage debt in the final two age groups follows from the mortgage-contract assumptions.

The pattern changes when households reach ages 50–59, the stage at which they receive stochastic inheritance and choose transfers to their own children. Mean liquid assets are then RMB 38,500 lower in B than in B^0 , while the home-equity gap narrows to approximately RMB 12,000. The reversal in liquid assets is consistent with parents devoting

resources to their children rather than retaining them on their own balance sheets. At ages 60–69 and 70–79, the mean asset and housing-equity profiles nearly coincide. End-of-stage mortgage debt is zero in both economies at these ages, reflecting the model’s mortgage maturities and restrictions on late-life borrowing.

These age profiles compare households that have spent their entire lives under different transfer regimes. The later-age differences consequently reflect current parental giving together with earlier transfer receipt, accumulated housing and financial positions, and the equilibrium inheritance distribution. Family support redistributes financing capacity across the life cycle: its largest balance-sheet advantages appear among young recipients and persist through early middle age, while the resources devoted to giving become visible when those households reach the parental stage.

5.4 Distributional and Welfare Consequences

Family transfers have limited effects on earnings inequality but larger effects on the distribution of consumption and wealth. Panel A of Table 7 shows that the working-age labor-income Gini is nearly unchanged between the two economies, consistent with the comparatively small adjustment in labor allocation documented in Section 5.2. Disposable-income inequality also changes little. Across all living households, however, the Gini coefficient of the consumption–housing composite falls from 0.386 without transfers to 0.346 with transfers. The net-worth distribution becomes less dispersed, and the share of households with nonpositive net worth falls from 11.99 to 5.85 percent.

Among young households, transfers broaden participation in ownership and reduce home-equity inequality. The home-equity Gini for ages 20–29 falls from 0.770 in B^0 to 0.722 in B . This measure includes renters at zero and therefore reflects both the prevalence of ownership and the distribution of equity holdings. Consistent with the ownership responses in Section 5.2, the gap in Region A ownership between the top 20 percent and bottom 80 percent of child talent narrows by 4.61 percentage points. Appendix Table D.3 reports lower inequality in the consumption–housing composite and home equity within each subsequent age group as well.

These reductions coexist with larger differences in children’s housing outcomes across parental wealth. Wealthier parents can provide more assistance, allowing their children to finance purchases beyond what their own earnings would support. Panel C of Table 7 compares children whose parents belong to the highest and lowest parental-wealth groups, using the benchmark wealth cutoffs in both economies. The gap in children’s Region A ownership is 11.16 percentage points with transfers, compared with 0.63 points without them. The corresponding gap in children’s mean home equity is approxi-

mately RMB 221,400 in B , compared with less than RMB 1,000 in B^0 .

Table 7: Distributional Consequences of Family Transfers

Outcome	B : transfers	B^0 : no transfers	$B - B^0$
Panel A. Income, consumption, and net worth			
Labor-income Gini, ages 20–69	0.463	0.464	–0.001
Disposable-income Gini, all ages	0.469	0.471	–0.001
Consumption–housing composite Gini, all ages	0.346	0.386	–0.040
Net-worth p90/p50, all ages	3.67	4.51	–0.83
Share with nonpositive net worth, all ages	5.85%	11.99%	–6.14 pp
Panel B. Young-household housing inequality, ages 20–29			
Home-equity Gini	0.722	0.770	–0.048
Region A ownership gap: top-20 minus bottom-80 talent	12.58 pp	17.18 pp	–4.61 pp
Panel C. Parental-wealth gaps in young households' outcomes			
Region A ownership gap: parental Q4 minus Q1	11.16 pp	0.63 pp	+10.53 pp
Mean home-equity gap: parental Q4 minus Q1, RMB 10,000	22.14	0.09	+22.04

Notes: Effects are $B - B^0$, calculated before rounding. Ownership gaps are percentage points in every column. Labor-income inequality covers ages 20–69; the other Panel A measures pool all six age groups. Young-household home-equity inequality includes renters at zero. Net worth equals end-of-stage liquid assets plus home equity. Parental groups use beginning-of-age-4 net worth and the wealth cutoffs calculated in B , applied unchanged in B^0 . Child outcomes are expected outcomes from the linked age-1 policies. Appendix D.3 gives the formulas and detailed weighting procedures.

I measure welfare using remaining-lifetime consumption-equivalent variation (CEV): the proportional adjustment to consumption and housing services in B^0 required to match self-lifetime welfare in B . Positive CEVs indicate welfare losses from eliminating transfers, while negative values indicate gains. Appendix D.4 provides the formal definition and calculation details.

Figure 9 shows that the welfare gains from transfers are concentrated early in life. The required compensation in B^0 is 23.33 percent for households aged 20–29, declines through middle age, and is approximately 3 percent for each age group from 50–59 onward. The all-living CEV is 10.78 percent.

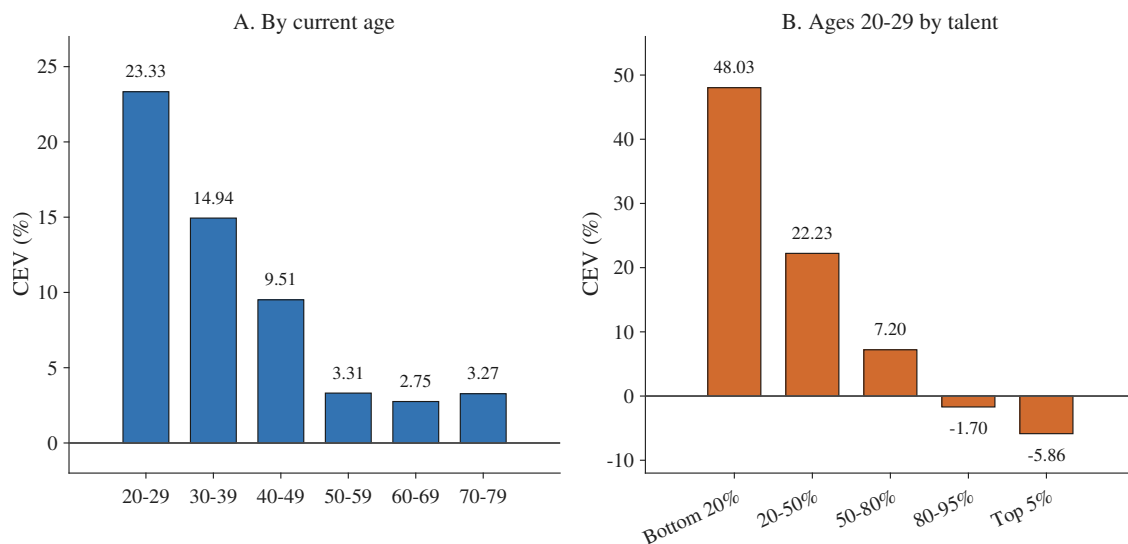


Figure 9: Remaining-Lifetime Welfare Effects of Family Transfers

Notes: Panel A reports consumption–housing CEVs by current age, and Panel B reports CEVs by talent among households aged 20–29. Positive values indicate the percentage increase in consumption and housing services required in B^0 to attain self-lifetime welfare in B ; negative values indicate higher self-lifetime welfare in B^0 . The comparison is between stationary equilibria. Appendix D.4 provides the welfare definition and calculation details.

The welfare response also varies substantially with talent. Among households aged 20–29, the CEV is 48.03 percent for the bottom talent quintile and declines to 7.20 percent for the 50th–80th percentiles. It becomes negative above the 80th percentile, reaching –5.86 percent for the top 5 percent. High-talent households receive relatively little parental assistance in the benchmark. In the no-transfer economy, they face lower purchase prices and no longer devote resources to inter vivos transfers to their own children later in life.

Differences across talent remain pronounced at the parental stage. Although average welfare at ages 50–59 is higher with transfers, households in the top 5 percent of the talent distribution have a CEV of –17.50 percent, consistent with the larger amounts given by high-talent parents in B . Appendix Figure D.2 reports the complete talent profiles at ages 30–79. The next section examines how public housing and place-based policies interact with these differences in family support and welfare incidence.

6 Public Policies and Private Family Finance

This section compares four policy instruments that operate on different margins of the household’s location and housing decisions. Relaxing the loan-to-value limit expands formal mortgage credit. Rental assistance lowers the cost of living in Region A without

subsidizing ownership. The talent subsidy rewards high-productivity households for locating in Region A, while the purchase voucher directly subsidizes Region A homeownership. Together, the experiments show how the general-equilibrium incidence of housing and place-based policies depends on whether the margin they relax overlaps with the financing role of the family.

Let B and B^0 denote the baseline equilibria with and without inter vivos family transfers. For a given policy, let P and P^0 denote the corresponding policy equilibria. I report the policy effects with and without private family finance,

$$\Delta_P^f(Y) = Y_P - Y_B, \quad \Delta_P^0(Y) = Y_{P^0} - Y_{B^0},$$

and their difference,

$$I_P(Y) = \Delta_P^f(Y) - \Delta_P^0(Y).$$

The interaction $I_P(Y)$ indicates whether family transfers make the policy response in outcome Y larger or smaller. When useful, I also compare the effect of transfers in the baseline, $B - B^0$, with their effect in the policy economy, $P - P^0$.

Welfare comparisons use the self-lifetime consumption–housing CEV defined in Appendix D.4, with positive values indicating a loss relative to the corresponding baseline.

Table 8 summarizes the direct effects of the four policies and their interactions with family finance. The policies differ in whether they primarily affect spatial access, ownership entry, leverage, or housing investment.

Table 8: Policy Effects and Interactions with Family Transfers

Outcome	LTV relaxation	Affordable rent	Talent subsidy	Purchase voucher
Panel A. Policy effects with family transfers, $P - B$				
Spending / tax revenue (%)	0.00	2.05	18.21	11.21
$\Delta \Pr(A)$, age 1 (pp)	0.131	1.675	0.908	0.448
$\Delta \Pr(oA)$, age 1 (pp)	0.181	-0.374	0.852	0.480
$\Delta [\Pr(oA) + \Pr(oB)]$, age 1 (pp)	0.081	-1.577	0.113	0.097
Δ aggregate output (%)	0.023	0.037	0.177	0.150
All-living welfare cost, CEV (%)	0.82	0.41	2.37	1.57
Panel B. Policy–transfer interactions				
$I_P[\Pr(A)]$, age 1 (pp)	-0.033	-0.911	0.108	-0.076
$I_P[\Pr(oA)]$, age 1 (pp)	-0.329	-0.072	0.028	-0.003
$I_P[\text{home equity}]$, age 1	0.311	-0.111	0.286	0.289

Notes: Panel A reports $P - B$. Panel B reports $I_P = (P - B) - (P^0 - B^0)$. Probability effects are percentage points. Home equity is an age-1 mean in units of RMB 10,000. A positive CEV means that the policy economy requires compensation to attain welfare in its corresponding baseline. CEVs are pairwise and are not differenced to form an interaction. The LTV experiment has no direct fiscal expenditure in the model, although financial-stability costs are outside the model.

6.1 LTV Relaxation: Formal Credit versus Family Finance

The LTV reform tests whether formal mortgage credit can substitute for parental support at the purchase-entry margin and whether family resources continue to shape leverage and home equity after purchase.

Minimum down-payment regulation is a central instrument of Chinese housing finance. In 2024, national regulators lowered the minimum down-payment floor for commercial mortgages, while allowing local authorities some discretion over implementation (People’s Bank of China and National Financial Regulatory Administration, 2024). I study a stronger, stylized reform that raises the maximum LTV ratio from 70 to 90 percent for new purchases in both regions. The reform lowers the minimum down payment from 30 to 10 percent. Existing mortgages are not refinanced, and the payment-to-income limit, mortgage rate, maturity, and amortization schedule remain unchanged.

The reform changes the financing of home purchases much more than location or tenure. With family transfers, age-1 Region A ownership rises by only 0.18 percentage points, homeownership in either region by 0.08 points, and Region A residence by 0.13 points. Aggregate output rises by only 0.023 percent. In contrast, mean age-1 mortgage debt rises by RMB 19,300 and home equity falls by RMB 12,700. Among age-1 Region A buyers, origination LTV rises from 46.9 to 58.4 percent, while the down-payment-to-income ratio falls from 46.3 to 36.2 percent. The higher ceiling therefore permits buyers to replace their initial equity contribution with mortgage borrowing rather than inducing substantial spatial reallocation.

The ownership response is larger when parental support is unavailable. LTV relaxation raises age-1 Region A ownership by 0.51 percentage points in B^0 , compared with 0.18 points in B , producing an interaction of -0.33 percentage points. Formal mortgage credit therefore substitutes for family finance in helping households cross the purchase threshold. It does not eliminate the balance-sheet role of parental resources. As Figure 10 shows, transfers reduce age-1 mortgage debt by RMB 6,900 under the baseline LTV limit and by RMB 12,500 under the higher ceiling. Their effect on home equity rises from RMB 75,900 to RMB 79,000. The conditional buyer statistics above describe purchase financing. Figure 10 reports means over all young households, including renters, so its balance-sheet effects reflect both financing decisions and selection into ownership.

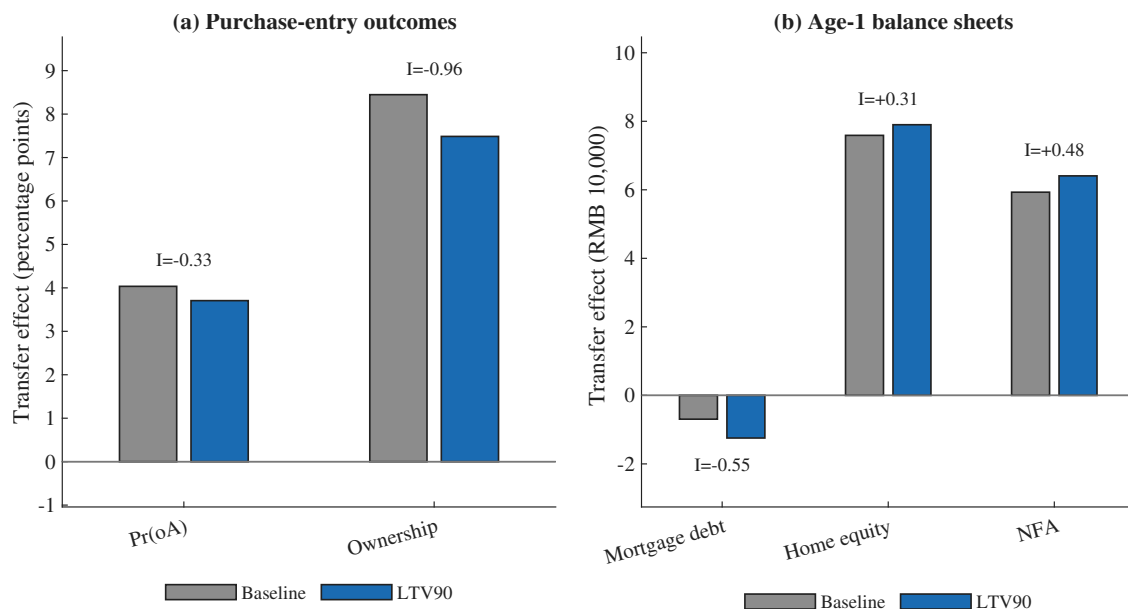


Figure 10: Family Transfers under Relaxed LTV Regulation

Notes: Gray bars report the effect of family transfers in the baseline, $B - B^0$, and blue bars report their effect under LTV90, $P - P^0$. Labels give the policy-transfer interaction $I_P = (P - P^0) - (B - B^0)$. Panel (a) reports percentage points; Panel (b) reports means over all age-1 households, including renters, in units of RMB 10,000.

Under the self-lifetime consumption-housing criterion, the reform produces an all-living welfare cost of 0.82 percent relative to the baseline, incorporating the equilibrium adjustment of housing markets and household allocations.¹ Detailed allocation, financing, and welfare results appear in Appendix F.1.

6.2 Rental Assistance: Spatial Access without Ownership

Rental assistance instead targets spatial access. I use it to assess whether public support reduces young households' dependence on family wealth for residence in the productive region while parental resources remain important for homeownership there.

China has promoted subsidized rental housing and monetary rental assistance as ways to broaden access to urban housing (General Office of the State Council of China, 2016). I consider a program for households at ages 1 and 2 whose potential Region A income is no higher than the baseline 60th percentile and who own no house at the beginning of the period. Eligible households renting in Region A receive a 20 percent discount on market rent for up to 70 square meters. Landlords receive the full market rent, the government pays the difference, and eligibility is reassessed in each period. Among eligible households, 23.5 percent take up the subsidy at age 1 and 21.5 percent at age 2. Total expenditure equals 2.05 percent of tax revenue.

¹The model abstracts from mortgage default, lender losses, risk-based pricing, and financial-stability consequences.

The policy expands Region A access through rental tenure. With family transfers, the age-1 rA share rises by 2.05 percentage points and total Region A residence rises by 1.68 points. Region A ownership instead falls by 0.37 points, while age-1 homeownership across both regions falls by 1.58 points. The increase in Region A renting is accompanied primarily by lower Region B shares: rB and oB decline by 0.47 and 1.20 percentage points, respectively, compared with a 0.37-point decline in oA . The net allocation therefore shifts toward residence in Region A, as well as toward rental tenure.

Figure 11 shows both the source of this reallocation and its interaction with private family finance. Without family transfers, the policy raises Region A residence by 2.59 percentage points, compared with 1.68 points when transfers are available. The resulting interaction is -0.91 points. Rental assistance therefore substitutes for parental resources as a route into Region A. The ownership margin is different. Transfers raise age-1 Region A ownership by approximately four percentage points both before and after the rental policy. Public rental support provides an alternative route into the productive labor market, but it does not replace family finance as a route into homeownership and housing wealth.

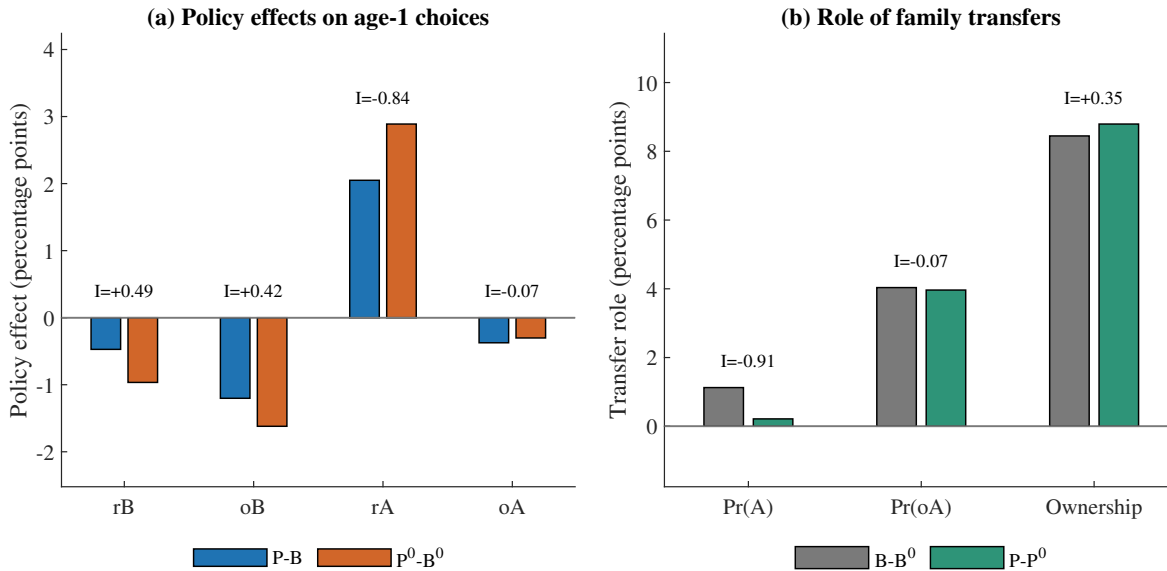


Figure 11: Affordable Renting and the Role of Family Transfers

Notes: Panel (a) reports the policy-induced change in each age-1 location–tenure choice with and without family transfers. The decline in both Region B choices shows that most of the increase in Region A renting reflects net spatial reallocation rather than only a shift from Region A ownership. Panel (b) compares the effect of family transfers in the baseline and rental-policy economies. Rental assistance substantially reduces the role of transfers in Region A residence while leaving their role in Region A ownership largely intact. Labels report the policy–transfer interaction. Effects are percentage points.

The increase in rental demand raises the Region A market rent by 3.02 percent, partly offsetting the statutory discount, while the Region A purchase price rises by only 0.40 percent. The increase in Region A’s working-age mass raises effective labor, while a decline in mean age-adjusted productivity among its workers from 1.1081 to 1.1005 partly

offsets this gain. Aggregate output increases by 0.037 percent. Despite a lower net rent burden, income-eligible age-1 households have a small remaining-lifetime welfare cost, with a CEV of 0.07 percent. Ineligible age-1 households receive no direct subsidy, face the higher market rent and lower fiscal transfer, and have a CEV of 0.23 percent.² The all-living welfare cost is 0.41 percent. Detailed eligibility, housing, and welfare results appear in Appendix F.2.

6.3 Talent Subsidies: Productive Sorting Rather than City Expansion

The talent subsidy shifts the focus to labor allocation by testing whether a place-based payment can attract high-productivity workers and whether its effectiveness depends on private family finance.

Chinese municipalities use settlement benefits, cash payments, and housing support to attract skilled workers (Shen and Li, 2022). I model a once-per-lifetime payment of RMB 300,000 to households in the top 20 percent of the talent distribution that choose Region A during either of the first two working-age stages. Both Region A renters and new Region A buyers qualify. The government finances the subsidy through the equilibrium adjustment in the universal transfer. Approximately 53 percent of eligible age-1 households claim the subsidy. Total spending equals 18.21 percent of tax revenue, making this the most fiscally costly of the four experiments.

The policy generates a sharp response at the talent cutoff. Among eligible age-1 households, Region A residence rises by 7.38 percentage points and Region A ownership by 6.41 points, while both outcomes decline slightly for the ineligible bottom 80 percent. Figure 12 shows that the response appears immediately at the first eligible talent node and is concentrated in oA . In the same-state, common-shock comparison of B and P in Appendix Figure F.6, approximately 83 percent of gross entries into oA originate from rB or oB choices.

²These welfare comparisons group households by income eligibility, including those who do not claim the subsidy.

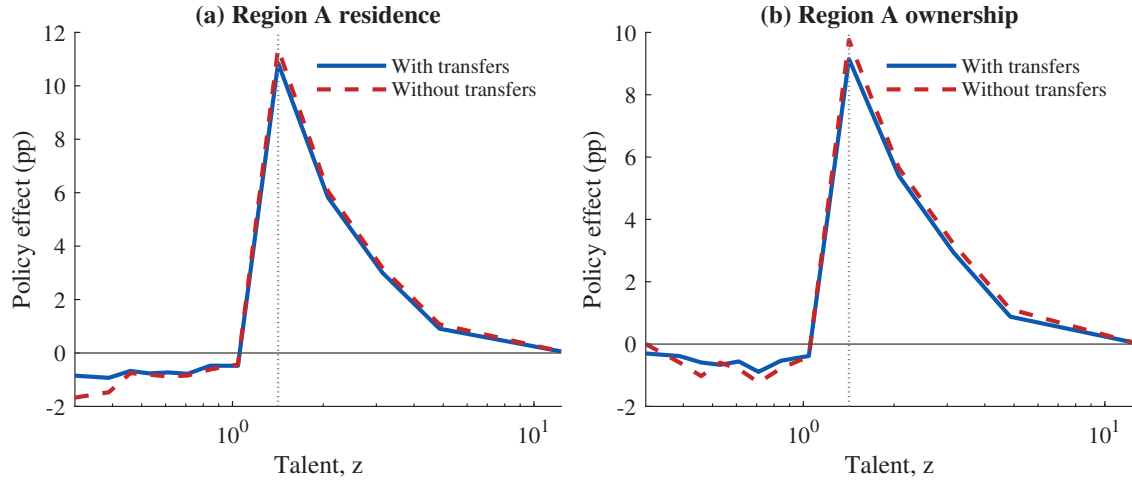


Figure 12: Talent Subsidy Effects around the Eligibility Cutoff

Notes: Solid blue lines report $P - B$, and dashed red lines report $P^0 - B^0$. Panel (a) shows the age-1 policy effect on Region A residence; Panel (b) shows the effect on Region A ownership. Effects are percentage points. The horizontal axis uses the baseline talent grid on a log scale, and the vertical dotted line marks the eligibility cutoff.

The aggregate stationary-equilibrium effect is selective rather than expansionary. Age-1 Region A residence rises by 0.91 percentage points and ϕA by 0.85 points. The increase in housing demand raises the Region A purchase price by 3.32 percent and its rent by 1.29 percent, while financing the program lowers the universal transfer. These equilibrium adjustments are associated with lower Region A residence among older cohorts. Consequently, the all-age Region A population share falls by 0.58 percentage points even though eligible young workers enter. Because the inflow is highly productive, Region A effective labor rises by 1.31 percent and aggregate output by 0.18 percent. The policy therefore changes who lives in Region A more than how many households live there.

Family transfers add little to this targeted relocation. The policy raises age-1 Region A residence by 0.91 percentage points with transfers and by 0.80 points without them; the corresponding ϕA effects are 0.85 and 0.82 points. The small interactions are consistent with the selection results in Section 5.2: parental transfers affect ownership primarily below the top of the talent distribution, whereas this policy targets high-earning households.

The welfare burden outside the targeted group is smaller with family transfers. Top-quintile age-1 households gain substantially, with a CEV of -7.35 percent. The bottom 80 percent face higher housing costs and a lower universal transfer, with a welfare cost of 2.23 percent with family transfers, compared with 3.76 percent without them. This comparison reflects the full transfer and no-transfer environments, not the isolated effect of additional parental giving induced by the subsidy. Detailed age profiles, transition diagnostics, and incidence results appear in Appendix F.3.

6.4 Purchase Vouchers: Ownership Location, Housing Scale, and Capitalization

The purchase voucher directly targets Region A ownership and housing investment. It allows me to assess whether assistance changes where talented young households own and how much housing they acquire, and whether parental transfers remain important once the purchase cost is subsidized.

Chinese cities often combine talent-attraction programs with assistance for home purchases (Nanjing Municipal Government, 2023). I model a once-per-lifetime voucher equal to 20 percent of the purchase value of a Region A home. Households qualify at ages 1 or 2 if they belong to the top talent quintile, have not claimed previously, and make a new oA purchase. The regulatory LTV ceiling applies to the full market value, while the purchase expenditure and down payment are based on the discounted cost. Recipients retain the full resale value and face no repayment or clawback. Voucher payments are financed through the equilibrium adjustment in the universal transfer. Take-up among households with unused eligibility is 28.6 percent at age 1 and 38.5 percent at age 2, and total expenditure equals 11.21 percent of tax revenue.

The voucher affects both the location and scale of ownership. Age-1 oA rises by 0.48 percentage points, while oB falls by 0.38 points. Consequently, age-1 homeownership across both regions rises by only 0.10 points. A common-taste-shock comparison in Appendix Figure F.9 attributes approximately 73 percent of gross inflows into oA to households coming from Region B. The policy therefore changes where eligible households own more than whether they own at all.

Voucher recipients also purchase larger homes and finance them differently from baseline top-quintile Region A buyers. As Table 9 shows, age-1 recipients buy 67.6 square meters, compared with 58.1 square meters among baseline top-quintile Region A buyers. After applying the voucher, their mean purchase expenditure is approximately RMB 1.016 million, below the RMB 1.067 million paid by the baseline comparison group despite the larger dwelling. Their mean down payment is RMB 191,900, compared with RMB 382,600 in the baseline buyer group, while their mean origination mortgage is RMB 824,400, compared with RMB 684,100. Mean end-of-age-1 home equity is also higher among recipients.

Part of the subsidy is capitalized into the Region A purchase price. Higher owner demand raises P_A by 2.27 percent, while lower rental demand reduces R_A by 1.21 percent. For a fixed dwelling, the price adjustment reduces the effective discount from the statutory 20 percent to 18.18 percent, eroding 9.09 percent of the original discount. In

Table 9: Home-Purchase Voucher: Buyer Financing and Price Capitalization

	Ages 20–29	Ages 30–39	Ages 20–39	
Panel A. Program scale				
Recipient share within age (%)	5.73	5.49	5.61	
Take-up among unused eligible households (%)	28.64	38.49	32.74	
Mean voucher per recipient	25.41	25.40	25.40	
Spending / tax revenue (%)	5.72	5.49	11.21	
Panel B. Financing of age-1 Region A buyers			Panel C. Price capitalization	
	Baseline top-20 oA	Voucher recipients	Policy value	
Housing size (m ²)	58.06	67.61	Statutory discount (%)	20.00
Net purchase expenditure	106.67	101.63	Region A price increase, P_A (%)	2.27
Down payment	38.26	19.19	Effective discount after GE (%)	18.18
Origination mortgage	68.41	82.44	Share of discount capitalized (%)	9.09
Mortgage payment / ten-year income	0.278	0.367		
End-of-age-1 home equity	51.01	59.95		

Notes: Panel B is a selected-group comparison, not a matched causal effect. Voucher recipients are endogenous policy takers; the comparison group is top-quintile age-1 Region A buyers in *B*. Voucher amounts, purchase expenditure, down payments, mortgage debt, and home equity are in units of RMB 10,000. Recipients retain full resale proceeds without repayment or clawback. The LTV ceiling applies to market value, while purchase expenditure and the down payment are based on the discounted cost. The capitalized share is the fraction of the statutory discount eroded by the equilibrium price increase.

the stationary equilibrium, Region A residence rises at eligible ages but falls at later ages, leaving the all-age Region A share 0.34 percentage points lower. Nevertheless, the high-talent inflow raises Region A effective labor by 1.12 percent and aggregate output by 0.15 percent.

The voucher's discrete purchase effect is nearly independent of family-transfer availability. It raises age-1 *oA* by 0.480 percentage points with transfers and by 0.483 points without them, yielding an interaction of approximately zero. Although transfer availability barely changes the voucher's effect on the ownership share, the policy modestly increases the equilibrium contribution of transfers to mean age-1 housing wealth and home equity. Recipients gain substantially, with a CEV of -7.17 percent, as reported in Appendix Table F.10, whereas the bottom 80 percent have a CEV of 1.44 percent because they receive no voucher and bear part of the higher purchase price and fiscal cost. Full financing, transition, transfer-response, and welfare results appear in Appendix F.4.

The four experiments show that the relationship between public policy and family finance depends on the margin targeted. A higher LTV ceiling substitutes for parental funds at the purchase threshold while leaving transfers important for leverage and equity; rental assistance substitutes for family finance in Region A access but not ownership. Because the talent subsidy and purchase voucher target high-earning households, parental

transfers add little to their discrete location and ownership responses and matter more for housing investment. For the two targeted subsidies, welfare costs are concentrated outside the eligible group through equilibrium housing-price and fiscal adjustments. Public and family finance are therefore neither uniformly substitutes nor complements.

7 Conclusion

Family transfers primarily reshape housing tenure and the distribution of housing wealth, with more modest effects on productive-city residence and aggregate output. This finding emerges from a quantitative life-cycle spatial equilibrium model calibrated to urban China, in which parents choose financial support for their adult children. Rental housing allows young workers to reside in the productive region without meeting the financing requirements for purchase. Although transfers increase the region's working population, the accompanying decline in average age-adjusted productivity partly offsets the contribution of this population expansion to effective labor.

Parental support expands ownership principally among young households with lower own earning capacity. In the transfer economy, young homeowners in the productive region have lower mean labor earnings but also use less mortgage leverage. Transfers narrow the ownership gap across children's talent groups and reduce inequality in young households' home equity. At the same time, wealthier parents provide larger amounts of assistance, widening the differences in children's ownership and equity across parental-wealth groups. The expansion of ownership among lower-earning young adults therefore coexists with a stronger relationship between housing outcomes and family background.

Family transfers raise aggregate welfare, with gains concentrated among young and lower-talent households. Assistance at the beginning of adult life supports consumption and liquidity and affects subsequent saving, mortgage repayment, and housing-equity accumulation. Some high-talent households nevertheless have lower self-welfare in the transfer economy: they receive comparatively little assistance when young, face higher purchase prices, and devote resources to their own children later in life.

Households that receive no parental transfers are also affected by the resulting changes in house prices and rents. Transfer-financed purchase demand raises equilibrium house prices, increasing the value of existing housing holdings and the acquisition costs faced by prospective buyers without parental assistance. The small decline in Region A rents instead reduces expenditure for households that access its labor market through rental housing.

The policy experiments distinguish the roles of public support and family resources. Relaxing mortgage borrowing limits partly substitutes for parental funds in financing home purchase, while family transfers continue to affect young households' leverage and equity. Rental assistance reduces dependence on parental support for residence in the productive region without materially reducing its role in ownership. The location and ownership effects of the talent subsidy and purchase voucher are similar with and without family transfers, reflecting their focus on households with relatively high earnings. Family support nevertheless affects the resulting distribution of housing wealth and welfare across eligible and ineligible households. At the scales considered, both targeted programs raise aggregate output but reduce age-weighted self-welfare: gains among eligible households are accompanied by fiscal costs and housing-price adjustments borne more broadly across the population.

Acknowledgements

I thank Martin Shu, Nathaniel Baum-Snow, Diego Daruich, Li Gan, Xican (Sean) Xi, Kristina Manyшева, Yu Yang, Jian Wang, Ting Chen, Shaoshuang Yang, Yuanning Liang, Yige Duan, Mark Roberts, Soyeon Lee, Linyi Cao, and Xue Qiao for helpful comments and discussions.

Funding. This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Declaration of Generative AI and AI-Assisted Technologies in the Manuscript Preparation Process

The author used ChatGPT and Codex (OpenAI) for language polishing and manuscript organization. The author reviewed and edited all AI-assisted revisions and takes full responsibility for the final manuscript.

References

Abbott, Brant, Giovanni Gallipoli, Costas Meghir, and Giovanni L. Violante, "Education Policy and Intergenerational Transfers in Equilibrium," *Journal of Political Economy*, 2019, 127 (6), 2569–2624.

- Adermon, Adrian, Mikael Lindahl, and Daniel Waldenström**, “Intergenerational Wealth Mobility and the Role of Inheritance: Evidence from Multiple Generations,” *The Economic Journal*, 2018, 128 (612), F482–F513.
- Bank of China**, “Individual First-Hand Residential Mortgage Loans,” 2008. Product terms specify a maximum maturity of 30 years; in Chinese.
- , “RMB Deposit Rates, October 24, 2015,” October 2015.
- Behrens, Kristian, Gilles Duranton, and Frédéric Robert-Nicoud**, “Productive Cities: Sorting, Selection, and Agglomeration,” *Journal of Political Economy*, 2014, 122 (3), 507–553.
- Benhabib, Jess, Alberto Bisin, and Shenghao Zhu**, “The Distribution of Wealth and Fiscal Policy in Economies With Finitely Lived Agents,” *Econometrica*, 2011, 79 (1), 123–157.
- Biasi, Barbara**, “School Finance Equalization Increases Intergenerational Mobility,” *Journal of Labor Economics*, 2023, 41 (1), 1–38.
- Boar, Corina and Danial Lashkari**, “Occupational Choice and the Intergenerational Mobility of Welfare,” Working Paper 29381, National Bureau of Economic Research October 2021.
- Brandsaas, Eirik Eylands**, “Illiquid Homeownership and the Bank of Mom and Dad,” Finance and Economics Discussion Series 2025-094, Board of Governors of the Federal Reserve System 2025.
- Bryan, Gharad and Melanie Morten**, “The Aggregate Productivity Effects of Internal Migration: Evidence from Indonesia,” *Journal of Political Economy*, 2019, 127 (5), 2229–2268.
- Campbell, John Y. and João F. Cocco**, “Household Risk Management and Optimal Mortgage Choice,” *The Quarterly Journal of Economics*, 2003, 118 (4), 1449–1494.
- Chen, Kaiji and Yi Wen**, “The Great Housing Boom of China,” *American Economic Journal: Macroeconomics*, 2017, 9 (2), 73–114.
- Chyn, Eric and Diego Daruich**, “An Equilibrium Analysis of the Effects of Neighborhood-Based Interventions on Children,” *American Economic Review*, 2025, 115 (12), 4476–4522.

- Colas, Mark, Sebastian Findeisen, and Dominik Sachs**, “Optimal Need-Based Financial Aid,” *Journal of Political Economy*, 2021, 129 (2), 492–533.
- Daysal, N. Meltem, Michael F. Lovenheim, and David N. Wasser**, “The intergenerational transmission of housing wealth,” Working Paper 31669, National Bureau of Economic Research September 2023.
- De Nardi, Mariacristina**, “Wealth Inequality and Intergenerational Links,” *The Review of Economic Studies*, 2004, 71 (3), 743–768.
- Engelhardt, Gary V. and Christopher J. Mayer**, “Intergenerational Transfers, Borrowing Constraints, and Saving Behavior: Evidence from the Housing Market,” *Journal of Urban Economics*, 1998, 44 (1), 135–157.
- Fang, Hanming and Jin Feng**, “The Chinese Pension System,” in Marlene Amstad, Guofeng Sun, and Wei Xiong, eds., *The Handbook of China’s Financial System*, Princeton, NJ: Princeton University Press, 2020, pp. 421–444.
- , **Quanlin Gu, Wei Xiong, and Li-An Zhou**, “Demystifying the Chinese housing boom,” *NBER Macroeconomics Annual*, 2016, 30 (1), 105–166.
- Favilukis, Jack, Pierre Mabilie, and Stijn Van Nieuwerburgh**, “Affordable housing and city welfare,” *The Review of Economic Studies*, 2023, 90 (1), 293–330.
- , **Sydney C. Ludvigson, and Stijn Van Nieuwerburgh**, “The macroeconomic effects of housing wealth, housing finance, and limited risk sharing in general equilibrium,” *Journal of Political Economy*, 2017, 125 (1), 140–223.
- Galor, Oded and Daniel Tsiddon**, “Technological Progress, Mobility, and Economic Growth,” *American Economic Review*, 1997, 87 (3), 363–382.
- Garriga, Carlos, Aaron Hedlund, Yang Tang, and Ping Wang**, “Rural-Urban Migration, Structural Transformation, and Housing Markets in China,” *American Economic Journal: Macroeconomics*, 2023, 15 (2), 413–440.
- General Office of the Guangzhou Municipal Government**, “Work Plan for Accelerating the Development of the Housing Rental Market in Guangzhou,” 2017. Guangzhou Government Document No. 29.
- General Office of the State Council of China**, “Opinions on Accelerating the Cultivation and Development of the Housing Rental Market,” 2016. State Council Document No. 39.

- Gilraine, Michael, James Graham, and Angela Zheng**, “Public Education and Intergenerational Housing Wealth Effects,” Working Paper 31345, National Bureau of Economic Research 2023.
- Greenwald, Daniel L. and Adam Guren**, “Do Credit Conditions Move House Prices?,” *American Economic Review*, 2025, 115 (10), 3559–3596.
- Guiso, Luigi and Tullio Jappelli**, “Private Transfers, Borrowing Constraints and the Timing of Homeownership,” *Journal of Money, Credit and Banking*, 2002, 34 (2), 315–339.
- He, Zhiguo, Zehao Liu, Xinle Pang, Yang Su, and Kunru Zou**, “Household Migration and Collateral Constraint: Cash-Based Housing Resettlement in China,” Working Paper 34982, National Bureau of Economic Research 2026.
- Hsieh, Chang-Tai and Enrico Moretti**, “Housing constraints and spatial misallocation,” *American Economic Journal: Macroeconomics*, 2019, 11 (2), 1–39.
- Huang, Wenbin**, “Government Land Regulations and Housing Supply Elasticity in Urban China,” *China & World Economy*, 2022, 30 (4), 122–148.
- IMF**, “IMF Engagement on Pension Issues in Surveillance and Program Work,” Technical Notes and Manuals 2022/004, International Monetary Fund June 2022.
- Kaplan, Greg, Kurt Mitman, and Giovanni L. Violante**, “The Housing Boom and Bust: Model Meets Evidence,” *Journal of Political Economy*, 2020, 128 (9), 3285–3345.
- Kennan, John and James R. Walker**, “The Effect of Expected Income on Individual Migration Decisions,” *Econometrica*, 2011, 79 (1), 211–251.
- Lee, Hyojung, Dowell Myers, Gary Painter, Johanna Thunell, and Julie Zissimopoulos**, “The Role of Parental Financial Assistance in the Transition to Homeownership by Young Adults,” *Journal of Housing Economics*, 2020, 47, 101597.
- Mabille, Pierre**, “The Missing Homebuyers: Regional Heterogeneity and Credit Contractions,” *The Review of Financial Studies*, 2023, 36 (7), 2756–2796.
- Meng, Lina, Xiao Xiao, and Yinggang Zhou**, “Housing boom and household migration decision: new evidence from China,” *The Journal of Real Estate Finance and Economics*, 2023, 67 (3), 453–479.
- Meng, Shasha, Fenghua Pan, and Fulong Wu**, “Intergenerational financial support for homeownership and co-residence in Chinese cities,” *Cities*, 2023, 137, 104310.

- Morten, Melanie and Jaqueline Oliveira**, “Paving the Way to Development: Costly Migration and Labor Market Integration,” Working Paper 22158, National Bureau of Economic Research April 2016.
- Nanjing Municipal Government**, “Nanjing Talent Housing Measures,” 2023. Effective February 12, 2023; in Chinese.
- Nishiyama, Shinichi**, “Bequests, Inter Vivos Transfers, and Wealth Distribution,” *Review of Economic Dynamics*, 2002, 5 (4), 892–931.
- People’s Bank of China**, “China Monetary Policy Report, Quarter Three, 2017,” Technical Report, People’s Bank of China November 2017. In Chinese.
- , “China Monetary Policy Report, Quarter Two, 2024,” Technical Report, People’s Bank of China August 2024.
- **and National Financial Regulatory Administration**, “Notice of the People’s Bank of China and the National Financial Regulatory Administration on Optimizing the Policy on Minimum Down Payment Ratios for Personal Housing Loans,” September 2024. Issued September 24, 2024.
- Saiz, Albert**, “The Geographic Determinants of Housing Supply,” *The Quarterly Journal of Economics*, 2010, 125 (3), 1253–1296.
- Shen, Yang and Bingqin Li**, “Policy Coordination in the Talent War to Achieve Economic Upgrading: The Case of Four Chinese Cities,” *Policy Studies*, 2022, 43 (3), 443–463.
- Zhang, Muyang and Jie Chen**, “Unequal School Enrollment Rights, Rent Yields Gap, and Increased Inequality: The Case of Shanghai,” *China Economic Review*, 2018, 49, 229–240.

Appendix A. Model Details and Numerical Solution

This appendix provides additional model details and describes how I solve the model numerically and compute the steady-state equilibrium. The numerical procedure follows three blocks: (i) solve the household dynamic programs given prices, (ii) propagate the cross-sectional distribution until it is stationary, and (iii) (when solving for general equilibrium) update prices and the universal transfer until markets clear.

A.1 Household Problems and Feasibility Constraints

The main text states the common renter, buyer, and continuing-owner budgets and the associated borrowing constraints. Here I specialize the Bellman equation and budget for four cases that cover entry, moving versus staying, transfer giving, and retirement.

State transitions. The option-specific transitions are

$$\begin{aligned} j = r\ell : \quad (\ell', \pi') &= (\ell, r), & h' = m' = q' &= 0, \\ j = o\ell : \quad (\ell', \pi', h') &= (\ell, o, h), & (m', q') &= (\vartheta_{K_g} M, \max\{K_g - 1, 0\}), \\ j = k\ell_- : \quad (\ell', \pi', h') &= (\ell_-, o, h_-), & (m', q') &= (\vartheta_q m, \max\{q - 1, 0\}). \end{aligned}$$

Thus, renting resets the housing and mortgage states, while keeping rules out moving, resizing, selling, and refinancing. Across all choices, consumption is strictly positive and liquid assets satisfy $x' \geq 0$. Renters and new buyers also satisfy the minimum housing-size requirement $h \geq \underline{h}$.

For every feasible action at ages 40–49, the continuation expectation integrates over the inheritance ι received and the child productivity z_c observed upon entry into ages 50–59. The next-period state is

$$s^4(j) = (z, x' + \iota/R_s, \ell'(j), \pi'(j), h'(j), m'(j), q'(j); z_c).$$

The housing and mortgage arguments follow the action-specific transitions above. This inheritance and child-productivity treatment applies equally to renters, new buyers, and keepers.

Example 1: A Region B entrant choosing Region A. Consider an age-1 household with state (z, f, B) . It begins without liquid assets, housing, or mortgage debt. Conditional on

renting in Region A, it solves

$$v_{rA}^1(z, f, B) = \max_{c>0, h, x' \geq 0} \left\{ u^1(c, h, rA) + \beta \bar{V}^2(z, x', A, r, 0, 0, 0) \right\}$$

$$\text{s.t. } c + x' + R_A h + \kappa_{A,1}^{\text{entry}} + \kappa_{rA,1}^{\text{rent}} = Y_A^{1,\text{disp}}(z) + f.$$

Conditional on buying in Region A, it solves

$$v_{oA}^1(z, f, B) = \max_{c>0, h, x' \geq 0, M \geq 0} \left\{ u^1(c, h, oA) + \beta \bar{V}^2(z, x', A, o, h, \vartheta_{K_1} M, \max\{K_1 - 1, 0\}) \right\}$$

$$\text{s.t. } c + x' + P_A h - M + \Pi_{K_1}(M) + \kappa_{A,1}^{\text{entry}} = Y_A^{1,\text{disp}}(z) + f,$$

$$M \leq \theta_{\text{LTV}} P_A h, \quad \Pi_{K_1}(M) \leq \theta_{\text{PTI}} y_A^1(z).$$

The renter cannot borrow and pays both the entry cost and the recurring rental friction. The buyer pays the entry cost but not the rental friction. Family transfers help finance the equity contribution required by the LTV limit but are excluded from the gross-income base used to calculate the PTI ceiling.

Example 2: An incumbent Region B owner moving or staying. Let $g \in \{2, 3, 5\}$ and $s^g = (z, x, B, o, h_-, m, q)$, and define sale equity by $\mathcal{E}_B = (1 - \delta_h) P_B h_- - m$. Conditional on renting in Region A, the household solves

$$v_{rA}^g(s^g) = \max_{c>0, h, x' \geq 0} \left\{ u^g(c, h, rA) + \beta \mathbb{E} \left[\bar{V}^{g+1}(s^{g+1}(rA)) \right] \right\}$$

$$\text{s.t. } c + x' + R_A h + \kappa_{A,g}^{\text{entry}} + \kappa_{rA,g}^{\text{rent}} = Y_A^{g,\text{disp}}(z) + R_s x + \mathcal{E}_B.$$

Conditional on purchasing a new Region A house, it solves

$$v_{oA}^g(s^g) = \max_{c>0, h, x' \geq 0, M \geq 0} \left\{ u^g(c, h, oA) + \beta \mathbb{E} \left[\bar{V}^{g+1}(s^{g+1}(oA)) \right] \right\}$$

$$\text{s.t. } c + x' + P_A h - M + \Pi_{K_g}(M) + \kappa_{A,g}^{\text{entry}} = Y_A^{g,\text{disp}}(z) + R_s x + \mathcal{E}_B,$$

$$M \leq \theta_{\text{LTV}} P_A h, \quad \Pi_{K_g}(M) \leq \theta_{\text{PTI}} y_A^g(z).$$

All continuation states follow the transitions defined above. If the household keeps its Region B house, it instead solves

$$v_{kB}^g(s^g) = \max_{c>0, x' \geq 0} \left\{ u^g(c, h_-, kB) + \beta \mathbb{E} \left[\bar{V}^{g+1}(s^{g+1}(kB)) \right] \right\}$$

$$\text{s.t. } c + x' + \Pi_q(m) = Y_B^{g,\text{disp}}(z) + R_s x.$$

Moving realizes sale equity and repays the old mortgage; keeping realizes no sale equity, incurs no mobility cost, and preserves the scheduled mortgage transition.

Example 3: A transfer-giving continuing owner. At $g = 4$, inheritance has already entered beginning liquid assets through $x_4 = x'_3 + \iota/R_s$ and is not added to the budget again. For state $s^4 = (z, x_4, \ell_-, o, h_-, m, q; z_c)$, a household that retains its existing home solves

$$v_{k\ell_-}^4(s^4) = \max_{\substack{c>0, x'\geq 0, \\ f\in\mathcal{F}}} \left\{ u^4(c, h_-, k\ell_-) \right. \\ \left. + \beta \bar{V}^5(z, x', \ell_-, o, h_-, \vartheta_q m, \max\{q-1, 0\}) \right. \\ \left. + \alpha_p \bar{V}^1(z_c, f, \ell_-) \right\} \\ \text{s.t. } c + x' + f + \Pi_q(m) = Y_{\ell_-}^{4,\text{disp}}(z) + R_s x_4, \quad f \geq 0.$$

The altruistic term values the linked entrant's optimized expected value. The parent's beginning-of-period region—not its newly selected region—is the entrant's origin. For a rent or purchase alternative, the Bellman equation retains the same altruistic term, while its budget includes sale equity and the corresponding mobility cost.

Example 4: A terminal-age renter, buyer, or continuing owner. At $g = 6$, a renter in region ℓ solves

$$v_{r\ell}^6(s^6) = \max_{c>0, h, x'\geq 0} \left\{ u^6(c, h, r\ell) + \omega_b \frac{(x' + \underline{e})^{1-\gamma}}{1-\gamma} \right\} \\ \text{s.t. } c + x' + R_\ell h + \kappa^6(r\ell; \ell_-) = Y_\ell^{6,\text{disp}}(z) + R_s x + \mathcal{E}(s^6).$$

Because terminal-age households cannot originate a mortgage, a buyer solves

$$v_{o\ell}^6(s^6) = \max_{c>0, h, x'\geq 0} \left\{ u^6(c, h, o\ell) + \omega_b \frac{[x' + (1 - \delta_h)P_\ell h + \underline{e}]^{1-\gamma}}{1-\gamma} \right\} \\ \text{s.t. } c + x' + P_\ell h + \kappa^6(o\ell; \ell_-) = Y_\ell^{6,\text{disp}}(z) + R_s x + \mathcal{E}(s^6).$$

Finally, a household that keeps its existing home solves

$$v_{k\ell-}^6(s^6) = \max_{c>0, x'\geq 0} \left\{ u^6(c, h_-, k\ell_-) + \omega_b \frac{[x' + (1 - \delta_h)P_{\ell-}h_- - \vartheta_q m + \underline{e}]^{1-\gamma}}{1 - \gamma} \right\}$$

$$\text{s.t. } c + x' + \Pi_q(m) = Y_{\ell-}^{6,\text{disp}}(z) + R_s x.$$

Each warm-glow estate must be nonnegative. A sale puts beginning home equity in current resources; otherwise, end-of-life home equity enters the estate.

A.2 Mortgage Amortization Schedule

The mortgage contract is fully amortizing with level monthly payments. Let r_m^{ann} denote the annual mortgage rate and $i_m = r_m^{\text{ann}}/12$ the corresponding monthly rate. At the beginning of a model period, $q \in \{0, 1, 2, 3\}$ denotes the number of ten-year payment blocks remaining. The mortgage-free state has $q = 0$, zero outstanding debt $m = 0$, and zero scheduled payments, $a_0(0) = \Pi_0(0) = 0$. Without new borrowing, its next-period mortgage state remains $m' = q' = 0$. The amortization formulas below apply only to $q \geq 1$, for which $120q$ monthly payments remain. For a beginning balance m , the level monthly payment is

$$a_q(m) = m \frac{i_m}{1 - (1 + i_m)^{-120q}}. \quad (\text{A.1})$$

Because one model period comprises 120 months, the mortgage payment entering the household's ten-year budget is

$$\Pi_q(m) = 120a_q(m) = \pi_q m, \quad \pi_q = \frac{120i_m}{1 - (1 + i_m)^{-120q}}. \quad (\text{A.2})$$

Thus, π_q is the total scheduled principal and interest paid during the current ten-year period per unit of beginning mortgage balance; it is not an interest rate.

After these 120 monthly payments, the remaining balance is

$$m' = \vartheta_q m, \quad \vartheta_q = \frac{(1 + i_m)^{120q} - (1 + i_m)^{120}}{(1 + i_m)^{120q} - 1}. \quad (\text{A.3})$$

The mortgage state then moves from q to $q - 1$. In particular, $\vartheta_1 = 0$ because a mortgage with one block remaining is fully repaid within the current model period.

With $r_m^{\text{ann}} = 0.0501$, and hence $i_m = 0.004175$, Table A.1 reports the coefficients used in the baseline.

Table A.1: Mortgage payment and remaining-balance coefficients

Remaining blocks q	Remaining maturity	π_q	ϑ_q
1	10 years	1.273373	0.000000
2	20 years	0.792610	0.622449
3	30 years	0.644920	0.813666

Notes: π_q aggregates the next 120 monthly payments relative to the beginning balance.
 ϑ_q is the fraction of that balance outstanding after the same payments.

A.3 Stochastic Inheritance Distribution

Terminal estates generated by age-6 households determine the inheritance distribution received upon entry into age group $g = 4$. For each terminal household, the estate is the nonnegative value defined in equation (24). Let F_e denote the mass-weighted cross-sectional distribution of these estates. I approximate F_e with five weighted-quantile bins whose cumulative-probability edges are

$$(q_0, \dots, q_5) = (0, 0.25, 0.50, 0.75, 0.90, 1).$$

The inheritance probabilities are fixed by the quantile-bin widths, $p_k^b = q_k - q_{k-1}$. Each inheritance node is the mass-weighted mean terminal estate within its bin:

$$b_k = \frac{1}{p_k^b} \int_{q_{k-1}}^{q_k} F_e^{-1}(u) du, \quad F_e^{-1}(u) = \inf\{e : F_e(e) \geq u\}.$$

The realized inheritance is $\iota = b_k$ with probability p_k^b . For discrete estates, probability mass crossing a quantile boundary is split between adjacent bins, with both portions retaining the same estate value. This preserves the prescribed bin masses. Thus, the probability vector is $\{p_k^b\}_{k=1}^5 = (0.25, 0.25, 0.25, 0.15, 0.10)$. In the calibrated baseline, the corresponding nodes are

$$\{b_k\}_{k=1}^5 = (7.5400, 12.6455, 20.6091, 37.3026, 98.7246),$$

in units of 10,000 RMB.

Inheritance is realized once, in the transition from age group $g = 3$ to $g = 4$. Because beginning-of-period liquid wealth yields the gross return R_s in the household budget, a draw b_k maps the chosen end-of-age-3 asset position into the age-4 state as

$$x_4 = x'_3 + \frac{b_k}{R_s}.$$

This makes current age-4 resources rise by exactly b_k . The age-3 continuation value integrates over the five inheritance nodes, and the stationary-distribution transition assigns the same probabilities while using linear interpolation over the liquid-asset grid. Inheritance is therefore a one-time resource shock rather than an additional persistent state.

The inheritance nodes are determined by a stationary fixed point. Given a candidate distribution $\mathcal{B}^{(n)} = \{b_k^{(n)}, p_k^b\}_{k=1}^5$, the household problem and stationary distribution imply conditional mean estates $\tilde{b}_k^{(n)}$ in the same quantile bins. The nodes are updated toward these means with damping parameter $\lambda_b = 0.5$, while the probabilities remain fixed:

$$b_k^{(n+1)} = (1 - \lambda_b)b_k^{(n)} + \lambda_b\tilde{b}_k^{(n)}, \quad k = 1, \dots, 5.$$

Iterations stop when the normalized node discrepancy defined in Appendix A.5 is no greater than 10^{-3} . The baseline inheritance gap is 5.56×10^{-4} . The inheritance draw is unconditional on the recipient's productivity and earlier family-transfer history. This preserves the stationary marginal estate distribution without introducing a separate dynastic-wealth state.

A.4 Housing Supply and Owner–Rental Allocation

This subsection provides the computational details behind the housing-supply system in equations (26)–(29). Let owner and renter demand for floor space in region ℓ be

$$D_{\ell,t}^{\text{own}} = \sum_{g=1}^6 \int h \, d\mu_{o\ell,t}^g, \quad D_{\ell,t}^{\text{rent}} = \sum_{g=1}^6 \int h \, d\mu_{r\ell,t}^g. \quad (\text{A.4})$$

At the reference equilibrium, total regional housing space, its rental share, and its owner-occupied share are backed out as

$$H_{\ell,0} = D_{\ell,0}^{\text{own}} + D_{\ell,0}^{\text{rent}}, \quad s_{\ell,0}^r = \frac{D_{\ell,0}^{\text{rent}}}{H_{\ell,0}}, \quad \Omega_{\ell,0} = 1 - s_{\ell,0}^r. \quad (\text{A.5})$$

Here $\Omega_{\ell,0}$ is the share of regional floor space used by owners, not the household home-ownership rate. Stationarity and the total-stock supply rule then imply

$$\bar{I}_\ell = \frac{\delta_h H_{\ell,0}}{P_{\ell,0}^{\rho_\ell}}. \quad (\text{A.6})$$

These backouts make the owner and rental supplies reproduce the four reference segment demands exactly at $(P_{\ell,0}, R_{\ell,0})$.

The log-odds rule in equation (28) has the closed form

$$Q_{\ell,t} \equiv \left[\frac{R_{\ell,t}/P_{\ell,t}}{R_{\ell,0}/P_{\ell,0}} \right]^{\chi_\ell}, \quad s_{\ell,t}^r = \frac{s_{\ell,0}^r Q_{\ell,t}}{1 - s_{\ell,0}^r + s_{\ell,0}^r Q_{\ell,t}}. \quad (\text{A.7})$$

Thus, χ_ℓ is the elasticity of the rental-to-owner stock ratio with respect to the rent-to-price ratio. For an interior share,

$$\frac{\partial \log[s_\ell^r/(1 - s_\ell^r)]}{\partial \log(R_\ell/P_\ell)} = \chi_\ell, \quad \frac{\partial \log s_\ell^r}{\partial \log(R_\ell/P_\ell)} = \chi_\ell(1 - s_\ell^r), \quad \frac{\partial \log(1 - s_\ell^r)}{\partial \log(R_\ell/P_\ell)} = -\chi_\ell s_\ell^r. \quad (\text{A.8})$$

The first derivative is the parameter reported in the calibration. The other two show that the elasticity of either segment's share depends on its current size.

Combining the share rule with $H_{\ell,t} = \bar{I}_\ell P_{\ell,t}^{\rho_\ell} / \delta_h$ gives

$$S_{\ell,t}^{\text{rent}} = s_{\ell,t}^r H_{\ell,t}, \quad S_{\ell,t}^{\text{own}} = (1 - s_{\ell,t}^r) H_{\ell,t}. \quad (\text{A.9})$$

Holding rent fixed, the purchase-price elasticities of these supplies are

$$\frac{\partial \log S_\ell^{\text{rent}}}{\partial \log P_\ell} = \rho_\ell - \chi_\ell(1 - s_\ell^r), \quad \frac{\partial \log S_\ell^{\text{own}}}{\partial \log P_\ell} = \rho_\ell + \chi_\ell s_\ell^r. \quad (\text{A.10})$$

The purchase price therefore changes both total stock and its division across uses, whereas rent changes the owner–rental division without directly changing total stock. With $\chi_A = \chi_B = 1$, the reference rental shares are 0.438 in Region A and 0.207 in Region B. A one-percent increase in the rent-to-price ratio consequently raises the rental share itself by approximately 0.562 percent in Region A and 0.793 percent in Region B.

The four housing-market residuals used by the equilibrium solver are scaled demand–supply gaps,

$$\mathcal{R}_\ell^{\text{own}} = \frac{D_\ell^{\text{own}}}{S_\ell^{\text{own}}} - 1, \quad \mathcal{R}_\ell^{\text{rent}} = \frac{D_\ell^{\text{rent}}}{S_\ell^{\text{rent}}} - 1, \quad \ell \in \{A, B\}. \quad (\text{A.11})$$

Equilibrium requires all four residuals to equal zero. For numerical stability, the implementation bounds the raw rental share from (A.7) to $[0.02, 0.98]$ and records whenever a bound is reached.

A.5 Numerical Solution and Stationary Distribution

Given prices, wages, the universal transfer, and a candidate inheritance distribution, I solve the six age-specific household problems on finite grids. For each feasible location–

tenure option, the code constructs tensors over the relevant states and controls, recovers consumption from the corresponding budget in equations (17)–(20), evaluates current and continuation utility, and maximizes over the continuous-control grid. Choices with $c \leq 0$ or that violate equation (19) receive an infeasible value. Location–tenure probabilities are then computed using a numerically stable log-sum-exp calculation.

Table A.2 summarizes the baseline grids. The grids for liquid assets, housing, and transfers are nonuniform, with more nodes near the financing margins relevant for young households. The productivity grid has a probability-weighted mean of one. In addition, the state space contains four location–tenure statuses, two parental-origin states for age-1 households, and age-specific mortgage-duration and mortgage-ratio states.

Table A.2: Baseline Numerical Grids

Grid object	Number of nodes	Range or discrete values
Productivity, z	13	[0.3003, 12.3146]
Liquid assets, x and x'	10	[0, 500]
Housing size, h	13	[10, 200]
Inter vivos transfer, f	13	[0, 300]
New mortgage as a fraction of the LTV ceiling	5	{0, 0.25, 0.50, 0.75, 1}
Mortgage duration, q	4	{0, 1, 2, 3} ten-year blocks
Inheritance, b	5	endogenous; [7.5400, 98.7246] in the baseline

Notes: Liquid assets, transfers, and inheritance are in units of 10,000 RMB; housing is in square meters. The asset nodes are {0, 1, 2, 5, 10, 20, 40, 80, 200, 500}, the housing nodes are {10, 15, 20, 30, 40, 50, 60, 80, 100, 125, 150, 175, 200}, and the transfer nodes are {0, 1, 2, 5, 10, 20, 30, 50, 80, 120, 160, 220, 300}. The new-mortgage grid is defined over $M/(\theta_{LTV} P_\ell h)$: origination debt equals the listed fraction multiplied by the LTV ceiling. Candidates that violate the PTI limit are excluded; the grid is not rescaled to that limit.

The terminal-age problem is solved first. Given a previous guess for the age-1 inclusive value, I solve the continuation problems backward. The age-4 altruistic term links the transfer-giving household to the value of its age-1 heir, so I iterate this backward recursion until the maximum absolute change in the linked value-function fixed point is below 10^{-5} . The update uses damping of 0.7 and allows at most 500 iterations. Shape-compatible value functions from a nearby calibration trial, policy case, or equilibrium-price iteration provide warm starts. In the accepted baseline, the final value residual is 2.44×10^{-6} .

Given the policy functions, I iterate probability mass over the finite state space until the cross-sectional distribution becomes stationary. The transition operator moves probability mass from age 1 through age 6 and then maps age-4 transfer choices and parental origins into the next cohort of age-1 households. Inheritance transitions use linear inter-

polation on the liquid-asset grid. The six age-specific state spaces contain, respectively, 338, 17,160, 30,680, 398,840, 30,680, and 3,640 points, or 481,338 potential states in total. At the baseline stationary distribution, 40,422 state records carry positive mass, and their policies generate 198,992 option-specific flow records. These counts, rather than a number of simulated households, determine the numerical population size.

Let $\mu_1^{(n)}(z, f, o)$ denote the entrant distribution after iteration n , where o is parental origin. Distribution iteration stops when

$$d_\mu^{(n)} = \max_{z, f, o} \left| \mu_1^{(n+1)}(z, f, o) - \mu_1^{(n)}(z, f, o) \right| < 10^{-4},$$

with a maximum of 500 iterations. The accepted baseline converges in four iterations with $d_\mu = 3.23 \times 10^{-5}$.

Finally, the inheritance nodes are updated from the terminal-estate distribution, with fixed quantile probabilities, as described in Appendix A.3. For current nodes b_k and their model-implied conditional means $\tilde{b}_k$, the inheritance gap is

$$d_b = \frac{\max_k |\tilde{b}_k - b_k|}{\max\{1, K^{-1} \sum_{k=1}^K |b_k|\}}, \quad K = 5.$$

The update uses damping of 0.5 and stops when $d_b \leq 10^{-3}$, with at most 20 iterations. The accepted baseline inheritance gap is 5.56×10^{-4} . The implementation also checks probability normalization and the discrepancy between current and implied bin masses. Under the fixed quantile construction, these discrepancies vanish up to numerical rounding; the probabilities are not independently endogenous.

A.6 Aggregation and Equilibrium Residuals

Given a converged stationary distribution, I integrate household outcomes using state measures for the full population. Each age-specific measure has mass 1/6 across the two regions; age weights are already incorporated. The saved within-age distributions are normalized to one and receive this weight once when constructing these measures. Working-age effective labor and output in region $\ell \in \{A, B\}$ are

$$L_\ell = \sum_{g=1}^5 \int G_g z d\mu_\ell^g(z), \tag{A.12}$$

$$Y_\ell = \text{TFP}_\ell L_\ell, \quad Y = Y_A + Y_B. \tag{A.13}$$

These output measures are productivity-normalized. For ratios with monetary numerators, I use $Y^{\text{mon}} = wY = 10w^{\text{ann}}Y = 50Y$, measured in 10,000 RMB over a ten-year stage. Spending, aggregate family transfers, and wealth are therefore divided by monetary output; wealth/output ratios use ten-year, not annual, output. The common monetary scale cancels in percentage output changes.

Only ages 1–5 supply labor. The exogenous age and talent distributions fix total effective labor, while household location choices determine its regional allocation. Because labor supply is inelastic and production is linear, there is no separate labor-market residual. Percentage changes in aggregate output relative to a reference equilibrium are reported as

$$\% \Delta Y = \left(\frac{Y^{\text{policy}}}{Y^{\text{base}}} - 1 \right) \times 100. \quad (\text{A.14})$$

Housing demand is aggregated separately for owners and renters in each region. The resulting four demand measures are compared with the owner and rental supplies defined in Appendix A.4; their scaled residuals are given by equation (A.11).

Financial market. The gross return on liquid assets and the mortgage-contract terms are fixed externally. The financial sector elastically absorbs household liquid assets and supplies mortgage credit at these terms, so the model imposes no bond- market-clearing residual. Aggregate liquid assets net of mortgage debt are reported as net financial assets rather than used as an equilibrium condition.

Government budget. Tax revenue equals the common labor-income tax rate τ_l times gross labor income over ages 1–5. Let M denote total stationary population mass. Total government spending is

$$\text{GovernmentSpending} = \text{PensionSpending} + MT + \text{PolicyOutlays},$$

where policy outlays are zero in the baseline. Family transfers and inheritances are private flows and do not enter this expression. The fiscal residual used by the equilibrium solver is

$$\mathcal{R}^G = \frac{\text{TaxRevenue}}{\text{GovernmentSpending}} - 1. \quad (\text{A.15})$$

A.7 General-Equilibrium Solver

For each counterfactual, the housing-supply intercepts and reference rental shares backed out in Appendix A.4, together with the elasticities ρ_ℓ and χ_ℓ , remain fixed. I initialize the

solver from the associated reference equilibrium and jointly update (P_A, P_B, R_A, R_B, T) and the inheritance nodes, keeping their quantile probabilities fixed. Household values are warm-started from the preceding equilibrium iteration, but every candidate requires a new household solution and stationary distribution.

To state the update rule, define $q_{\ell,n}^j = D_{\ell,n}^j / S_{\ell,n}^j$ for $j \in \{\text{own}, \text{rent}\}$ and $\text{clip}(u; \bar{u}) = \max\{-\bar{u}, \min(u, \bar{u})\}$. At iteration n , prices and rents satisfy

$$\log P_{\ell,n+1} = \log P_{\ell,n} + \text{clip}(d_{P,n} \log q_{\ell,n}^{\text{own}}; 0.10), \quad (\text{A.16})$$

$$\log R_{\ell,n+1} = \log R_{\ell,n} + \text{clip}(d_{R,n} \log q_{\ell,n}^{\text{rent}}; 0.10). \quad (\text{A.17})$$

Thus, excess demand raises the corresponding purchase price or rent. The government-budget-implied transfer and its update are

$$T_n^* = \frac{\text{TaxRevenue}_n - \text{PensionSpending}_n - \text{PolicyOutlays}_n}{M}, \quad (\text{A.18})$$

$$T_{n+1} = T_n + \text{clip}(d_{T,n}(T_n^* - T_n); 5). \quad (\text{A.19})$$

The inheritance nodes are updated in the same outer iteration using the rule in Appendix A.3; the quantile-bin probabilities remain fixed.

The initial damping coefficients are $d_{P,0} = d_{R,0} = 0.25$ and $d_{T,0} = 0.50$. If a model evaluation fails, or if the joint residual rises in three consecutive iterations, the solver halves the damping coefficients, subject to lower bounds of 0.02, 0.02, and 0.05, respectively. The caps in equations (A.16)–(A.19) prevent a single iteration from changing a log price or rent by more than 0.10 or T by more than five model units.

Let

$$d_{\text{GE}} = \max \left\{ \max_{\substack{\ell \in \{A,B\} \\ j \in \{\text{own}, \text{rent}\}}} |q_{\ell}^j - 1|, |\mathcal{R}^G| \right\}. \quad (\text{A.20})$$

An equilibrium is accepted only when $d_{\text{GE}} \leq 10^{-3}$, $d_b \leq 10^{-3}$, and the inner value-function and stationary-distribution iterations have converged. I also verify the government budget in levels. General-equilibrium iterations are capped at 50. If this cap is reached, the solver retains the best inner-converged iteration for diagnosis but does not label it as converged.

A.8 Computation Time

The reported results were computed with 100 MATLAB workers on a dual-socket Intel Xeon Platinum 8280 server. Starting from stored value-function warm starts, resolving the

baseline inheritance fixed point required nine full household and distribution solutions and 41.0 minutes. The no-transfer general equilibrium then converged in 13 iterations and 19.7 minutes; its individual warm-started iterations took between 54 and 158 seconds. Thus, computing the complete baseline and no-transfer pair took approximately 61 minutes.

Appendix B. Data and Calibration

The quantitative analysis uses the 2017 China Household Finance Survey (CHFS). I use individual records to construct the regional population shares and earnings process, and household records to construct tenure, purchase-price, and rent moments. All statistics use the corresponding CHFS individual or household survey weights. This appendix describes the samples and variable construction, then maps the resulting moments into the calibrated objects.

B.1 Survey, Geography, and Samples

The CHFS reports demographic characteristics, labor income, education, location, housing tenure, dwelling size, self-assessed home value, and rent. I merge the individual and household files with the survey’s city identifiers and weights. The analysis excludes rural observations because rural housing is often obtained through village-allocated homestead land rather than the market prices and mortgage contracts represented in the model. The restriction also aligns the model’s talent-based earnings process with urban labor markets; the model does not represent land-based rural production.

Region A comprises 18 Tier 1 and Tier 2 cities: Beijing, Shanghai, Guangzhou, Shenzhen, Chengdu, Hangzhou, Wuhan, Chongqing, Nanjing, Tianjin, Suzhou, Xi’an, Changsha, Shenyang, Qingdao, Zhengzhou, Dalian, and Ningbo. Region B contains the remaining urban cities. Age is survey year minus birth year. The first five empirical bins are 20–29 through 60–69; the terminal bin pools respondents aged 70 and older and maps to the model’s sixth life-cycle stage.

Regional population shares and the earnings process use individual records and individual weights. Tenure, purchase prices, and rents use the household head, the household housing record, and household weights. The estimation sample for the working-age earnings process contains 24,465 individuals with positive labor income and valid age, region, education, and survey weight. The final purchase-price samples contain 8,401 owner households in Region A and 10,626 in Region B; the rent samples contain 2,165 and 1,470 renter households, respectively. Sample sizes differ because each moment re-

quires its own valid income or housing fields.

B.2 Location and Tenure Targets

I first compute the weighted share of individuals living in Region A within each age group. The raw shares are 31.8, 33.2, 29.4, 30.9, 34.9, and 34.5 percent from the youngest to the oldest group. I combine these estimates with weighted homeownership rates among household heads in each age–region cell. Because some conditional tenure cells are small, the calibration uses rounded life-cycle profiles rather than treating sampling variation in every cell as a separate target. Table B.1 reports the three profiles used to construct the 24 joint location–tenure targets.

Let A_g denote the target Region A share and let $O_{\ell g}$ denote the target ownership rate conditional on region ℓ at age g . The four target shares are

$$\begin{aligned} s_{rB,g} &= (1 - A_g)(1 - O_{Bg}), & s_{oB,g} &= (1 - A_g)O_{Bg}, \\ s_{rA,g} &= A_g(1 - O_{Ag}), & s_{oA,g} &= A_gO_{Ag}. \end{aligned}$$

This construction makes the regional and conditional-tenure content of the targets explicit and guarantees that the four shares sum to one within age.

Table B.1: Construction of Age-Specific Location–Tenure Targets

Age	Target components			Implied joint shares			
	$\Pr(A)$	$\Pr(o B)$	$\Pr(o A)$	rB	oB	rA	oA
20–29	35.0	60.0	37.0	26.00	39.00	22.05	12.95
30–39	35.0	85.0	60.0	9.75	55.25	14.00	21.00
40–49	30.0	85.0	70.0	10.50	59.50	9.00	21.00
50–59	30.0	85.0	70.0	10.50	59.50	9.00	21.00
60–69	35.0	85.0	70.0	9.75	55.25	10.50	24.50
70 and older	35.0	85.0	70.0	9.75	55.25	10.50	24.50

Notes: Entries are percentages. The component profiles are rounded benchmarks constructed from CHFS-weighted region and tenure profiles. The implied shares are the targets reported in Appendix Table B.5. The terminal survey group maps to the model stage labeled 70–79.

B.3 Earnings, Regional Productivity, and Talent

Individual labor income is the sum of the two reported monthly labor-income components. I drop observations for which both components are missing or refused, set a single missing component to zero, and retain positive total income. Monthly income is converted to annual units of 10,000 RMB and winsorized at the first and ninety-ninth per-

centiles before estimation. No additional trimming is applied in the earnings regressions, which use the individual survey weights and working ages 20–69.

With mean productivity normalized to one, I set $w^{\text{ann}} = 5$, or RMB 50,000 before age and regional adjustments; the ten-year model wage is $w = 50$.

To recover the age profile and permanent talent distribution, I estimate

$$\log y_i = a + \sum_{g \neq 2} d_g \mathbb{1}\{g_i = g\} + d_A \mathbb{1}\{\ell_i = A\} + b_e \text{edu}_i + u_i, \quad (\text{B.1})$$

where education categories are mapped into years of schooling and ages 30–39 are the reference group. Exponentiating the age coefficients gives

$$(G_1, \dots, G_5) = (0.844, 1.000, 0.968, 0.911, 0.670).$$

I set $G_6 = 0$ because the last stage is retired. Permanent talent retains both the schooling component and the residual. Specifically, I remove only the constant, age effect, and region effect from log income, exponentiate, and normalize the weighted mean of z_i to one.

A weighted maximum-likelihood fit of $F(z) = \exp[-(s_z/z)^{\theta_z}]$ to these normalized observations gives $\theta_z = 1.9266$ and $s_z = 0.5430$. The baseline uses 13 conditional-bin means: nine decile bins through the ninetieth percentile and four upper-tail bins with probabilities 0.05, 0.03, 0.01, and 0.01. The grid spans 0.3003 to 12.3146 and is renormalized so that its probability-weighted mean is exactly one. Productivity remains fixed over the life cycle, and the benchmark assigns each heir an independent draw from this distribution. Appendix E.3 allows positive intergenerational persistence in talent.

A separate weighted log-earnings regression with age-group and Region A indicators provides the regional earnings premium used to set $\text{TFP}_A = 1.37$ relative to $\text{TFP}_B = 1$.

The taste-shock scale is then calibrated to an equilibrium working-age gross-income ratio of 2.0 between Regions A and B. This ratio is a sorting target, not the raw regional mean in the cleaned earnings sample: the latter is 1.404, while the age-and-education-adjusted regional coefficient in (B.1) implies 1.275. Keeping these objects separate prevents the exogenous regional productivity premium from absorbing the endogenous sorting governed by σ_ν .

I set the pension to one-half of the location-independent earnings base wG_4z , corresponding to $\rho_{\text{pen}} = 0.50$. This replacement rate is close to the 46 percent average pension replacement ratio reported for China in 2017 by Fang and Feng (2020) and lies within the 40–60 percent adequacy range in IMF (2022).

The labor-income tax is proportional: all working-age households pay 10 percent of gross labor earnings. This assigned rate, $\tau_l = 0.10$, remains fixed across counterfactuals.

Appendix Figure B.1 shows the weighted distributional fit used to construct the productivity grid.

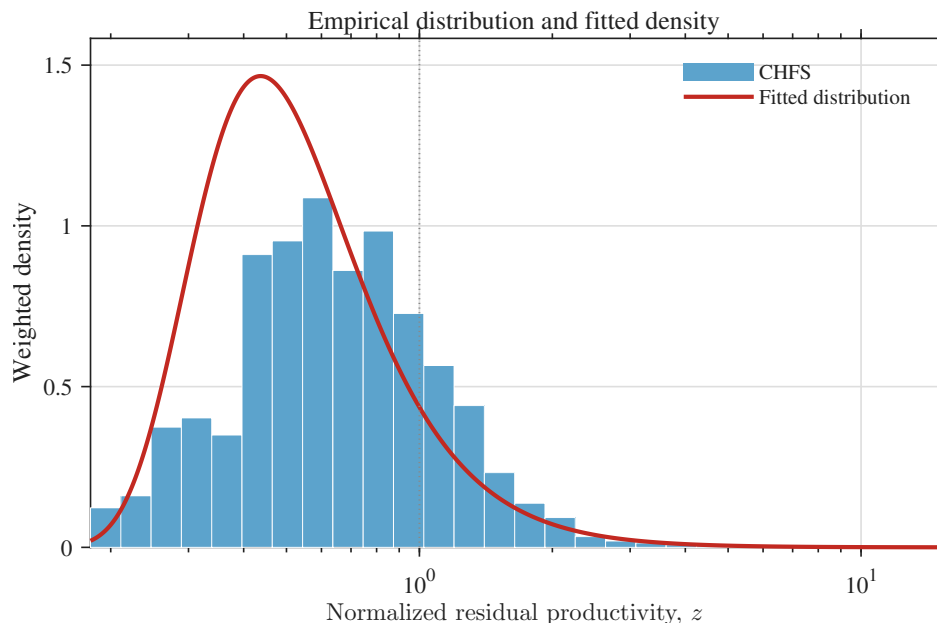


Figure B.1: Residual Productivity Distribution and Fréchet Fit

Notes: The figure plots the CHFS-weighted distribution of normalized residual productivity and the fitted mean-one Fréchet density. The dotted vertical line marks $z = 1$.

B.4 Productivity Sorting across Location and Tenure

I use conditional productivity across location–tenure groups only as a qualitative sorting discipline. The main sample consists of urban household heads aged 20–29 with positive labor income and valid productivity, tenure, and household weights. The alternative sample assigns the household’s tenure to every positive-income worker aged 20–29 and uses individual survey weights. Table B.2 reports both definitions together with the stationary age-1 model means.

The stable empirical restriction is that Region A renters are more productive than Region B owners. The difference is 9.0 percent among household heads and 28.2 percent in the all-worker sample, while the model implies 10.1 percent. The sensitivity of the magnitude to the unit of observation motivates disciplining the ordering and its approximate strength rather than fitting the complete set of conditional means. Appendix E.5 shows how this sorting diagnostic changes when the recurring rental friction is removed.

Table B.2: Young-Household Productivity by Location and Tenure

Choice	CHFS household heads	CHFS all workers	Baseline model
rB	1.407 (114)	1.112 (355)	0.908
oB	1.367 (211)	1.008 (2,271)	0.907
rA	1.490 (322)	1.291 (662)	0.999
oA	1.529 (192)	1.078 (1,336)	1.419

Notes: Entries are survey-weighted or stationary-mass-weighted conditional mean talent. Unweighted CHFS sample sizes are in parentheses. CHFS talent is normalized to have mean one in the full working-age estimation sample; the model talent grid has an ex ante mean of one. The conditional levels and gaps are not quantitative calibration targets.

B.5 Purchase Prices and Rents

For owner households, I use the reported value and floor area of the primary residence. I replace interval-coded values by their bracket midpoints and winsorize reported values at the weighted first and ninety-ninth percentiles. I remove floor-area observations above the weighted 99.5th percentile because a small number of reported dwellings have extremely large areas and would otherwise distort the regional price moments. The regional purchase-price moment is the household-weighted mean of each owner's unit price,

$$P_\ell^{\text{data}} = \frac{1}{10,000} \frac{\sum_{i \in \mathcal{O}_\ell} w_i (V_i / H_i)}{\sum_{i \in \mathcal{O}_\ell} w_i}. \quad (\text{B.2})$$

Here $\ell \in \{A, B\}$ indexes the region, $\mathcal{O}_\ell$ is the set of sampled owner households in region ℓ , w_i is household i 's CHFS sampling weight, V_i is the winsorized value of its primary residence in RMB, and H_i is its floor area in square meters. The factor $1/10,000$ converts the resulting unit price into 10,000 RMB per square meter. This yields $P_A = 1.8373$ and $P_B = 0.4747$ in units of 10,000 RMB per square meter, or RMB 18,373 and RMB 4,747 per square meter.

For renter households, I require positive monthly rent and floor area and apply the same weighted 99.5th-percentile floor-area trim. The model-period rent per square meter is

$$R_\ell^{\text{data}} = \frac{10 \times 12}{10,000} \frac{\sum_{i \in \mathcal{R}_\ell} w_i (\text{rent}_i^{\text{mo}} / H_i)}{\sum_{i \in \mathcal{R}_\ell} w_i}. \quad (\text{B.3})$$

The resulting ten-year rents are $R_A = 0.3108$ and $R_B = 0.1576$ in model units. For reference, the corresponding weighted mean dwelling sizes are 99.34 and 123.16 square meters for owners in Regions A and B, and 53.80 and 65.85 square meters for renters. These size statistics are descriptive and are not calibration targets.

I map these four price and rent moments into the reference housing-supply system

using model-implied regional floor-space demand. The owner-space share $\Omega_{\ell,0}$ is owner-occupied floor space divided by total regional floor space, not the household homeownership rate. The backout gives $\Omega_{A,0} = 0.562$ and $\Omega_{B,0} = 0.793$, corresponding to reference rental shares of 0.438 and 0.207. Given $\delta_h = 0.140$ and the assigned supply elasticities, the stationary stock condition gives $\bar{I}_A = 1.386$ and $\bar{I}_B = 9.497$. These objects reproduce owner and renter floor-space demand at the four target prices and rents; Appendix A.4 provides the backout equations and segment-market residuals.

The supply elasticities are assigned rather than estimated from this sample.³

B.6 Calibration Map and Fit

Table B.3 summarizes how the empirical objects discipline the model. The 18 nonnormal-ized location–tenure constants match three independent shares per age; the taste-shock scale matches the regional income ratio; and the owner-space shares and construction shifters reproduce the four purchase-price and rent targets. The universal transfer balances the government budget, and the inheritance nodes close the terminal-estate fixed point. The Region A-renter versus Region B-owner productivity ordering provides a qualitative discipline for the recurring rental friction. The remaining externally assigned parameters in Tables 1 and B.4 are not chosen to match these quantitative moments.

Table B.3: Empirical Objects and Calibration Discipline

Object	Construction or target	Model discipline
Age–location–tenure shares	Weighted urban CHFS profiles, smoothed as in Table B.1	$\{\xi_{oB}^g, \xi_{rA}^g, \xi_{oA}^g\}_{g=1}^6$
Age earnings and permanent talent	Weighted log-earnings regression and mean-one Fréchet fit	$\{G_g\}, \theta_z, s_z$, and the 13-point z grid
Regional earnings premium	Weighted age-adjusted log-earnings regression	TFP_A/TFP_B
Young rA – oB productivity ordering	Weighted conditional means under the two samples in Table B.2	Qualitative discipline for $\{\kappa_{rA,g}^{\text{rent}}\}_{g=1}^6$
Working-age A/B gross-income ratio	Equilibrium target of 2.0	σ_ν
Regional purchase prices and rents	Equations (B.2) and (B.3)	$\Omega_{A,0}, \Omega_{B,0}, \bar{I}_A, \bar{I}_B$
Government budget	Payments equal tax revenue in the stationary equilibrium	T
Terminal-estate distribution	Conditional mean estates within fixed quantile bins	Inheritance nodes

Notes: Positive-transfer incidence is reported as a model outcome and is not targeted. Housing-size moments and conditional productivity levels are also untargeted.

³Saiz (2010) estimates metropolitan housing-supply elasticities using demand variation, terrain and water coverage, and land-use regulation. Mabilie (2023) uses these estimates in a quantitative spatial housing model.

Table B.4 collects numerical and institutional parameters omitted from the compact main calibration table. I set the bequest shift to $\underline{e} = 10^{-6}$, ensuring that warm-glow utility remains finite at zero terminal estate.

Table B.4: Additional Calibration and Institutional Parameters

Parameter	Explanation	Value	Source or role
θ_z	Fréchet shape	1.9266	CHFS earnings
s_z	Fréchet scale	0.5430	CHFS earnings
$\underline{e}$	Bequest shift	10^{-6}	Numerical regularization
N_b	Inheritance bins	5	Appendix A.3
$\{p_k^b\}_{k=1}^5$	Inheritance probabilities	(0.25, 0.25, 0.25, 0.15, 0.10)	Fixed quantile-bin widths
$\underline{h}$	Minimum housing size	10 m ²	25% of urban average

Notes: Monetary values are in units of 10,000 RMB. The inheritance nodes are updated to match the stationary terminal-estate distribution; mortgage payment and balance coefficients follow from the mortgage rate and remaining maturity.

Appendix Table B.5 reports the age-specific utility constants and the location–tenure shares they target.

Table B.5: Age-Specific Location–Tenure Utility Constants and Target Shares

Age group	Alternative-specific constants			Target shares (percent)			
	ξ_{oB}^g	ξ_{rA}^g	ξ_{oA}^g	rB	oB	rA	oA
20–29	0.00688	-0.00073	0.00746	26.00	39.00	22.05	12.95
30–39	0.02025	0.00611	0.02369	9.75	55.25	14.00	21.00
40–49	0.01961	0.00048	0.02206	10.50	59.50	9.00	21.00
50–59	0.01339	0.00031	0.01192	10.50	59.50	9.00	21.00
60–69	0.01129	0.00716	0.01828	9.75	55.25	10.50	24.50
70–79	0.01646	0.02146	0.04220	9.75	55.25	10.50	24.50

Notes: The utility constant for renting in Region B is normalized to $\xi_{rB}^g = 0$ for every age group. The remaining constants are measured in flow-utility units. Target shares sum to 100 percent within each age group.

All calibrated blocks, value functions, and the stationary distribution meet the tolerances in Appendix A. The maximum share error is 0.0169, the working-age A/B income ratio is 1.989, and the budget and inheritance gaps are 8.8×10^{-4} and 5.6×10^{-4} .

Appendix C. Additional Baseline Equilibrium Results

This appendix supplements Section 4 with additional evidence from the calibrated baseline equilibrium B . It reports location–tenure sorting over the life cycle, conditional responses to parental transfers, talent composition within each choice, housing affordability, and household balance sheets. The transfer-response profiles vary the amount re-

ceived while holding baseline prices, rents, T , continuation values, and parental-origin weights fixed. No re-solved no-transfer equilibrium is used in this appendix.

C.1 Location–Tenure Sorting over the Life Cycle

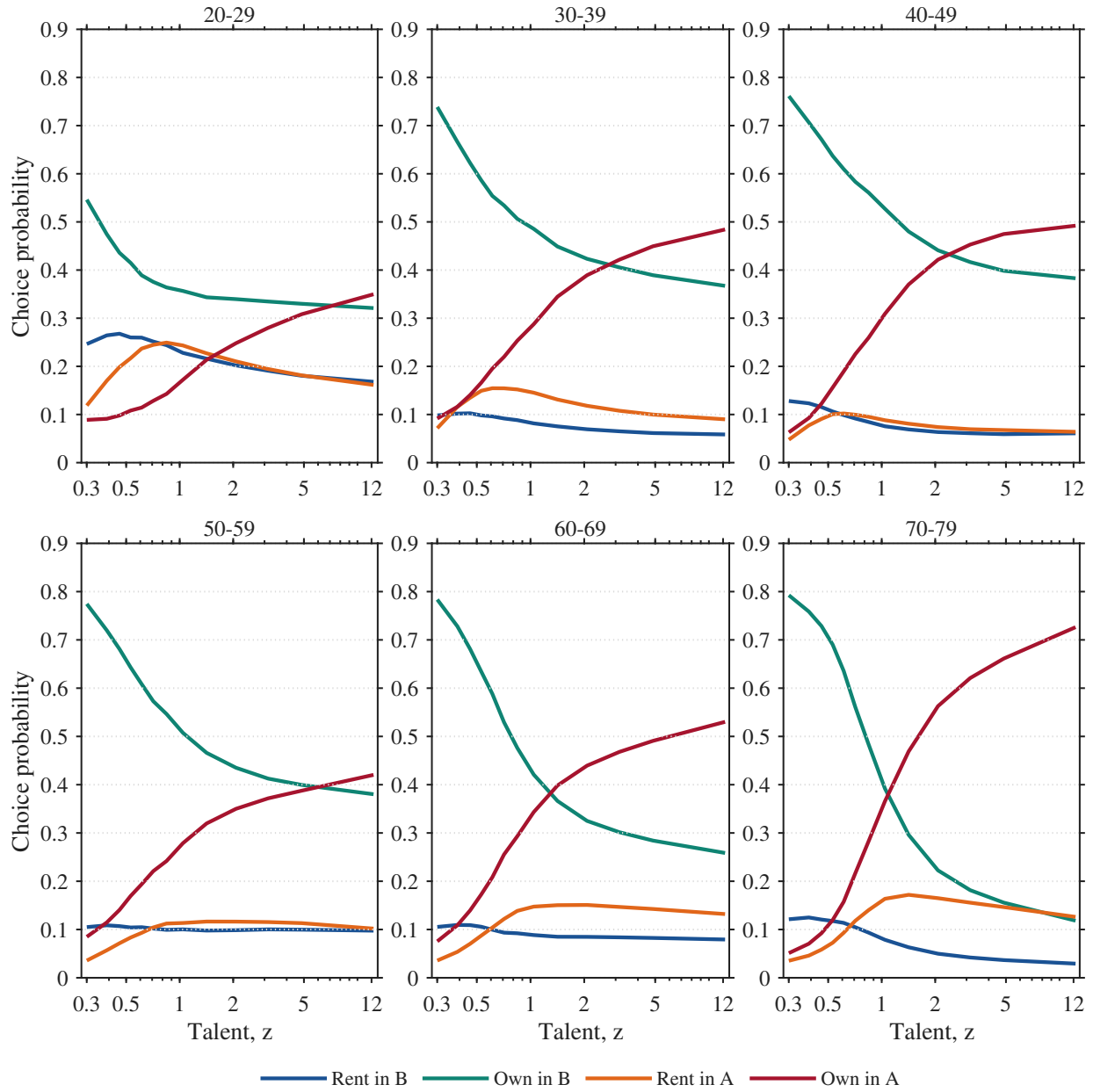


Figure C.1: Location–Tenure Choices by Talent and Age

Notes: Each panel reports baseline choice probabilities for renting and owning in Regions A and B conditional on talent for one age group. All panels use common vertical-axis limits, and the talent axis is logarithmic.

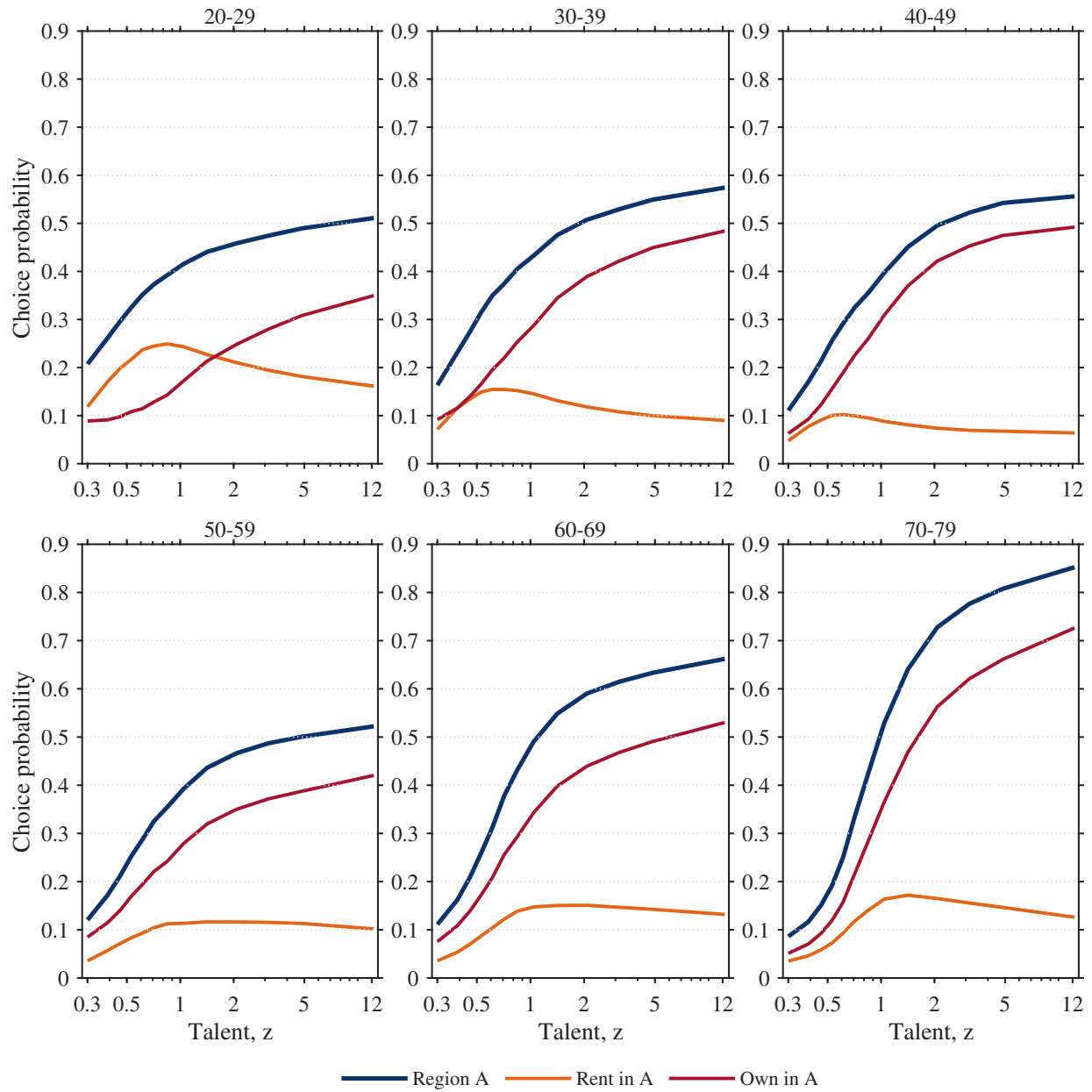


Figure C.2: Region A Location Choices by Talent and Age

Notes: Each panel reports baseline choice probabilities conditional on talent for one age group. Region A residence includes both renters and owners. All panels use common vertical-axis limits, and the talent axis is logarithmic.

C.2 Talent Composition by Location and Tenure

Table C.1: Baseline Mean Talent by Age and Location–Tenure Choice

Age	$E[z \mid rB]$	$E[z \mid oB]$	$E[z \mid rA]$	$E[z \mid oA]$
20–29	0.908	0.907	0.999	1.419
30–39	0.882	0.870	0.956	1.399
40–49	0.834	0.865	0.953	1.451
50–59	0.979	0.858	1.141	1.345
60–69	0.929	0.787	1.203	1.419
70–79	0.755	0.701	1.242	1.673

Notes: Entries are stationary-mass-weighted mean talent within each baseline age and location–tenure choice. Ages 70–79 are included as a sorting diagnostic but do not supply labor.

C.3 Housing Affordability over the Life Cycle

Table C.2 extends the age-1 affordability evidence in Table 3 across all six ages. All ratios use income earned over a ten-year model stage.

Table C.2: Baseline Housing Affordability and Financing over the Life Cycle

Age	Region A share (%)	Owner share (%)	Mean housing size	House value / income, owners	Rent / income, renters	Mortgage payment / income, owners	End-stage LTV, owners	Mean home equity
20–29	35.34	54.66	49.15	0.699	0.256	0.180	0.318	12.84
30–39	35.48	77.81	46.97	0.623	0.233	0.112	0.206	22.54
40–49	30.84	81.92	48.60	0.661	0.244	0.071	0.139	27.50
50–59	30.33	80.71	53.96	0.771	0.242	0.023	0.020	31.13
60–69	35.13	79.72	48.40	0.991	0.301	0.007	0	30.26
70–79	34.86	79.62	41.16	1.457	0.533	0	0	26.35

Notes: Entries are stationary baseline means. Housing size is in square meters, and home equity is in units of 10,000 RMB. House-value, rent, and mortgage-payment ratios use ten-year gross income and are conditional on the tenure group shown in the column heading. LTV is the owner-weighted mean of end-of-stage mortgage debt divided by market house value, for both new buyers and keepers; Table 3 instead reports origination LTV. Mean home equity includes renters at zero. Income at ages 70–79 is pension income.

C.4 Conditional Responses to Parental Transfers

Figure C.3 reports the full talent–transfer surface underlying Figure 6. Figure C.4 separates all four choices for the low-talent example, showing which tenure alternatives account for the non-monotone response of Region A residence.

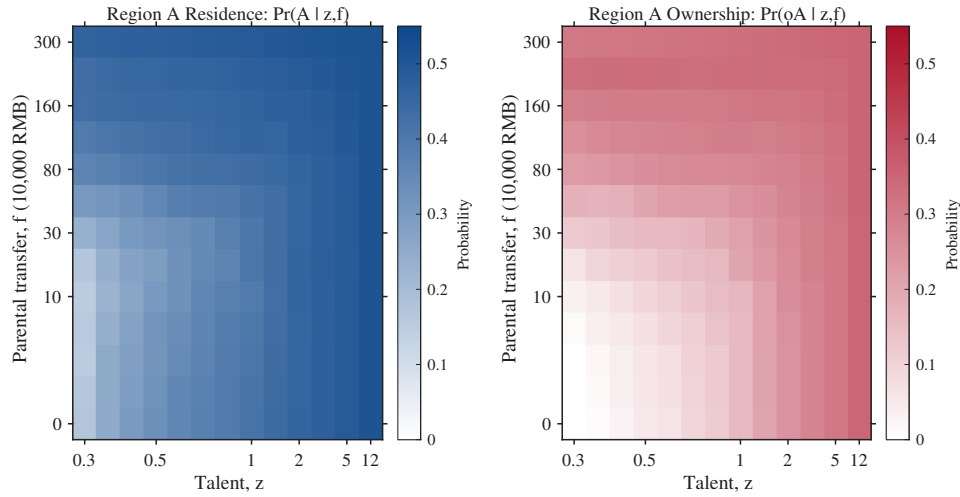


Figure C.3: Full Region A Policy Surface by Talent and Parental Transfer

Notes: Age-1 probabilities on the full 13-by-13 grid use the same fixed environment and origin weights as Figure 6. Colors measure choice probabilities, not equilibrium frequencies. Grid points are equally spaced visually; ticks give actual values. Transfers are in units of 10,000 RMB.

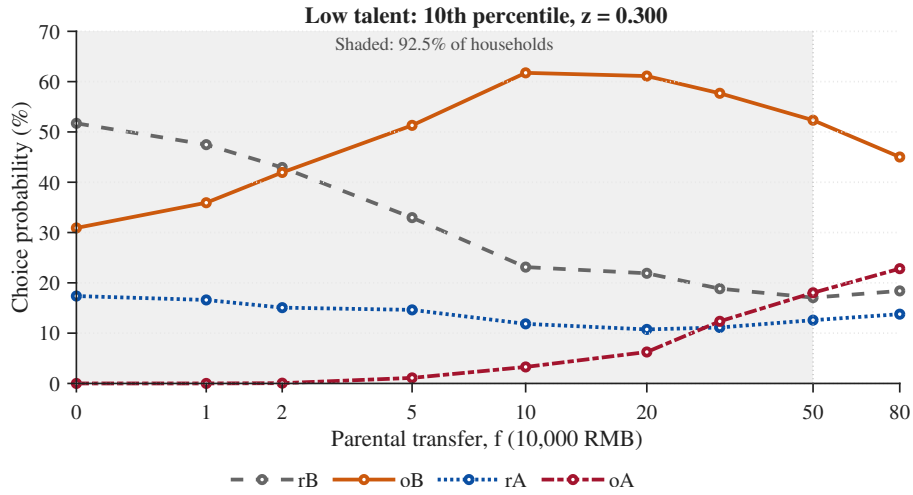


Figure C.4: Low-Talent Location–Tenure Responses to Parental Transfers

Notes: The low-talent example and plotting conventions match Figure 6. From $f = 0$ to $f = 2$, oB rises by 11.03 percentage points, while rB and rA fall by 8.78 and 2.31 points; oA rises by 0.057 points. The initial decline in A residence thus accompanies greater B ownership, not greater B renting.

C.5 Life-Cycle Household Balance Sheets

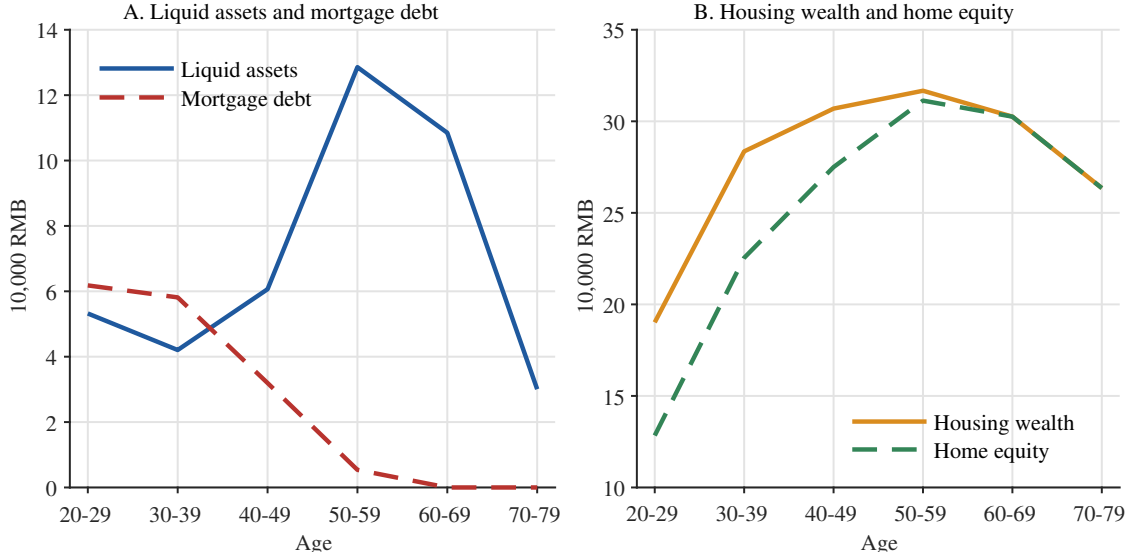


Figure C.5: Life-Cycle Household Balance Sheets

Notes: Panel A reports mean liquid assets and mortgage debt by age. Panel B reports mean housing wealth and home equity. All monetary values are in units of 10,000 RMB. Balances are measured at the end of each age after current location, tenure, saving, and mortgage decisions. Zero end-of-stage mortgage debt in the final two age groups reflects the mortgage maturities and restrictions on late-life borrowing.

Appendix D. No-Transfer Counterfactual: Additional Decomposition, Distributional, and Welfare Results

This appendix defines the decomposition, distributional, and welfare measures used in Section 5 and reports additional results from the re-solved no-transfer equilibrium. All differences are measured as $B - B^0$.

D.1 Residence–Ownership and Effective-Labor Decompositions

Residence–ownership decomposition. For economy $e \in \{B, B^0\}$, let $A_e = \Pr(A)_e$ denote the Region A population share and let $q_e = \Pr(\text{own} \mid A)_e$ denote the ownership rate among Region A residents. Since $\Pr(oA)_e = A_e q_e$, the difference in Region A ownership has the exact symmetric decomposition

$$\Delta oA \equiv \Pr(oA)_B - \Pr(oA)_{B^0} = \underbrace{\frac{q_B + q_{B^0}}{2}(A_B - A_{B^0})}_{\mathcal{S}: \text{spatial component}} + \underbrace{\frac{A_B + A_{B^0}}{2}(q_B - q_{B^0})}_{\mathcal{T}: \text{tenure component}}.$$

The spatial component $\mathcal{S}$ attributes the change in oA to the change in Region A residence while evaluating ownership within Region A at its average across the two equilibria. The tenure component $\mathcal{T}$ attributes the change to ownership conditional on Region A residence while evaluating the Region A population share at its cross-equilibrium average. When $\Delta oA \neq 0$, the tenure share reported in Table 4 is $100\mathcal{T}/\Delta oA$ percent.

Effective-labor decomposition. For economy $s \in \{B, B^0\}$, let $\mu_A^{g,s}$ denote the measure of age- g households residing in Region A. These measures include stationary age weights: the total mass of each age across both regions is $1/6$. Working-age mass, mean age-adjusted productivity, and effective labor in Region A are

$$M_A^s = \sum_{g=1}^5 \int d\mu_A^{g,s}, \quad \overline{Gz}_A^s = \frac{\sum_{g=1}^5 \int G_g z d\mu_A^{g,s}}{M_A^s}, \quad L_A^s = M_A^s \overline{Gz}_A^s. \quad (\text{D.1})$$

The residence-mass component evaluates the change in worker mass at average mean efficiency, while the composition component evaluates the change in mean efficiency at average worker mass. Their sum equals the effective-labor difference exactly:

$$\Delta L_A^{\text{mass}} = \frac{1}{2} \left(\overline{Gz}_A^B + \overline{Gz}_A^{B^0} \right) \left(M_A^B - M_A^{B^0} \right), \quad (\text{D.2})$$

$$\Delta L_A^{\text{comp}} = \frac{1}{2} \left(M_A^B + M_A^{B^0} \right) \left(\overline{Gz}_A^B - \overline{Gz}_A^{B^0} \right), \quad (\text{D.3})$$

$$L_A^B - L_A^{B^0} = \Delta L_A^{\text{mass}} + \Delta L_A^{\text{comp}}. \quad (\text{D.4})$$

Since working ages comprise $5/6$ of the stationary population, the Region A residence share among working-age households is $M_A^s/(5/6)$. Thus the 1.18-percentage-point change in full-population mass corresponds to a 1.41-point change in this conditional residence share. The same aggregation applies to Region B. Regional output is $Y_\ell^s = \text{TFP}_\ell L_\ell^s$, and aggregate output is $Y_A^s + Y_B^s$. Table 5 reports these quantities and the two decomposition components.

D.2 Additional Talent and Choice Results over the Life Cycle

Table D.1: Mean Talent by Age and Location–Tenure Choice

Age	Economy	$E[z \mid rB]$	$E[z \mid oB]$	$E[z \mid rA]$	$E[z \mid oA]$
20–29	B	0.908	0.907	0.999	1.419
	B^0	0.816	0.972	0.941	1.771
30–39	B	0.882	0.870	0.956	1.399
	B^0	0.798	0.875	0.905	1.561
40–49	B	0.834	0.865	0.953	1.451
	B^0	0.770	0.866	0.942	1.578
50–59	B	0.979	0.858	1.141	1.345
	B^0	0.962	0.853	1.144	1.391
60–69	B	0.929	0.787	1.203	1.419
	B^0	0.940	0.775	1.233	1.449
70–79	B	0.755	0.701	1.242	1.673
	B^0	0.764	0.686	1.296	1.683

Notes: Entries are stationary-mass-weighted mean talent within each age and location–tenure choice. Economy B includes family transfers and B^0 sets inter vivos transfers to zero. Ages 70–79 are included as a sorting diagnostic but do not supply labor.

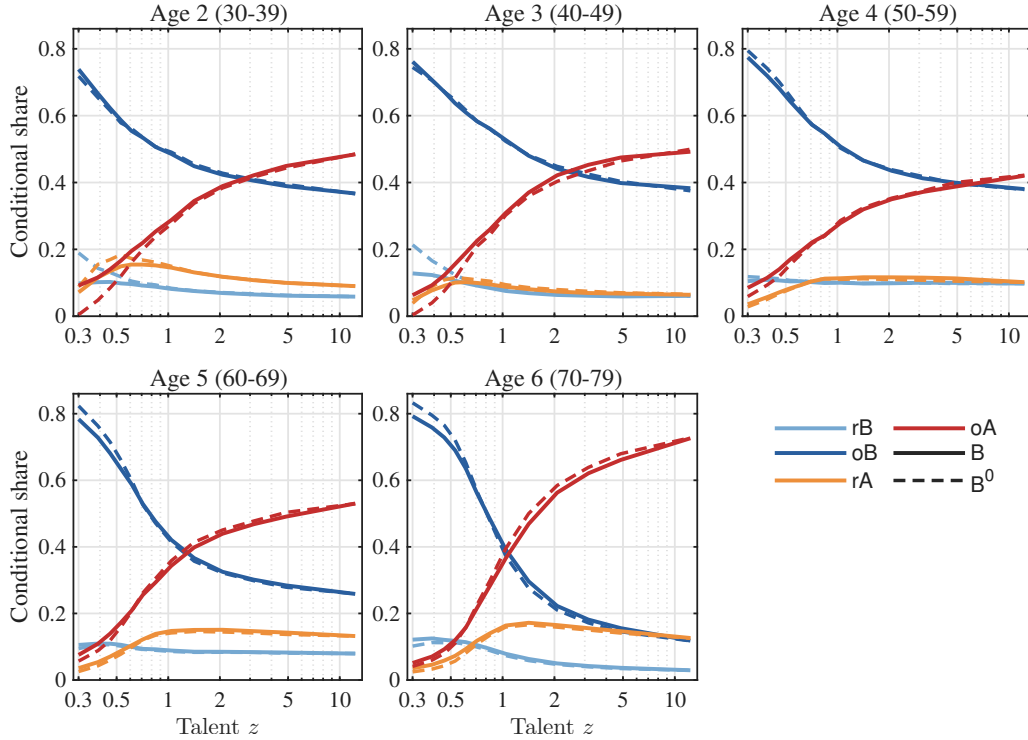


Figure D.1: Family Transfers and Location–Tenure Choices, Ages 30–79

Notes: Each panel reports choice probabilities conditional on talent for one age group. Colors identify rB , oB , rA , and oA . Solid lines denote the baseline equilibrium B , and dashed lines denote the no-transfer equilibrium B^0 . The talent axis is logarithmic.

Table D.2: Young-Household Choices and Region A Purchase Financing

Outcome	B : transfers	B^0 : no transfers
Panel A. Location–tenure shares at ages 20–29		
$\Pr(rB)$	24.31%	29.84%
$\Pr(oB)$	40.35%	35.94%
$\Pr(rA)$	21.03%	23.94%
$\Pr(oA)$	14.31%	10.27%
$\Pr(A)$	35.34%	34.22%
$\Pr(B)$	64.66%	65.78%
$\Pr(oA) + \Pr(oB)$	54.66%	46.21%
Panel B. Financing of Young Region A Purchases With and Without Family Transfers		
Housing size (square meters)	37.674	34.660
Purchase value	69.217	61.162
Down payment	37.961	18.711
Origination mortgage	31.256	42.452
Mean origination LTV	46.9%	68.8%
End-of-stage home equity	43.785	26.621

Notes: Panel A reports shares among all households aged 20–29. Panel B conditions on young households making a new Region A purchase in each economy, using stationary state mass times the probability of choosing oA as weights. Housing size is in square meters; monetary quantities are in RMB 10,000 per buyer. Origination mortgage is recovered from the saved post-amortization balance. Down payment equals purchase value minus origination mortgage. Mean origination LTV averages individual mortgage-to-purchase-value ratios; it is not the ratio of mean mortgage to mean purchase value. End-of-stage home equity subtracts the post-amortization debt from the equilibrium value of the property. The same B/B^0 buyer comparisons appear alongside the LTV-policy economies in Appendix Table F.3.

D.3 Distributional Measures and Within-Generation Results

Let i index the stationary allocation cells used for a reported group in economy $s \in \{B, B^0\}$, and let ω_i^s denote the corresponding stationary mass. For any outcome x_i^s , define total group mass $W_s = \sum_i \omega_i^s$ and the weighted mean $\bar{x}_s = W_s^{-1} \sum_i \omega_i^s x_i^s$. The weighted Gini coefficient and weighted quantile are

$$G_s(x) = \frac{\sum_i \sum_j \omega_i^s \omega_j^s |x_i^s - x_j^s|}{2W_s^2 \bar{x}_s},$$

$$Q_s(p; x) = \inf \left\{ q : \frac{1}{W_s} \sum_i \omega_i^s \mathbb{1}\{x_i^s \leq q\} \geq p \right\}.$$

The Gini is zero under complete equality and increases as the distribution becomes more unequal. The net-worth p90/p50 ratio is $Q_s(0.90; NW)/Q_s(0.50; NW)$, and the share with nonpositive net worth is $W_s^{-1} \sum_i \omega_i^s \mathbb{1}\{NW_i^s \leq 0\}$. End-of-period home equity and net worth are

$$HE_i^s = \mathbb{1}\{i \text{ owns}\} (P_{\ell_i}^s h_i^s - m_i'^s), \quad NW_i^s = x_i'^s + HE_i^s,$$

so renters enter the home-equity distribution at zero. Labor-income inequality pools ages 1–5. Disposable-income, consumption–housing, and net-worth statistics pool all six ages, with disposable income and the composite defined in equations (16) and (2). Age-specific statistics use the same definitions within each age group.

The age-1 talent gap for outcome y is $E_s[y \mid z \text{ in top } 20\%] - E_s[y \mid z \text{ in bottom } 80\%]$. For the dynastic gaps, beginning-of-age-4 parental wealth is

$$NW_{p,i}^{\text{beg},s} = x_{p,i}^s + \mathbb{1}\{i \text{ owns}\} \left[(1 - \delta_h) P_{\ell_{p,i}}^s h_{p,i}^s - m_{p,i}^s \right].$$

Quartile cutoffs are computed from the weighted parent-wealth distribution in B and applied unchanged in both economies. The reported Q4–Q1 gap is the weighted mean linked child outcome among Q4 parents minus the corresponding mean among Q1 parents. Linked child outcomes average over child-talent realizations and age-1 policy probabilities.

Among households aged 20–29, the share with nonpositive net worth is 24.87 percent in B and 50.02 percent in B^0 , a reduction of 25.15 percentage points. These within-age shares differ from the all-age measure in Table 7.

Table D.3 extends the within-generation comparison beyond young households. Income inequality moves little at later ages, whereas inequality in the consumption–housing composite and home equity falls throughout the life cycle.

D.4 Self-Lifetime Welfare and Consumption-Equivalent Variation

Welfare criterion. For a household currently in age group g , self-lifetime welfare in economy $s \in \{B, B^0\}$ includes utility through the end of life:

$$\mathcal{W}_g^s = \mathbb{E}_s \left[\sum_{a=g}^6 \beta^{a-g} \left\{ u^a(c_a, h_a, j_a) + \mathbb{1}\{a = 6\} \omega_b \frac{(e_6 + \underline{e})^{1-\gamma}}{1-\gamma} \right\} \right]. \quad (\text{D.5})$$

Here u^a is the deterministic flow utility in (1), including location–tenure utility, and e_6 is the terminal estate in (24). The criterion includes terminal warm-glow utility but excludes logit taste utility and the age-4 altruistic child-value term. The latter exclusion prevents the linked child’s welfare from entering both the parent’s and the child’s welfare.

CEV construction. Let λ_g^{c+h} denote the consumption–housing CEV. It solves

$$\mathcal{W}_g^{B^0} \left((1 + \lambda_g^{c+h})c, (1 + \lambda_g^{c+h})h; e_6 \right) = \mathcal{W}_g^B. \quad (\text{D.6})$$

Table D.3: Within-Generation Distributional Measures at Ages 30–79

Outcome	30–39	40–49	50–59	60–69	70–79
Panel A: Baseline (B)					
Labor-income Gini	0.456	0.459	0.456	0.464	—
Disposable-income Gini	0.452	0.455	0.452	0.459	0.421
Composite $\mathcal{C}(c, h)$ Gini	0.355	0.351	0.319	0.332	0.346
Home-equity Gini	0.598	0.590	0.528	0.544	0.572
Net-worth p90/p50	3.95	3.87	2.37	2.73	2.90
Share with nonpositive net worth (%)	3.56	6.67	0.01	0.00	0.00
Panel B: No transfer (B^0)					
Labor-income Gini	0.457	0.460	0.458	0.466	—
Disposable-income Gini	0.454	0.457	0.454	0.460	0.422
Composite $\mathcal{C}(c, h)$ Gini	0.392	0.391	0.364	0.366	0.375
Home-equity Gini	0.642	0.637	0.566	0.570	0.589
Net-worth p90/p50	4.00	4.40	2.82	2.98	3.75
Share with nonpositive net worth (%)	9.15	12.79	0.00	0.00	0.00
Panel C: Transfer effect ($B - B^0$)					
Labor-income Gini	−0.002	−0.002	−0.001	−0.002	—
Disposable-income Gini	−0.002	−0.002	−0.001	−0.002	0.000
Composite $\mathcal{C}(c, h)$ Gini	−0.038	−0.040	−0.045	−0.034	−0.029
Home-equity Gini	−0.043	−0.047	−0.038	−0.027	−0.018
Net-worth p90/p50	−0.05	−0.53	−0.45	−0.25	−0.85
Share with nonpositive net worth (pp)	−5.59	−6.11	+0.01	0.00	0.00

Notes: Statistics are calculated separately within each age group using stationary-mass weights. Home-equity Ginis include renters at zero. Net worth is end-of-period liquid assets plus home equity. Panel C reports $B - B^0$; changes in the nonpositive-net-worth share are percentage points. Labor-income inequality is not defined for retired households aged 70–79. Entries are calculated before rounding.

Consumption and housing services are scaled proportionally along the B^0 allocation, while location–tenure choices and terminal estates remain unchanged. Deterministic location–tenure constants and terminal warm-glow utility are therefore unscaled. Positive CEVs measure the compensation required in B^0 to attain self-lifetime welfare in B ; negative CEVs indicate higher self-lifetime welfare in B^0 .

Population and group weights. The expectation in (D.5) uses each equilibrium’s stationary distribution and policy probabilities. Saved flow masses incorporate the option probabilities once, and weights are normalized within the reported age and talent group. Current ages are the six ten-year groups from 20–29 through 70–79. Talent profiles use the same percentile groups in both economies: bottom 20 percent, 20th–50th, 50th–80th, 80th–95th, and top 5 percent. These comparisons describe stationary equilibria. Welfare

Table D.4: Self-Lifetime Consumption-Equivalent Welfare Effects

Group	$c + h$ CEV (%)	Consumption-only CEV (%)
<i>Panel A: Remaining lifetime by age</i>		
20–29	23.33	40.99
30–39	14.94	25.28
40–49	9.51	15.27
50–59	3.31	4.99
60–69	2.75	4.09
70–79	3.27	4.92
All living	10.78	17.24
<i>Panel B: Age 20–29 by talent</i>		
Bottom 20%	48.03	100.72
20–50%	22.23	37.88
50–80%	7.20	11.86
80–95%	-1.70	-2.67
Top 5%	-5.86	-10.13

Notes: Positive values are the percentage increase in the indicated B^0 inputs required to attain welfare in B . The main $c + h$ measure scales consumption and housing services; the consumption-only measure leaves housing unchanged. Both measures include deterministic location–tenure utility and terminal warm-glow utility, exclude logit entropy and the age-4 altruistic child-value term, and leave terminal estates unscaled.

differences among older households reflect current giving, earlier transfer receipt, accumulated positions, and the equilibrium environment; they do not separately identify the welfare cost of being a donor.

All-living aggregation. The all-living CEV is obtained by applying stationary age weights to welfare before solving for common compensation; it is not the weighted average of age-specific CEVs. Each current age receives weight $1/6$, and the same proportional compensation is applied to consumption and housing in all six remaining-lifetime welfare calculations.

Alternative welfare measure. The consumption-only diagnostic λ_g^c scales consumption along the B^0 allocation while leaving housing services and terminal estates unchanged. All other components and weights match the main criterion. Baseline and policy comparisons use the same CES utility function and welfare convention. All occurrences of B and B^0 use the same saved baseline equilibria; the replication registry records their source hashes.

Detailed results. Table D.4 reports both welfare measures by current age and, for ages 20–29, by talent. Figure D.2 reports the consumption–housing CEV profiles by talent at ages 30–79.

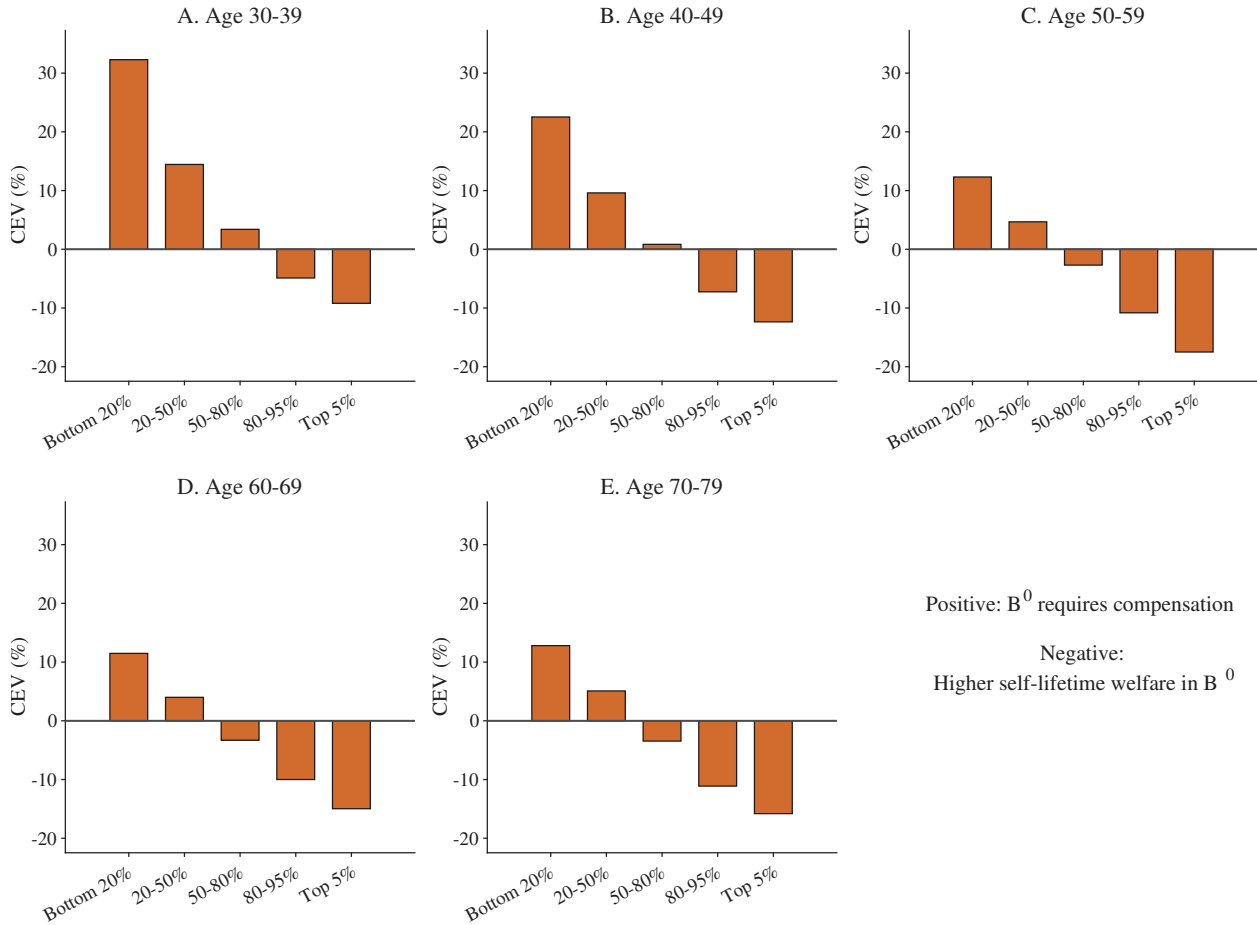


Figure D.2: Self-Lifetime Welfare Effects by Talent at Ages 30–79

Notes: Panels report remaining-lifetime consumption–housing CEV by current-age talent group. Positive values are the percentage increase in consumption and housing services required in B^0 to attain welfare in B ; negative values indicate higher self-lifetime welfare in B^0 . All panels use a common vertical scale. The welfare convention is identical to Figure 9.

Appendix E. Sensitivity Analysis

This appendix tests whether the distinction between spatial access and ownership survives alternative preferences, parental altruism, housing-supply conditions, mobility frictions, rental access, and intergenerational talent persistence. It then evaluates numerical accuracy by refining one state-space dimension at a time.

E.1 Sensitivity Protocol

Each economic sensitivity is evaluated as a pair of stationary equilibria. When a primitive changes the baseline household problem, I recalibrate the 18 age-specific location-tenure constants, T , and σ_v to the same 24 age-choice shares and the same Region A-to-Region B income ratio. Except for the predetermined parameter under examination, all other predetermined parameters remain at their benchmark values; prices and rents remain fixed at their calibration targets, and inheritance is updated endogenously. The housing-supply and owner-rental conversion cases do not change the calibrated baseline allocation and therefore reuse the benchmark transfer equilibrium.

An accepted transfer equilibrium must have a maximum age-choice share error no larger than 0.02, an income-ratio error no larger than 2 percent, a government-budget residual no larger than 10^{-3} , an inheritance gap no larger than 10^{-3} , and converged value functions and stationary distribution. For each accepted transfer economy, I back out housing supply from its demand and solve a new no-transfer equilibrium with $f = 0$, jointly updating prices, rents, T , and the inheritance distribution. Cases that miss the no-transfer GE tolerance are reported as provisional rather than converged.

All effects below are transfer equilibrium minus no-transfer equilibrium. The primary mechanism is *preserved* when transfers raise the age-1 Region A share and raise the age-1 oA share by still more. It is *mixed* when only one condition holds and *overturned* when neither holds. The tenure share uses the decomposition in Appendix D.1. The tables also report the age-1 home-equity effect and the aggregate-output effect normalized by the benchmark transfer effect. Separate talent-group diagnostics classify low- and middle-talent access gains as broad or uneven. A verdict is marked fragile when its defining margin is smaller than the 0.245 percentage-point numerical envelope established in Appendix E.6.

E.2 Preferences and Parental Altruism

The preference tests vary the CES curvature from its benchmark value $\epsilon = -2$ to -3 and -1 . The altruism tests set α_p to 0.25 and 0.55. Table E.1 reports the resulting transfer effects.

Both curvature changes preserve the mechanism. Raising α_p increases the oA , home-equity, and welfare effects, whereas lowering it attenuates them.

Table E.1: Sensitivity to Preferences and Parental Altruism

Specification	$\Delta \text{Pr}(A)$	$\Delta \text{Pr}(oA)$	Tenure share	ΔHE_1	$\Delta Y/\Delta Y_{\text{bench}}$	Mechanism
Benchmark	1.126	4.035	90.2%	7.59	1.00	Preserved
$\epsilon = -3$	1.186	3.811	89.8%	6.77	0.97	Preserved
$\epsilon = -1$	1.385	3.849	87.4%	8.62	0.88	Preserved
$\alpha_p = 0.25$	0.574	2.768	93.1%	5.25	0.67	Preserved ^a
$\alpha_p = 0.55$	1.207	4.490	91.3%	8.93	1.04	Preserved

Notes: Probability effects are percentage points. Tenure share is in percent; ΔHE_1 is in units of 10,000 RMB. The output column divides each aggregate-output effect by the benchmark effect.

^a The no-transfer GE objective is 1.27×10^{-3} , slightly above the 10^{-3} tolerance; the verdict is provisional.

E.3 Intergenerational Talent Persistence

The benchmark draws a young adult's talent independently of parental talent. I instead use the parent-child transition matrix

$$\Pi(r) = (1 - r)\mathbf{1}p' + rI,$$

where p is the stationary talent distribution and r governs persistence. This specification preserves the marginal talent distribution. I consider $r = 0.30$ and $r = 0.50$, recalibrate the baseline, and resolve the no-transfer equilibrium in each case. Table E.2 reports the resulting transfer effects.

Table E.2: Sensitivity to Intergenerational Talent Persistence

Specification	$\Delta \text{Pr}(A)$	$\Delta \text{Pr}(oA)$	Tenure share	ΔHE_1	$\Delta Y/\Delta Y_{\text{bench}}$	Mechanism
Independent ($r = 0$)	1.126	4.035	90.2%	7.59	1.00	Preserved
Moderate ($r = 0.30$)	0.889	3.146	90.4%	5.83	0.76	Preserved
High ($r = 0.50$)	0.754	2.545	90.1%	4.73	0.55	Preserved

Notes: Units and columns follow Table E.1. All three specifications satisfy the calibration and equilibrium-convergence criteria.

Persistence attenuates the spatial, ownership, home-equity, and output effects but leaves their ordering unchanged. Even at $r = 0.50$, transfers raise Region A residence by 0.75 percentage points and oA by 2.55 points, with 90 percent of the ownership response operating through tenure. The central mechanism therefore does not depend on independent parent and child talent draws.

E.4 Housing Supply and Owner–Rental Conversion

These tests vary the housing-supply elasticities around the benchmark $(\rho_A, \rho_B) = (0.65, 0.95)$ and the owner–rental conversion elasticities around $(\chi_A, \chi_B) = (1, 1)$. They change the GE response to removing transfers without changing the calibrated transfer allocation. Table E.3 reports the results.

Table E.3: Sensitivity to Housing Supply and Owner–Rental Conversion

Specification	$\Delta \text{Pr}(A)$	$\Delta \text{Pr}(oA)$	Tenure share	ΔHE_1	$\Delta Y/\Delta Y_{\text{bench}}$	Mechanism
Benchmark	1.126	4.035	90.2%	7.59	1.00	Preserved
$(\rho_A, \rho_B) = (0.45, 0.70)$	1.179	4.022	89.7%	7.65	0.98	Preserved ^a
$(\rho_A, \rho_B) = (1.00, 1.50)$	1.012	4.063	91.2%	7.59	0.97	Preserved
$(\rho_A, \rho_B) = (1.80, 2.70)$	0.971	4.099	91.7%	7.58	0.99	Preserved
$(\chi_A, \chi_B) = (0, 0)$	1.180	3.794	88.9%	7.51	0.93	Preserved
$(\chi_A, \chi_B) = (2, 2)$	1.047	4.113	91.1%	7.61	0.99	Preserved

Notes: Units and columns follow Table E.1. In the ρ rows, $\chi_A = \chi_B = 1$; in the χ rows, $(\rho_A, \rho_B) = (0.65, 0.95)$.

^a The no-transfer GE objective is 1.46×10^{-3} ; this directional result is provisional.

The ownership effect remains close to four percentage points throughout these tests, and the tenure margin explains 89–92 percent of it. Supply elasticity and conversion flexibility change equilibrium prices and rents, but they do not materially alter the distinction between Region A access and ownership.

E.5 Mobility Costs and Region A Rental Frictions

The mobility tests multiply the benchmark Region A entry-cost schedule by 0.5 and 1.5. The rental-access tests eliminate or double the recurring Region A rental-friction schedule. These assumptions have the largest effect on the spatial margin because they directly change the cost of reaching Region A through renting. The zero-friction specification is also a mechanism-removal test of the benchmark sorting discipline, not an alternative estimate of the institutional cost. Table E.4 summarizes these comparisons.

Removing the rental friction substantially weakens the model’s sorting fit. Mean age-1 talent among Region A renters falls from 0.999 in the benchmark to 0.944, while the mean among Region B owners rises from 0.907 to 0.923. The $rA-oB$ gap therefore contracts from 10.1 to 2.3 percent, well below the 9.0 percent gap among young household heads. The ordering does not reverse, but the zero-friction model no longer reproduces its empirical strength.

Lower entry costs and the removal of rental frictions also reverse the aggregate residence effect: transfers reduce age-1 Region A residence by 0.49 and 1.05 percentage

Table E.4: Sensitivity to Mobility Costs and Rental Frictions

Specification	$\Delta \Pr(A)$	$\Delta \Pr(oA)$	Tenure share	ΔHE_1	$\Delta Y/\Delta Y_{\text{bench}}$	Mechanism
Benchmark	1.126	4.035	90.2%	7.59	1.00	Preserved
Entry costs $\times 0.5$	-0.490	3.826	103.9%	7.54	0.82	Mixed
Entry costs $\times 1.5$	1.915	3.688	82.9%	6.96	1.02	Preserved ^a
Rental friction = 0	-1.050	3.901	108.3%	7.56	0.70	Mixed
Rental friction $\times 2$	2.872	3.710	73.0%	7.10	1.20	Preserved

Notes: Units and columns follow Table E.1. The benchmark entry-cost schedule is (4, 5, 6, 7, 8, 9); the benchmark rental-friction schedule is (2, 2, 2.5, 3, 3, 3).

^a The no-transfer GE objective is 1.10×10^{-3} ; this directional result is provisional.

points, respectively. In both cases, transfers still raise oA by almost four percentage points. The tenure share exceeds 100 percent because a negative spatial component partly offsets the positive tenure component. Doubling the rental friction instead increases the spatial effect to 2.87 percentage points. Thus, the ownership result is robust, while the sign and size of the spatial-access effect depend on the cost of renting and moving to Region A; the no-friction specification also provides a substantially poorer fit to observed location-tenure sorting.

E.6 Numerical Accuracy

The stationary distribution is deterministic, so simulation-seed tests are not applicable. I instead refine the transfer, talent, liquid-asset, housing, and inheritance grids separately and tighten the GE tolerance. Refining one dimension at a time avoids an infeasible simultaneous expansion of the state space. Table E.5 reports the resulting accuracy checks.

Table E.5: Numerical Accuracy of the Transfer Mechanism

Variant	n_z	n_f	n_x	n_h	n_b	$\Delta \Pr(A)$	$\Delta \Pr(oA)$	$ d_A $	$ d_M $
Benchmark	13	13	10	13	5	1.126	4.035	0.000	0.000
Denser transfer grid	13	22	10	13	5	1.100	4.053	0.027	0.044
22-point talent grid	22	13	10	13	5	1.353	4.017	0.227	0.245
13-point asset grid	13	13	13	13	5	1.091	4.042	0.035	0.042
25-point housing grid	13	13	10	25	5	1.332	4.096	0.206	0.145
Seven inheritance bins	13	13	10	13	7	1.124	4.036	0.002	0.003
Nine inheritance bins	13	13	10	13	9	1.121	4.035	0.005	0.005
GE tolerance 5×10^{-4}	13	13	10	13	5	1.126	4.035	0.001	0.000

Notes: Probability effects and discrepancies are percentage points. n_b is the number of inheritance bins. d_A is the change in $\Delta \Pr(A)$ relative to the benchmark; d_M is the change in the tenure margin $\Delta \Pr(oA) - \Delta \Pr(A)$. All variants converge. The housing-grid case is conditionally recalibrated because the refined grid moves a targeted share by more than 0.2 percentage points.

The largest numerical discrepancy is 0.245 percentage points, generated by the 22-

point talent grid. Every refinement preserves a positive Region A effect and an ownership effect that is substantially larger. Across the 13 converged economic specifications, 11 preserve the full mechanism and two are mixed; none overturns both conditions. The mixed cases are lower entry costs and zero rental friction, and both spatial reversals exceed the numerical envelope. Every converged specification also produces positive age-1 home-equity and aggregate-output effects, and the all-living welfare effect remains positive, ranging from 6.10 to 12.21 percent.

Appendix F. Policy Experiments: Additional Results

This appendix reports the full allocation, financing, distributional, and welfare results underlying the four policy experiments in Section 6. Throughout, $P - B$ denotes the policy effect with family transfers and $P^0 - B^0$ the corresponding effect without them.

F.1 LTV Relaxation

Tables F.1, F.2, F.3, and F.4 report aggregate effects, talent-specific responses, buyer financing, and welfare, respectively. Figure F.1 shows the life-cycle effects. The policy raises the maximum LTV from 70 to 90 percent for new purchases in both regions.

Table F.1: LTV Relaxation: Full General-Equilibrium Results

Outcome	Baseline B	Policy effect with transfers $P - B$	Policy effect without transfers $P^0 - B^0$	Policy–transfer interaction
Panel A. Age-1 location and ownership (percentage points)				
$\Pr(A)$	35.342	0.131	0.164	-0.033
$\Pr(oA)$	14.308	0.181	0.510	-0.329
$\Pr(oA) + \Pr(oB)$	54.660	0.081	1.042	-0.961
Panel B. Transfers and age-1 balance sheets				
$\Pr(f > 0)$ (%)	52.906	-0.898	0.000	-0.898
Mean transfer	18.831	-0.444	0.000	-0.444
Mortgage debt	6.181	1.934	2.485	-0.551
Home equity	12.838	-1.272	-1.582	0.311
Net financial assets	-0.858	-2.005	-2.481	0.476
Panel C. Prices and aggregate outcomes				
P_A	1.837	-0.003	-0.004	0.001
Aggregate output	0.846	0.00019	0.00051	-0.00032

Notes: For ordinary outcomes, the interaction is $(P - B) - (P^0 - B^0)$. Probability changes are percentage points. Transfers and balance-sheet variables are in units of RMB 10,000. Output effects are level changes; the policy raises output by 0.023% with transfers and 0.061% without transfers. Pairwise welfare results appear in Table F.4.

Table F.2: LTV Relaxation: Age-1 Location and Ownership Effects by Talent

Talent group	B	P	$P - B$	B^0	P^0	$P^0 - B^0$	Interaction
Panel A. Region A ownership, $\Pr(oA)$							
Bottom 20%	8.993	8.916	-0.076	0.589	0.921	0.332	-0.409
Middle 60%	12.726	12.931	0.205	8.919	9.498	0.580	-0.375
Top 20%	24.370	24.735	0.366	24.019	24.497	0.478	-0.112
All age 1	14.308	14.489	0.181	10.273	10.783	0.510	-0.329
Panel B. Region A residence, $\Pr(A)$							
Bottom 20%	23.434	23.416	-0.018	19.626	19.742	0.116	-0.134
Middle 60%	35.888	36.040	0.152	35.397	35.545	0.148	0.004
Top 20%	45.611	45.829	0.218	45.262	45.523	0.261	-0.043
All age 1	35.342	35.473	0.131	34.216	34.380	0.164	-0.033

Notes: Levels are percentages; differences are percentage points. The interaction is $(P - B) - (P^0 - B^0)$.

Table F.3: LTV Relaxation: Financing of Age-1 Region A Buyers

Outcome	B	P	B^0	P^0
Panel A. Purchase scale and cash requirement				
$\Pr(oA)$ (%)	14.308	14.489	10.273	10.783
Housing size (m ²)	37.674	39.302	34.660	37.205
House value	69.217	72.106	61.162	65.515
Ten-year gross income	82.035	81.928	102.443	100.185
Origination mortgage	31.256	42.421	42.452	57.599
Down payment	37.961	29.685	18.711	7.916
Down payment / income (%)	46.274	36.233	18.264	7.901
Mortgage payment / income (%)	24.572	33.393	26.725	37.079
Panel B. Constraints and end-of-period position				
Mean origination LTV (%)	46.910	58.400	68.783	86.364
$\Pr(\text{LTV binds})$ (%)	29.009	30.171	56.237	39.023
$\Pr(\text{PTI binds})$ (%)	0.000	0.000	0.000	0.000
$\Pr(f > 0)$ (%)	50.616	49.003	0.000	0.000
Mean transfer	30.601	29.505	0.000	0.000
Transfer / down payment (%)	80.613	99.395	0.000	0.000
End liquid assets	1.632	1.576	0.000	0.000
End mortgage debt	25.432	34.517	34.541	46.867
End home equity	43.785	37.589	26.621	18.648
End net financial assets	-23.800	-32.941	-34.541	-46.867

Notes: $\Pr(oA)$ is measured among all households aged 20–29; the remaining statistics condition on young Region A buyers. Buyer weights equal stationary state mass times the probability of choosing oA . Down payment/income, mortgage payment/income, and transfer/down payment are ratios of weighted totals. Mean origination LTV averages individual mortgage-to-purchase-value ratios. Income is ten-year gross income; mean transfers include zero receipts. Origination debt is recovered from the saved post-amortization balance. Housing size is in square meters; monetary quantities are in units of 10,000 RMB.

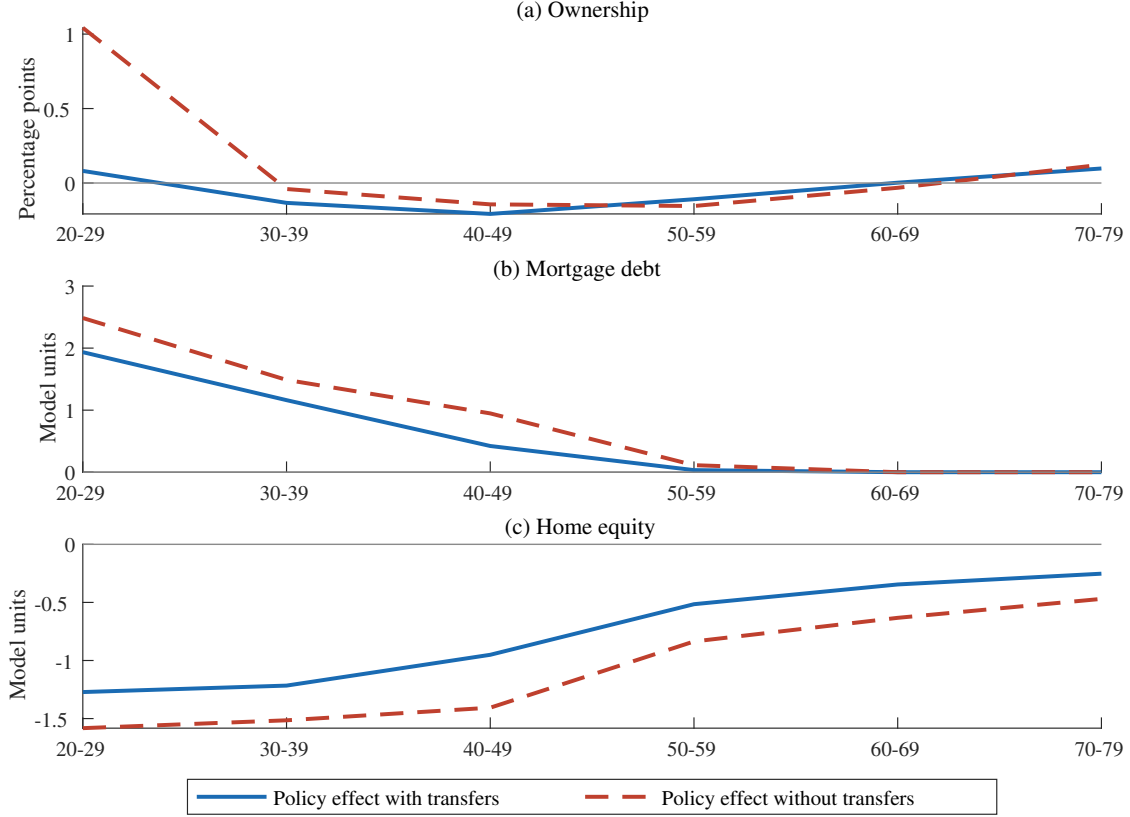


Figure F.1: Life-Cycle Effects of Relaxing the LTV Limit

Notes: Panel (a) reports the policy effect on homeownership in either region, $\Pr(oA) + \Pr(oB)$, within each age group, in percentage points; Panels (b) and (c) report effects on average mortgage debt and home equity in units of RMB 10,000. Solid blue curves show $P - B$, and dashed red curves show $P^0 - B^0$.

Table F.4: LTV Relaxation: Pairwise Consumption-Plus-Housing CEVs

Group	P relative to B	P^0 relative to B^0	P^0 relative to P
Age 1	0.366	-0.342	22.450
Bottom 20%	0.699	-0.326	46.454
Middle 60%	0.311	-0.335	15.051
Top 20%	-0.307	-0.474	-2.395
All living	0.816	0.992	10.981

Notes: Entries are percentages. Positive values mean that the first-named economy requires additional consumption and housing to attain welfare in the reference economy.

F.2 Affordable Rental Access

Tables F.5 and F.6 summarize program scale, equilibrium effects, and distributional and welfare incidence. Figures F.2 and F.3 show age-1 location–tenure shares by income eligibility and choice responses over the income distribution. Eligibility uses the potential Region A income threshold fixed from the baseline economy.

Table F.5: Affordable Rental Access: Full General-Equilibrium and Incidence Results

Outcome	B	$P - B$	$P^0 - B^0$	$P - P^0$	Interaction
Panel A. Age-1 location and tenure					
$\Pr(rB)$	24.306	-0.473	-0.966	-5.045	+0.493
$\Pr(oB)$	40.352	-1.202	-1.620	+4.830	+0.418
$\Pr(rA)$	21.034	+2.049	+2.888	-3.748	-0.839
$\Pr(oA)$	14.308	-0.374	-0.303	+3.964	-0.072
$\Pr(A)$	35.342	+1.675	+2.586	+0.215	-0.911
$\Pr(oA \cup oB)$	54.660	-1.577	-1.923	+8.793	+0.346
Panel B. Transfers and age-1 balance sheets					
Mean f	18.831	-0.153	+0.000	+18.678	-0.153
Age-1 mortgage debt	6.181	-0.101	-0.090	-0.706	-0.012
Age-1 home equity	12.838	-0.223	-0.112	+7.478	-0.111
Age-1 NFA	-0.858	+0.167	+0.080	+6.017	+0.087
Panel C. Prices, fiscal scale, and macro outcomes					
P_A	1.837	+0.007 (+0.40%)	+0.010 (+0.58%)	+0.070 (+3.94%)	-0.003
R_A	0.311	+0.009 (+3.02%)	+0.010 (+3.29%)	-0.002 (-0.68%)	-0.001
T	0.434	-0.081 (-18.60%)	-0.083 (-19.47%)	+0.008 (+2.42%)	+0.003
Policy spending/tax revenue (%)	0.000	+2.053	+2.063	-0.010	-0.010
Aggregate output	0.846	+0.000	+0.001	+0.001	-0.000
Group	P relative to B			P^0 relative to B^0	
Panel D. Pairwise consumption-plus-housing CEVs					
All age 1		+0.096			-0.220
Income-eligible age 1		+0.069			-0.306
Income-ineligible age 1		+0.230			+0.305
All living		+0.409			+0.307

Notes: Interactions equal $(P - B) - (P^0 - B^0)$. Probability levels are percentages and changes are percentage points; $oA \cup oB$ denotes ownership in either region. The four choice-share changes sum to zero and describe net allocation, not household transitions. Spending/tax revenue is in percent, with differences in percentage points. Parentheses beside prices and T give percentage changes relative to the comparison economy. Balance sheets are means over all age-1 households; monetary values are RMB 10,000. Positive CEVs indicate welfare costs under Appendix D.4. Income-eligible groups include households that do not claim the subsidy. Additional results appear in Table F.6.

Table F.6: Affordable Rental Access: Supplementary Program, Equilibrium, and Welfare Outcomes

Panel A. Program scale and recipient outcomes					
Outcome	Age 20–29		Age 30–39	Ages 20–39	
Population mass	0.167		0.167	0.333	
Mean recipient housing size	38.37		32.89	36.87	
Mean subsidized area (m ²)	37.14		32.77	35.94	
Pr($h > 70$ recipient)	0.040		0.011	0.032	
Mean market-rent payment	12.286		10.532	11.807	
Mean tenant net-rent payment	9.908		8.434	9.505	
Policy spending / output (%)	0.154		0.051	0.205	
Panel B. Age-1 distributional effects, $P - B$					
Outcome	Income eligible		Income ineligible	All age 1	
$\Delta \text{Pr}(\text{own} \mid A)$ (pp)	-4.086		+0.172	-2.843	
Δ mean housing size	+0.760		-0.425	+0.404	
CEV: P relative to B (%)	+0.069		+0.230	+0.096	
Panel C. General-equilibrium and transfer interactions					
Outcome	Baseline B	$P - B$	$P^0 - B^0$	$P - P^0$	Interaction
Pr(own A)	40.485	-2.843	-2.932	+10.550	+0.089
Pr($f > 0$)	52.906	-0.373	0.000	+52.533	-0.373
Age-1 liquid assets	5.322	+0.066	-0.009	+5.311	+0.075
Effective labor in A	0.309	+0.001	+0.002	+0.004	-0.001
Panel D. Pairwise consumption-plus-housing CEVs					
Group	B^0 relative to B			P^0 relative to P	
All age 1	+23.333			+23.023	
Income-eligible age 1	+28.065			+27.704	
Income-ineligible age 1	+0.113			+0.188	
All living	+10.777			+10.681	

Notes: Panels collect outcomes omitted from Table F.5. Probability effects are percentage points. Monetary quantities are in units of 10,000 RMB. Spending/output uses monetary ten-year output, $Y^{\text{mon}} = 50Y$, as defined in Appendix A.6. Age-specific spending is divided by aggregate output in the policy economy. CEV entries are percentages.

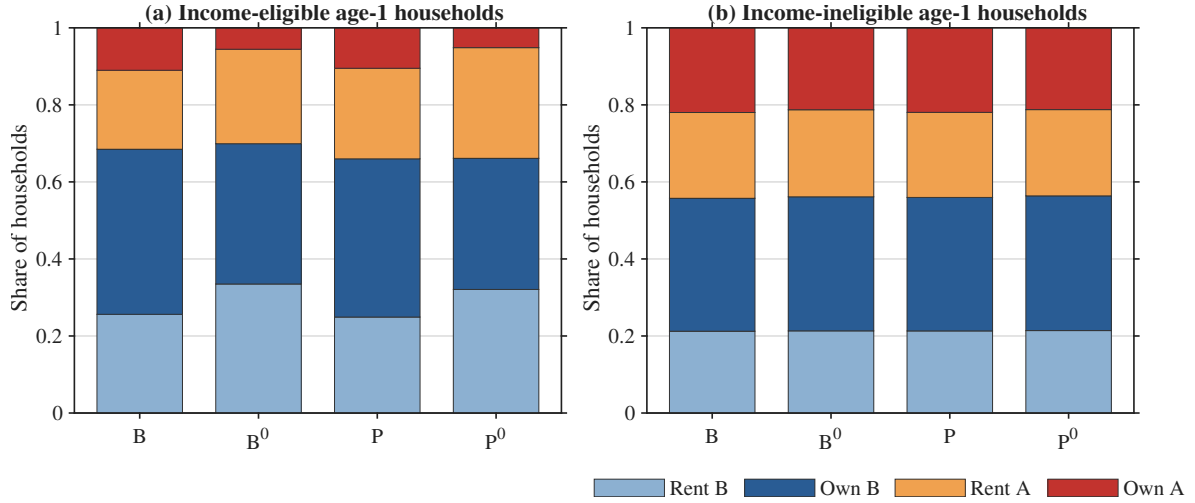


Figure F2: Affordable Rental Access and Age-1 Location-Tenure Choices
Notes: Panel (a) reports age-1 households whose potential Region A income is at or below the frozen baseline 60th-percentile threshold; Panel (b) reports the remaining age-1 households. B and B^0 are the baseline equilibria with and without family transfers, while P and P^0 are the corresponding affordable-rent equilibria. Each bar sums to one.

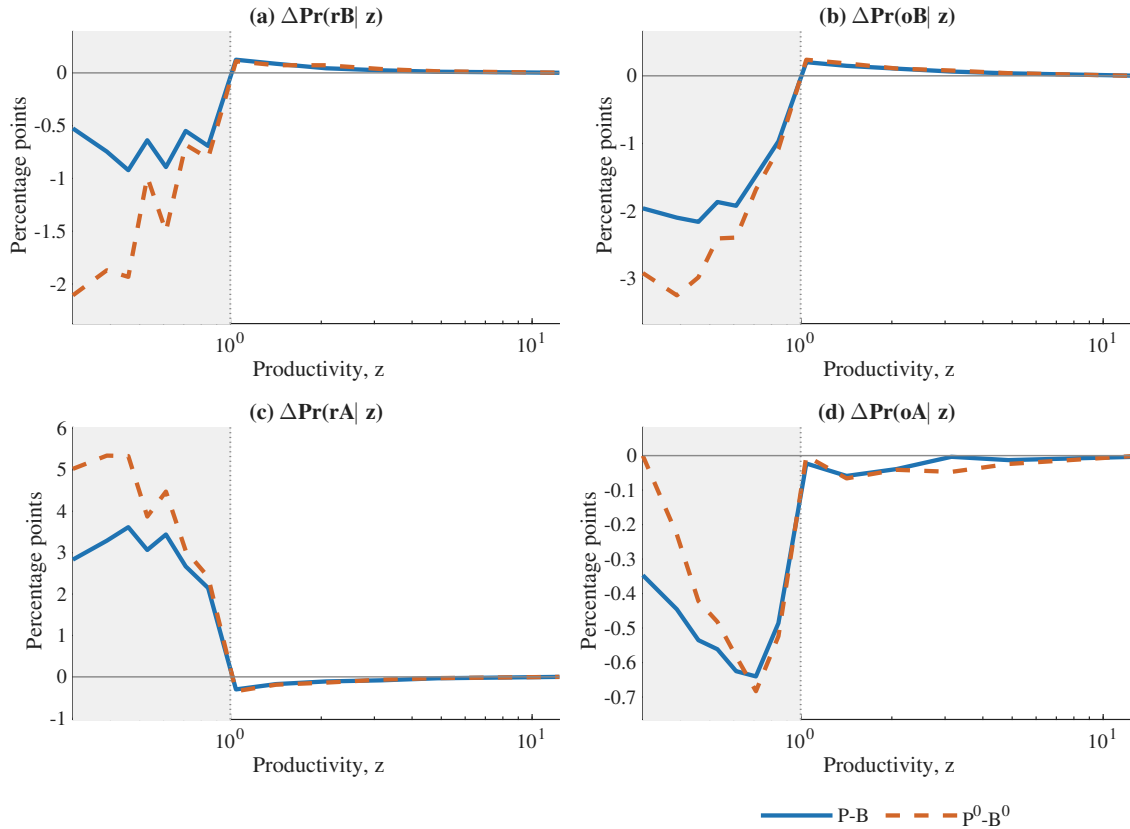


Figure F3: Affordable Rental Access: Age-1 Choice Responses over the Income Distribution

Notes: Solid blue lines report $P - B$, and dashed orange lines report $P^0 - B^0$. Panels (a)–(d) show percentage-point changes in rB , oB , rA , and oA . Gray shading marks the income-eligible range, and the dotted line marks the frozen baseline 60th-percentile cutoff.

F.3 Talent Attraction

Tables F.7, F.8, and F.9 report equilibrium effects, targeting and distributional incidence, and spatial reallocation by age, respectively. Figures F.4 and F.5 show talent-specific choice responses and the age profiles of residence and effective labor. Figure F.6 presents the common-shock choice-switching diagnostic.

Table F.7: Talent Subsidy: Full Spatial-Reallocation and Incidence Results

Outcome	Baseline B	Policy effect with transfers $P - B$	Policy effect without transfers $P^0 - B^0$	Effect of transfers in policy economy $P - P^0$	Additional policy effect due to transfers (interaction)
Panel A. Age-1 location and tenure					
$\Pr(rB)$	24.306	-0.170	+0.174	-5.881	-0.343
$\Pr(oB)$	40.352	-0.739	-0.974	+4.647	+0.236
$\Pr(rA)$	21.034	+0.057	-0.023	-2.830	+0.079
$\Pr(oA)$	14.308	+0.852	+0.823	+4.064	+0.028
$\Pr(A)$	35.342	+0.908	+0.800	+1.234	+0.108
$\Pr(oA) + \Pr(oB)$	54.660	+0.113	-0.151	+8.711	+0.264
Panel B. Cohort reallocation and aggregate composition					
Region A share, ages 1–2	35.413	+0.681	+0.563	+1.579	+0.118
Region A share, ages 3–6	32.792	-1.217	-1.262	+1.171	+0.045
Region A share, all ages	33.666	-0.584	-0.654	+1.307	+0.069
Mean G_gz in Region A	1.108	+0.026	+0.027	-0.032	-0.00185
Effective labor in A	0.309	+0.00405	+0.00327	+0.00584	+0.00078
Output in A	0.423	+0.00555	+0.00448	+0.00801	+0.00107
Aggregate output	0.846	+0.00150	+0.00121	+0.00216	+0.00029
Panel C. Transfers, balance sheets, prices, and fiscal effects					
Mean family transfer	18.831	+1.058	+0.00000	+19.889	+1.058
Age-1 mortgage debt	6.181	+0.854	+0.745	-0.585	+0.109
Age-1 home equity	12.838	+1.355	+1.069	+7.875	+0.286
Age-1 NFA	-0.858	+0.330	+0.028	+6.232	+0.302
P_A	1.837	+0.061	+0.059	+0.075	+0.00210
R_A	0.311	+0.00400	+0.00445	-0.00176	-0.00046
T	0.434	-0.761	-0.760	+0.00469	-0.00095
Policy spending / tax revenue (%)	0.000	+18.205	+18.109	+0.096	+0.096
Panel D. Pairwise consumption-plus-housing CEVs (percent)					
Group	P relative to B		P^0 relative to B^0		
All age 1	+1.340		+2.884		
Top 20 percent	-7.350		-8.014		
Bottom 80 percent	+2.227		+3.762		
All living	+2.367		+3.276		

Notes: Probability levels and effects are percent and percentage points; other rows are in levels unless indicated. Spending/tax revenue is in percent, with differences in percentage points. The interaction compares the policy effect with and without family transfers. It is distinct from $P - P^0$, which measures the total role of transfers in the policy economy. A positive interaction means that the policy has a larger effect when family transfers are available. Mean G_gz and effective labor use working ages 1–5. CEVs are nonlinear and reference-dependent and are therefore reported only as pairwise comparisons.

Table F.8: Talent Subsidy: Targeting, Take-Up, and Distributional Incidence

Age	Top-20 eligible mass	Recipient share within age	Take-up among eligible	Mean subsidy per recipient	Total policy spending	Spending/ tax revenue	ΔT
Panel A. Policy scale							
Age 1	0.0333	10.60%	52.99%	30.00	0.530	12.50%	-0.761
Age 2	0.0157	4.84%	51.44%	30.00	0.242	5.70%	-0.761
Ages 1–2 pooled	0.0490	7.72%	52.50%	30.00	0.772	18.21%	-0.761
Group	Expected direct subsidy	$\Delta \text{Pr}(A)$	$\Delta \text{Pr}(oA)$	ΔPr (owner)	Δ home equity	Δ mean family transfer	Pairwise CEV
Panel B. Age-1 incidence by talent eligibility							
Top 20%	15.897	+7.380	+6.411	+1.307	+5.809	-0.223	-7.350%
Bottom 80%	0.000	-0.710	-0.538	-0.185	+0.242	+1.379	+2.227%
All age 1	3.179	+0.908	+0.852	+0.113	+1.355	+1.058	+1.340%

Notes: The top-20 group is fixed by the probability-weighted baseline talent distribution. Recipient share within age divides recipients by the entire age group; take-up divides recipients by households that remain eligible for the once-per-lifetime payment. Probability effects are percentage points. Monetary quantities are in model units of 10,000 RMB. Bottom-80 households receive no direct subsidy but face equilibrium changes in housing costs, T , and family transfers. Positive CEV means that the policy economy requires additional consumption and housing to match its baseline reference.

Table F.9: Talent Subsidy: Spatial Reallocation by Age

Panel A. Location and tenure effects						
Age	$P - B$			$P^0 - B^0$		
	$\Delta \text{Pr}(A)$	$\Delta \text{Pr}(rA)$	$\Delta \text{Pr}(oA)$	$\Delta \text{Pr}(A)$	$\Delta \text{Pr}(rA)$	$\Delta \text{Pr}(oA)$
20–29	+0.908	+0.057	+0.852	+0.800	-0.023	+0.823
30–39	+0.454	+0.002	+0.453	+0.325	+0.087	+0.238
40–49	-0.494	-0.128	-0.366	-0.626	-0.074	-0.552
50–59	-0.946	-0.198	-0.749	-1.022	-0.195	-0.828
60–69	-1.547	-0.282	-1.266	-1.558	-0.272	-1.287
70–79	-1.881	-0.283	-1.598	-1.841	-0.258	-1.583
Panel B. Region A effective-labor effects						
Age	$\Delta L_A, P - B$			$\Delta L_A, P^0 - B^0$		
20–29	+0.00303			+0.00310		
30–39	+0.00266			+0.00236		
40–49	+0.00020			-0.00013		
50–59	-0.00072			-0.00089		
60–69	-0.00112			-0.00117		
70–79	0.00000			0.00000		

Notes: Probability effects are percentage points. Effective-labor entries are age-weighted contributions. Each comparison uses its own stationary equilibrium. The common-shock switching diagnostics are reported in Figure F.6.

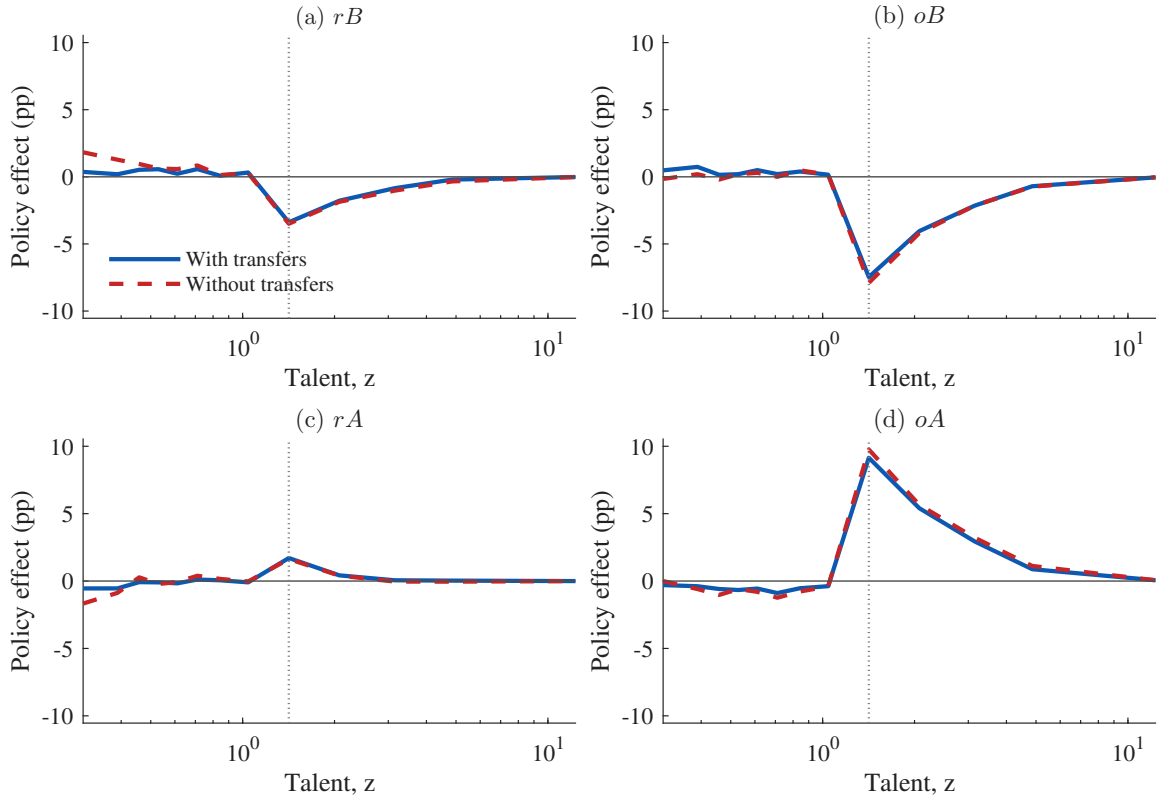


Figure F.4: Talent Subsidy: Age-1 Location–Tenure Responses over Talent
Notes: Solid blue lines report $P - B$, and dashed red lines report $P^0 - B^0$. Effects are percentage points. The vertical dotted line marks the first top-quintile eligible node.

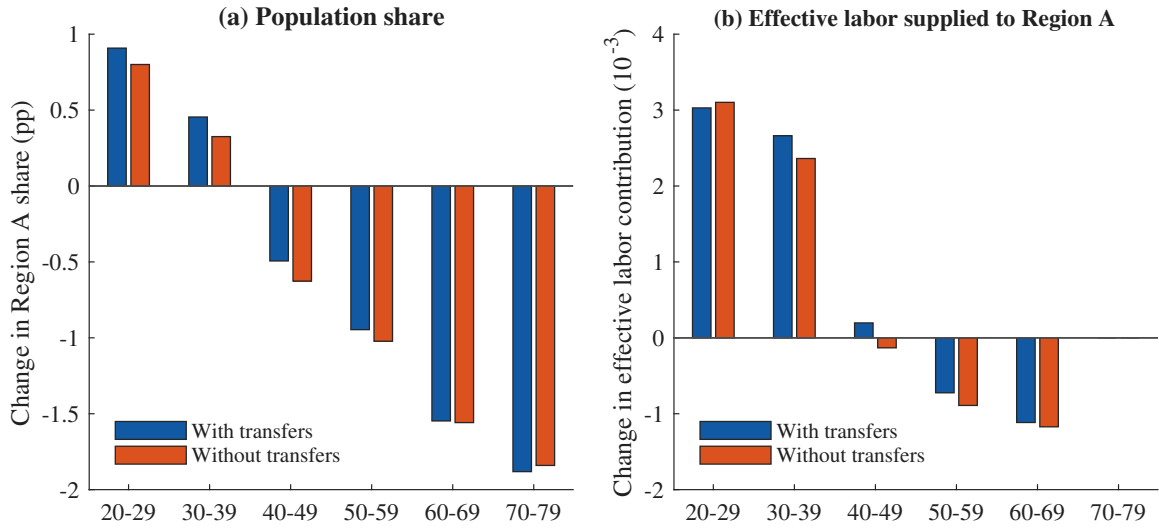


Figure F.5: Talent Subsidy: Region-A Population and Effective Labor by Age
Notes: The panels report policy-induced changes by age with family transfers, $P - B$, and without family transfers, $P^0 - B^0$. Population effects are percentage points; effective-labor effects are levels.

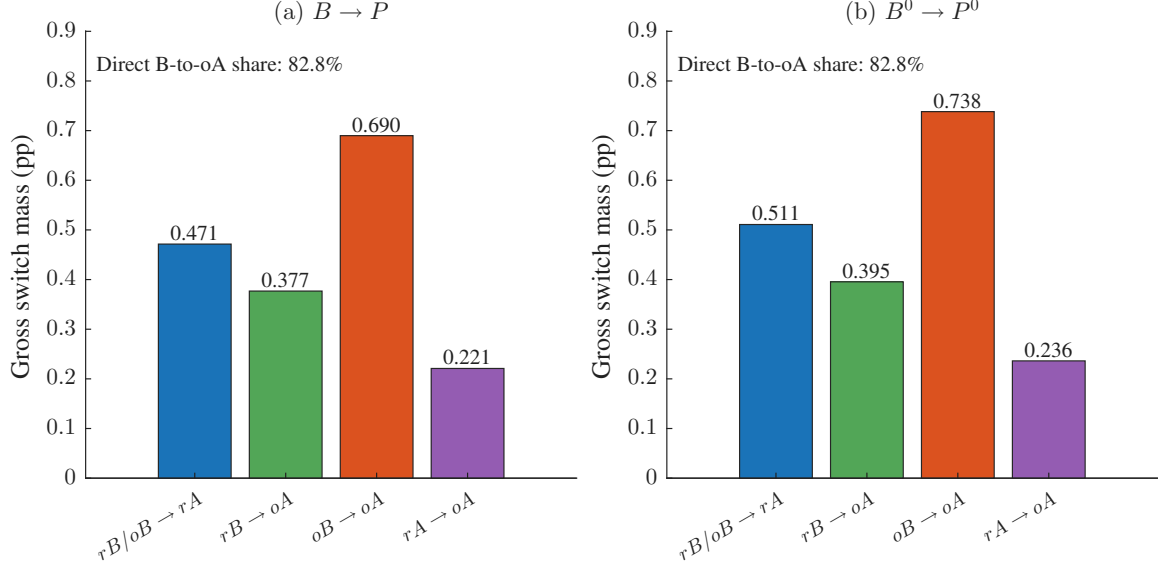


Figure F.6: Talent Subsidy: Model-Implied Age-1 Choice Switching

Notes: The bars report same-state switches generated by common Type-I extreme-value taste shocks. Panel (a) compares B with P using B age-1 state weights; Panel (b) compares B^0 with P^0 using B^0 weights.

F.4 Purchase Voucher

Tables F.10, F.11, F.12, and F.13 report equilibrium outcomes, buyer financing, age profiles, and housing-market and parental-transfer accounting, respectively. Figures F.7, F.8, F.9, and F.10 show cutoff responses, age profiles, same-state common-shock switching, and family-transfer effects.

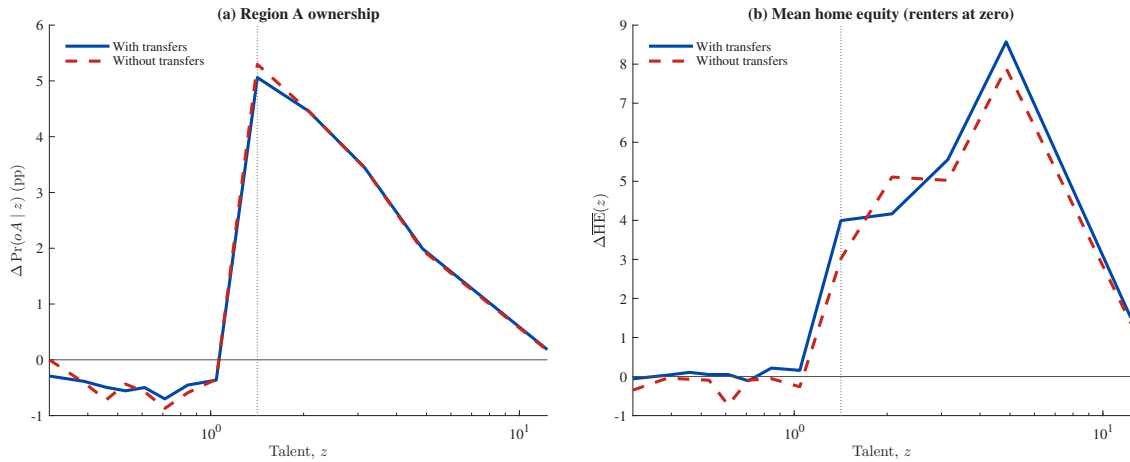


Figure F.7: Purchase-Voucher Effects around the Talent Cutoff

Notes: Solid blue lines report $P - B$, and dashed red lines report $P^0 - B^0$. Panel (a) shows the age-1 effect on Region A ownership. Panel (b) shows the effect on mean end-of-age-1 home equity, including renters at zero. Choice effects are percentage points, and home equity is in units of RMB 10,000. The vertical dotted line marks the first eligible top-quintile talent node.

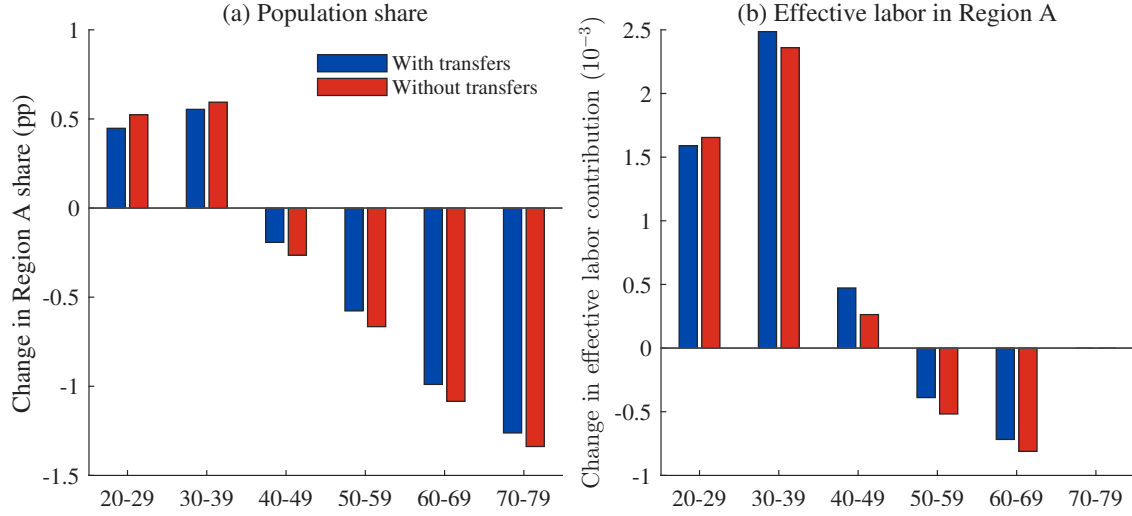


Figure F.8: Purchase Voucher: Region-A Population and Effective Labor by Age
Notes: The panels report policy-induced changes by age with and without family transfers. Population effects are percentage points; effective-labor effects are levels.

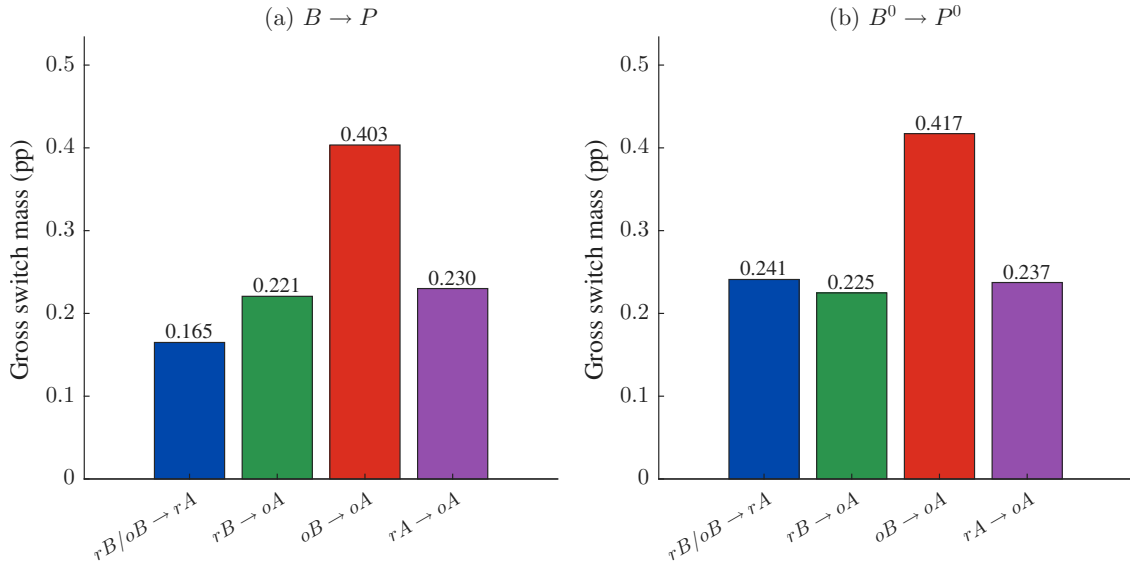


Figure F.9: Purchase Voucher: Model-Implied Age-1 Choice Switching
Notes: Bars report same-state choice switches generated from aligned age-1 states and common Type-I extreme-value taste shocks. Panel (a) compares B with P using B state weights; Panel (b) compares B^0 with P^0 using B^0 state weights.

Table F.10: Purchase Voucher: Full Allocation, Market, and Welfare Results

Outcome	Baseline B	Policy effect with transfers $P - B$	Policy effect without transfers $P^0 - B^0$	Transfer effect under policy $P - P^0$	Policy-transfer interaction
Panel A. Age-1 location and tenure					
$\Pr(rB)$	24.31	-0.06	+0.06	-5.66	-0.12
$\Pr(oB)$	40.35	-0.38	-0.58	+4.61	+0.20
$\Pr(rA)$	21.03	-0.03	+0.04	-2.98	-0.07
$\Pr(oA)$	14.31	+0.48	+0.48	+4.03	-0.00
$\Pr(A)$	35.34	+0.45	+0.52	+1.05	-0.08
$\Pr(oA) + \Pr(oB)$	54.66	+0.10	-0.10	+8.64	+0.19
Panel B. Age-1 housing and family finance					
Housing wealth	19.018	+1.893	+1.538	+7.249	+0.354
Mortgage debt	6.181	+0.972	+0.907	-0.629	+0.066
Home equity	12.838	+0.921	+0.632	+7.878	+0.289
NFA	-0.858	-0.772	-0.928	+6.087	+0.156
Mean family transfer	18.831	+0.550	+0.000	+19.381	+0.550
Panel C. Housing markets and fiscal incidence					
P_A	1.837	+0.042	+0.041	+0.073	+0.001
R_A	0.311	-0.004	-0.004	-0.001	+0.000
T	0.434	-0.465	-0.453	-0.007	-0.012
Policy spending / tax revenue (%)	0.00	+11.21	+10.84	+0.37	+0.37
Panel D. Spatial composition and productivity					
Working-age mass in Region A	0.279	-0.001	-0.001	+0.012	+0.000
Mean G_gz in Region A	1.108	+0.017	+0.018	-0.030	-0.000
Effective labor in Region A	0.309	+0.003	+0.003	+0.006	+0.000
Output in Region A	0.423	+0.005	+0.004	+0.008	+0.001
All-age Region A population share	33.67	-0.34	-0.37	+1.27	+0.04
Aggregate output	0.846	+0.001	+0.001	+0.002	+0.000
Panel E. Pairwise consumption-housing CEV (percent)					
Group	—	P relative to B	P^0 relative to B^0	—	—
All age 1	—	+0.99	+1.62	—	—
Top 20 percent	—	-3.39	-3.50	—	—
Bottom 80 percent	—	+1.44	+2.04	—	—
All living	—	+1.57	+2.14	—	—
Actual age-1 recipients	—	-7.17	-7.59	—	—

Notes: B and B^0 are the baseline equilibria with and without family transfers; P and P^0 apply the voucher in those environments. The interaction is $(P - P^0) - (B - B^0) = (P - B) - (P^0 - B^0)$. Choice and population effects are percentage points. Spending/tax revenue is in percent, with differences in percentage points. Balance-sheet outcomes are age-1 means; housing-market and macro outcomes are aggregate. Working ages are ages 1–5, and mean G_gz is conditional on working-age residence in Region A. Positive CEV means the evaluated economy requires that percentage increase in consumption and housing to match its reference. CEVs are nonlinear and reference-dependent and are therefore reported only as pairwise welfare comparisons. Actual-recipient rows use separately matched P - and P^0 -defined common-shock cohorts.

Table F.11: Purchase Voucher: Full Targeting, Buyer Financing, and Capitalization Results

	Age 20–29	Age 30–39	Ages 20–39	
Panel A. Eligibility, take-up, and fiscal scale				
Currently eligible with unused voucher (mass)	0.0333	0.0238	0.0571	
Recipient mass	0.0095	0.0092	0.0187	
Recipient share within age (%)	5.73	5.49	5.61	
Take-up among currently eligible unused (%)	28.64	38.49	32.74	
Mean voucher per recipient	25.41	25.40	25.40	
Voucher spending / tax revenue (%)	5.72	5.49	11.21	
Change in universal transfer, $T_P - T_B$	—	—	-0.465	
	B : top-20 oA buyers	P : voucher recipients	B^0 : top-20 oA buyers	
			P^0 : voucher recipients	
Panel B. Age-1 buyer financing				
Mean talent, z	2.882	2.736	2.897	2.741
Housing size (m ²)	58.06	67.61	56.35	66.46
Market house value	106.67	127.03	99.44	120.01
Voucher amount	0.00	25.41	0.00	24.00
Net purchase expenditure	106.67	101.63	99.44	96.01
Down payment after voucher	38.26	19.19	29.83	12.00
Origination mortgage	68.41	82.44	69.61	84.00
Regulatory LTV	0.657	0.665	0.700	0.700
Mortgage / net purchase cost	0.657	0.831	0.700	0.875
Mortgage payment / 10-year income	0.278	0.367	0.284	0.382
Mean family transfer	8.68	8.31	0.00	0.00
Transfer / down payment	0.227	0.433	0.000	0.000
End-of-age-1 home equity	51.01	59.95	42.80	51.65
Panel C. Price capitalization and effective discount				
Statutory discount				20.00 percent
P_A percentage change				2.27 percent
Effective fixed-unit discount after GE				18.18 percent
Discount erosion				1.82 percentage points
Share of statutory discount eroded				9.09 percent
Recipient net purchase-cost change				-4.73 percent
Recipient housing-size change				16.44 percent

Notes: The policy pays 20 percent of the full Region-A purchase value to top-quintile households making a new *oA* purchase at ages 20–39, at most once per lifetime. Age-2 eligibility uses the actual unused-voucher state. Regulatory LTV is measured against full market value; purchase expenditure and the down payment use the discounted cost, and mortgage debt cannot exceed that net cost. Recipients retain full resale proceeds without repayment or clawback. Monetary values are in units of 10,000 RMB. Recipient comparisons with baseline top-quintile *oA* buyers are descriptive and reflect selection into receipt.

Table F.12: Purchase Voucher: Spatial and Effective-Labor Effects by Age

Age	$P - B$			$P^0 - B^0$		
	$\Delta \Pr(A)$	$\Delta \Pr(oA)$	ΔL_A	$\Delta \Pr(A)$	$\Delta \Pr(oA)$	ΔL_A
20–29	+0.448	+0.480	+0.00159	+0.524	+0.483	+0.00165
30–39	+0.554	+0.580	+0.00249	+0.594	+0.442	+0.00236
40–49	-0.192	-0.246	+0.00047	-0.264	-0.387	+0.00026
50–59	-0.577	-0.573	-0.00039	-0.665	-0.667	-0.00052
60–69	-0.989	-0.967	-0.00072	-1.085	-1.038	-0.00081
70–79	-1.262	-1.221	0.00000	-1.338	-1.268	0.00000

Notes: Probability effects are percentage points and effective-labor entries are age-weighted contributions. Recipient financing is reported in Table F.11.

Table F.13: Purchase Voucher: Housing Markets and Parental-Transfer Responses

Panel A. Region A housing market

Economy	Owner demand	Rental demand	Total demand	Total supply	Rental share (%)	Rent / price
B	8.254	6.423	14.677	14.677	43.76	0.1692
P	8.505	6.392	14.897	14.893	42.91	0.1634
B^0	7.881	6.409	14.291	14.297	44.86	0.1769
P^0	8.126	6.379	14.505	14.513	43.98	0.1707

Panel B. Fixed-equilibrium parental-transfer response

Group	Mean Δf	Immediate MPT	Positive (%)	Zero (%)	Negative (%)
All linked children	-0.0363	-0.0249	0.11	99.63	0.27
Child top 20 percent	-0.1762	-0.0242	0.54	98.19	1.27
Child bottom 80 percent	-0.0013	—	0.00	99.99	0.01
Actual voucher recipients	-0.1869	-0.0074	0.52	98.17	1.31
Parent wealth Q1	0.0000	0.0000	0.00	100.00	0.00
Parent wealth Q4	-0.1428	-0.0839	0.30	98.70	1.00

Notes: Housing quantities are square meters per household in the full stationary population. Total demand sums owner and rental demand; total supply uses the partial-separation GE closure. The rental share is rental supply divided by total supply. Demand–supply gaps reflect the 10^{-3} market-clearing tolerance; all four regional housing markets satisfy it. Panel B reports $\Delta f = f^{\text{voucher}} - f^{\text{suppressed}}$, in RMB 10,000, at the fixed P equilibrium. Child voucher eligibility is suppressed at ages 1–2; these child problems and the linked age-4 parent problem are recomputed, holding prices, rents, T , and later continuation values fixed. Parents reoptimize giving and continuous controls at their P states, with their own discrete actions and action probabilities fixed. Immediate MPT is the ratio of the weighted transfer response to expected current age-1 child voucher payments in the reported group; it excludes prospective age-2 payments. Actual-recipient rows weight by receipt probability. Wealth is defined as in Appendix D.3, but quartile cutoffs use the beginning-of-age-4 distribution in P . Welfare results appear in Table F.10.

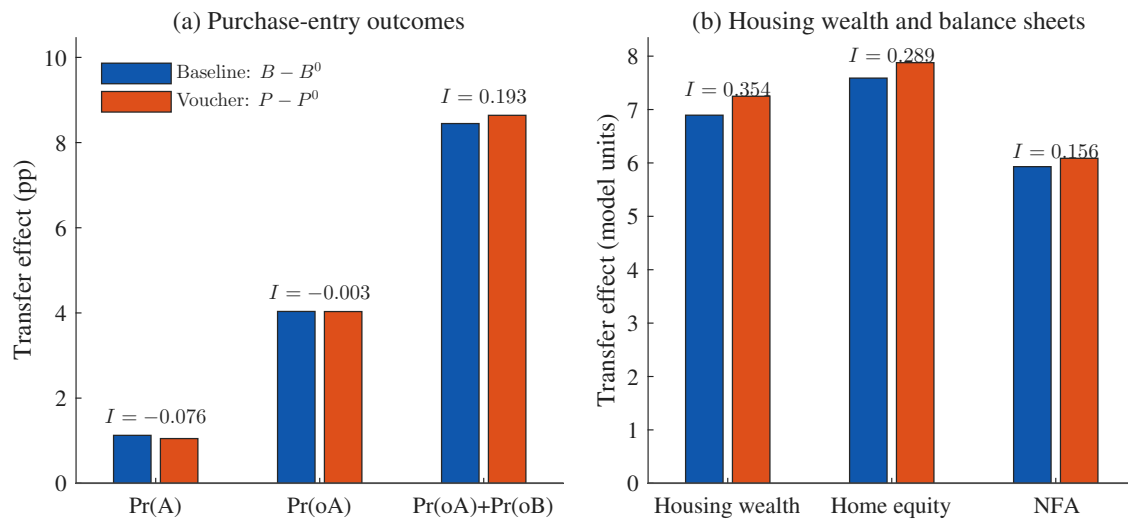


Figure F.10: Purchase Voucher: Family-Transfer Effects on Entry and Housing Wealth

Notes: Bars compare the family-transfer effect in the baseline, $B - B^0$, with its effect under the voucher, $P - P^0$. Numbers above paired bars report $(P - P^0) - (B - B^0)$.